Relax.js

A Deep Build Guide

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1. What we're building (and why)

1.1 The elevator pitch

Relax.js is a small frontend library built around a Virtual DOM (VDOM) and a component model. What makes this repository interesting is that it contains *two* rendering strategies:

1. A traditional VDOM runtime under `src/`.
2. An experimental compiler-assisted runtime (HRBR) designed to update the DOM through precomputed **slots**.

The goal of this book is not only to show *how to use* Relax.js, but to explain *how to build it*. We'll read the real source code, map it to a mental model, and connect every important concept to tests.

This isn't a conceptual "frameworks 101" book. It's a guided reverse-engineering and rebuild:

- You will learn the smallest set of abstractions needed to describe UI.
- You will see how those abstractions are represented in memory.
- You will trace how a rendering pass turns them into real DOM.
- You will follow how updates are scheduled and applied.
- You will use the test suite as the specification that locks behavior in place.

By the end, you should be able to (1) understand every important subsystem in this repository, and (2) rebuild something similar with confidence.

1.2 A tiny contract: what does Relax.js promise?

At a high level, Relax.js promises:

- You describe UI as a tree (VNode tree or compiled block).
- The runtime mounts it into a DOM host element.
- When state changes, only the necessary DOM updates happen.
- Components have lifecycle hooks and can emit/handle events.

Those promises are verified by tests under `src/__tests__/` and (for the HRBR runtime) under `runtime/__tests__/`.

1.2.1 A quick terminology primer

You'll see these words throughout the code and tests:

- **VNode**: a plain object that describes what DOM you want.
- **Mount**: create DOM from a VNode for the first time.
- **Patch**: update existing DOM by comparing old/new VNodes.
- **Destroy**: remove DOM nodes and detach listeners.
- **Host element**: the DOM element you mount the app into (often `document.body` in tests).
- **HRBR block**: a compiled representation: template HTML + dynamic *slots*.
- **Slot**: a precomputed "write location" (text/attr/prop/class/style/event) in a template.

1.3 Public API entry points (code references)

The *public* exports live in `src/index.ts`:

```
export { createApp } from './app'
export { defineComponent } from './component'
export { DOM_TYPES, h, hFragment, hSlot, hString } from './h'
export { nextTick } from './scheduler'
```

This tells you the learning order:

1. `createApp()` (app lifecycle)
2. `defineComponent()` (component model)
3. `h()/hFragment()` (VNode model)
4. `mountDOM()/patchDOM()/destroyDOM()` (DOM mechanics)
5. `nextTick()` (async scheduling in the VDOM runtime)

1.4 Why a second runtime? (HRBR in one paragraph)

The VDOM approach is flexible, but it has overhead: you allocate VNodes, diff lists, and interpret the tree every update. HRBR is the opposite trade-off:

- The compiler extracts a stable template (static HTML) once.
- Dynamic values become **slots** with stored DOM paths.
- Updates become *direct writes* to known nodes.

You'll see HRBR show up in VDOM code via `DOM_TYPES.HRBR` and `hBlock()` in `src/h.ts`.

1.4.1 What problem does HRBR try to solve?

If a UI subtree is structurally stable (same elements in the same shape, only values change), a classic VDOM update does extra work:

- allocate and traverse VNodes on every update
- interpret the tree to find where to write
- diff children arrays (sometimes expensive)

HRBR moves some of that cost earlier ("compile time"):

- build a static DOM template once
- record paths to dynamic locations (slots)
- update by writing directly to those DOM nodes

That doesn't make VDOM obsolete. It creates a spectrum: VDOM is the flexible general case; HRBR is an optimized fast path when the structure is known.

1.5 How we'll prove correctness throughout this book

This repository uses **Vitest** and **jsdom**. Our rule is simple: every time we explain a behavior, we should be able to point to a unit test.

Examples:

- Mounting basics: `src/__tests__/mount-dom.test.ts`
- Patching/diffing: `src/__tests__/patch-dom.test.ts`
- Components + lifecycle: `src/__tests__/component*.test.ts`
- HRBR block runtime: `runtime/__tests__/block.test.ts`

1.5.1 What "tests as spec" means in this repository

You should treat tests here as *contracts*. When you change runtime behavior, you don't just "fix failing tests"—you decide whether the contract changed.

Most tests follow the same simple pattern:

1. Set up a small DOM (usually `document.body`).
2. Build a VNode tree (or HRBR block).
3. Mount, patch, or destroy.
4. Assert on resulting `innerHTML` or element identity.

That gives you two superpowers as a reader:

- You can infer semantics without reading every implementation detail.
- You can reproduce bugs by writing a tiny test first.

1.6 Walkthrough: the smallest end-to-end loop (with real code)

Let's walk through the smallest app lifecycle you can observe in one sitting.

1.6.1 createApp(): mount and unmount

The application API is defined in `src/app.ts`. The critical part is that `createApp()` remembers three pieces of state:

- `parentEl`: where we mounted
- `isMounted`: guard against double mount/unmount
- `vdom`: the root VNode tree

When you mount:

1. it throws if already mounted
2. it creates the root VNode by calling `h(RootComponent, props)`
3. it calls `mountDOM(vdom, parentEl)`

When you unmount:

1. it throws if not mounted
2. it calls `destroyDOM(vdom)`
3. it resets internal references

This is a small but important architectural choice: *the App owns the root VNode*. That makes "unmount" a reliable cleanup boundary.

1.6.2 The test that proves it works

Open `src/__tests__/app.test.ts`. It reads like a story:

- It creates an app via `createApp(App, { todos: [...] })`.
- It mounts to `document.body`.
- It waits for `nextTick()` to flush async work.
- It asserts a stable HTML snapshot via `singleHtmlLine`...`.

Notice the discipline here: the tests make DOM updates predictable by explicitly awaiting `nextTick()`. That foreshadows Chapter 10, where we explain why scheduling exists and how it interacts with DOM patching.

1.7 Walkthrough: patching is "same element, new values"

Relax's VDOM runtime is built around a simple principle:

> If the old and new nodes are the "same kind" of node, we reuse the existing DOM element and update attributes/children. If they are different, we replace.

The test suite encodes this principle over and over. For example, `src/__tests__/patch-dom.test.ts` starts with extremely small cases:

- **no change**: patching identical trees should reuse `el`.
- **change the root node**: patching `div -> span` replaces the root element.
- **patch text**: `"foo" -> "bar"` updates only the Text node.

Those tests tell you what to look for when reading `src/patch-dom.ts`: element identity reuse is a correctness requirement, not an optimization.

1.8 Mini-exercise (with intent)

Before moving on, do this intentionally (don't just skim):

1. Read `src/app.ts` and answer: what runtime calls does `createApp()` delegate to?
2. Read `src/__tests__/app.test.ts` and find every `await nextTick()`. For each one, explain what DOM change it is waiting for.
3. Read the first 5 tests in `src/__tests__/patch-dom.test.ts`. For each, write a one-sentence rule (e.g., "patching text updates textContent without replacing parent elements").

1.9 Online resources (optional, but good context)

This book is grounded in the Relax.js repository, but it helps to know the broader ecosystem it draws from. These are references you can read alongside chapters; they are not prerequisites.

- Virtual DOM background (high-level): React — Render and Commit.
- The diffing strategy commonly used in VDOM children reconciliation: Neil Fraser, "An O(ND) Difference Algorithm and Its Variations".
- Fine-grained reactivity conceptual background (signals): SolidJS docs: Signals.
- Scheduling and keeping UIs responsive: MDN: requestAnimationFrame, MDN: MessageChannel.
- Testing DOM code in Node: Vitest Guide, jsdom.

If you'd like, we can make these "real" citations later by adding a bibliography system (BibTeX or biblatex) to `book/main.tex`.

2. Repo tour & project constraints

2.1 What you're looking at

This repository is intentionally small, but it isn't a single "runtime". It is a collection of cooperating subsystems, each with its own contract and its own tests.

At a high level, you should think of Relax.js as three products living in one repo:

1. **Relax VDOM runtime** (the teaching-friendly baseline) under `src/`.
2. **HRBR runtime** (experimental, compiled fast-path) under `runtime/`.
3. **Compiler** (experimental transform from TSX/JSX subset to HRBR blocks) under `compiler/`.

Everything else exists to support one of these (docs, benchmarks, build tooling).

2.1.1 Directory map (first pass)

This repository is split into a few major areas:

- `src/`: the Relax.js Virtual DOM runtime.
- `runtime/`: an experimental runtime (HRBR) based on signals, scheduling, and compiled blocks.
- `compiler/`: a compiler/transform that converts a JSX subset into HRBR block definitions.
- `benchmarks/`: a browser harness to compare performance.
- `docs/`: design notes and whitepapers.

Two more folders matter when you're thinking about correctness:

- `src/`$_{tests/}$$:theVDOMruntimespec.$
- `runtime/`$_{tests/}$$and$`compiler/`$_{tests/}$ $: theHRBR + compilerspec.$

2.2 The constraint that shapes everything: DOM correctness in jsdom

Almost every implementation choice is constrained by one hard rule:

> The runtime must be correct in a headless DOM (jsdom), and tests are the contract.

That affects:

- Attribute vs property behavior (inputs/selects are tricky in jsdom).
- Event binding and cleanup.
- Diff algorithms (keyed moves must preserve node identity).

2.2.1 Why jsdom is a meaningful constraint

jsdom is *not* a full browser, and that is exactly why it is useful here. If your runtime depends on unspecified browser behavior or timing quirks, your test suite will become flaky.

Instead, tests in this repo focus on:

- DOM structure (`innerHTML`)
- identity (whether an element is reused or replaced)
- cleanup (event listeners and references are removed)

This shapes file boundaries: many helpers exist to make DOM updates deterministic and assertable.

2.3 How to read code in this repo (recommended path)

When you see a source file, immediately find its tests. That's the spec.

Recommended reading order:

1. `src/index.ts` (public exports)
2. `src/h.ts` (VNode model)
3. `src/mount-dom.ts` (mount)
4. `src/patch-dom.ts` (patch)
5. `src/destroy-dom.ts` (cleanup)
6. `src/component.ts` (components)

2.3.1 A second reading path: follow a user action

The fastest way to build intuitions is to follow a single user interaction end-to-end. For example: "typing in an input and clicking a button adds a todo".

You can trace that across:

- event normalization: `src/events.ts`
- component event binding: `src/dispatcher.ts` and `src/component.ts`
- scheduling: `src/scheduler.ts` (`nextTick()`)
- patching: `src/patch-dom.ts`

Doing this with the tests open beside you is the theme of this book.

2.4 A concrete example: createApp() is tiny on purpose

extttcreateApp() exists so the user has a simple lifecycle API. The full file is short: `src/app.ts`.

Key behavior:

- It creates a root VNode with `h(RootComponent, props)`.
- It mounts it once.
- It can unmount and fully cleanup.

2.4.1 Code reference

See `src/app.ts`. The central calls are `mountDOM(...)` and `destroyDOM(...)`.

2.4.2 Test reference (what proves the contract)

The high-level application lifecycle is specified by `src/`$_{tests}$$_{/app.test.ts.Thistestfileshowsafullmount-to-interactionflow:}$

- `It mounts the app.`
- `It waits for async work with await nextTick().`
- `It simulates user input by setting input.value and dispatching an input event.`
- `It simulates clicks via button.click().`
- `It asserts DOM structure with singleHtmlLine;..`.

`That single test is a great "reading harness" for learning how Relax wires components + events + patching.`

2.5 Project constraint: small API surface

`Relax exports only a few primitives (h, fragments, components, scheduler). Everything else is internal. That keeps the library small and also makes the tests more important.`

2.5.1 Where to see the supported surface area

`The public exports are in src/index.ts and mirrored by the exports map in package.json.`

`In other words:`

- `if it's not in src/index.ts, it's not a stable public API`
- `tests are free to depend on internals, but users shouldn't`

`This book leans on that design: we can explain the internals without promising them as user-facing contracts.`

2.6 Tests: how this repo measures correctness

This repository uses Vitest with a jsdom environment. The configuration lives in vitest.config

- test patterns: **/$_{tests}$$_{/**/*.test.ts(x)}$
- reporters: verbose (good for learning)
- environment: jsdom

2.6.1 The single most important helper: singleHtmlLine

Many tests assert against HTML strings. If you try to compare raw multi-line template literals to innerHTML, whitespace will fight you.

This repo solves that in src/$_{tests}$$_{/utils.ts}$$with$singleHtmlLine():

- it collapses whitespace
- it removes spaces between tags (>+< -> ><)
- it trims

This is a micro-lesson in test design: you want assertions that are strict about structure, but forgiving about irrelevant formatting.

2.6.2 Where to learn semantics fastest (test tour)

If you're new to the codebase, these tests teach the contracts in the right order:

- src/$_{tests}$$_{/mount-dom.test.ts}$$:mountbehavior,fragmentbehavior,saving$vdom.el$references.$
- src/$_{tests}$$_{/patch-dom.test.ts}$$:identityreusevsreplacement,attributeupdates,keyedchildren,andedgecases.$
- src/$_{tests}$$_{/destroy-dom.test.ts}$$:cleanupandidempotency,especially"removeeventlisteners".$
- src/$_{tests}$$_{/component.test.ts}$$:componentlifecycle,props/stateupdates,andeventbindingtothehostcomponent.$
- src/$_{tests}$$_{/jsx-react-syntax.test.tsx}$$:TSXcompatibilitycontract(className/onClick/fragments/components).$

Each of these will be revisited later in the book when we dive into the implementation files they protect.

2.7 Mini-exercise

Open these tests and write down what they prove (one sentence each). Then locate the implementation file they exercise.

- src/$_{tests}$/$_{app.test.ts}$
- src/$_{tests}$/$_{component.test.ts}$
- src/$_{tests}$/$_{mount-dom.test.ts}$
- src/$_{tests}$/$_{destroy-dom.test.ts}$

If you can answer these, you're ready for Part II.

2.8 Online resources (optional, citation-ready)

This chapter is about reading *this repo*, but a few external references help explain why these constraints and structures are common.

- DOM living standard (event/bubbling/capturing model): WHATWG DOM Standard.
- Event handling basics and differences between properties vs attributes: MDN: addEventList MDN: HTMLElement.
- Testing DOM in Node: Vitest Guide, jsdom.
- Module packaging and exports maps: Node.js docs: exports.

Later, if you want formal citations in PDF output, we can add a bibliography system to book/main.tex and convert these links into \cite{...} entries.

3. TypeScript + JSX prerequisites

3.1 The goal

Relax.js supports writing UI in TSX/JSX with a React-like syntax. But remember: JSX is only syntax. It becomes function calls (or, in TypeScript's modern mode, calls to a small "JSX runtime" module).

In Relax, the important function is still h() from src/h.ts. The TSX layer exists to make authoring more ergonomic and familiar.

This chapter answers four practical questions:

1. What does TypeScript emit for TSX in this repo?
2. Where is the TSX runtime implemented (and what does it do)?
3. Which React-like conveniences are supported (and which are not)?
4. How do tests lock the TSX contract in place?

3.2 How TSX works in this repository (TypeScript configuration)

The TSX behavior in this repo is primarily defined in tsconfig.json:

- jsx: "react-jsx"
- jsxImportSource: "relax-jsx"

This combination means:

- You *do not* configure a global JSX factory function.
- TypeScript emits imports from relax-jsx/jsx-runtime (and sometimes relax-jsx/jsx-dev-runti

So the contract becomes very clean: "TSX syntax must compile into objects that Relax can mount/patch/destroy."

3.3 Two JSX styles youll see

3.3.1 Direct VDOM authoring

You can write VNodes manually:

```
import { h, hFragment } from 'relax.js'

const view = hFragment([
  h('h1', {}, ['Hello']),
  h('button', { on: { click: () => console.log('clicked') } }, ['Click'])
])
```

3.3.2 React-like TSX authoring

TypeScript can compile TSX into calls to a configured JSX runtime. This repo uses a React-like authoring style where:

- className maps to Relaxs class.
- onClick maps to Relaxs event prop form (internally becomes on: { click: fn }).
- Fragments (<>...</>) are supported.

There's an important nuance here:

> Relax supports both "native" Relax props (like class and on: { click: ... }) *and* React-like compatibility props (like className and onClick).

This design keeps the underlying runtime simple (it only needs to understand one internal prop shape) while letting TSX authoring feel familiar.

3.4 The Relax JSX runtime (the real implementation)

The TSX runtime lives in relax-jsx/jsx-runtime.ts. It's intentionally tiny.

At a high level it does four jobs:

1. Normalizes children to an array.
2. Applies React-like prop compatibility:
   - className -> class
   - onClick/onInput/... -> on: { click/input/... }
3. Threads the optional key through into Relax props.
4. Delegates to Relax's real VDOM constructors: exttth(type, props, children) or hFragment(ch

3.4.1 Fragments: why Fragment is exported

In the modern TS transform, fragments compile to a special "Fragment" value. Relax's JSX runtime exports:

- export const Fragment = hFragment as any

Then jsxImpl() checks whether the element type is Fragment and returns hFragment(children). That makes fragment VNodes indistinguishable from ones created explicitly via hFragment().

3.4.2 Dev runtime: jsxDEV

Tooling sometimes emits jsxDEV calls in development builds. This repo includes relax-jsx/jsx-d which simply re-exports Fragment, jsx, and jsxs, and provides jsxDEV(...) as an alias to jsx(...).

This is not a performance hack; it's a compatibility bridge so Vite/Vitest can run TSX code in different modes.

3.5 Tests as spec

The definitive spec for supported TSX/JSX features lives in:

- src/$_{tests}$$_{/jsx-tsx.test.ts}$
- src/$_{tests}$$_{/jsx-react-syntax.test.tsx}$

3.5.1 Test 1: TSX can mount via the Relax JSX runtime

extttsrc/$_{tests}$$_{/jsx-tsx.test.ts}$$isthesmallestend-to-endproof:$

- It imports and runs examples/jsx/hello.tsx.
- That example defines a component whose render() returns TSX.
- It mounts using createApp(), waits nextTick(), and returns innerHTML.

If this test fails, it usually means one of these broke:

- TS didn't resolve relax-jsx/jsx-runtime the way we expect (config/alias problem).
- The TSX runtime produced VNodes in a shape mountDOM() doesn't handle.

3.5.2 Test 2: React-like conveniences are supported

extttsrc/$_{tests}$$_{/jsx-react-syntax.test.tsx}$$isafeaturematrix.It'sintentionallywrittenas"contracts"inacommentatthetopofthefile.$
It specifies (and proves) things like:

- intrinsic elements accept id, className, and text children
- fragments (<>...</>) work
- onClick attaches a DOM click handler
- list rendering via items.map(...) works (including key)
- patching TSX trees updates DOM without reconstructing the whole host
- TSX works for Relax components created via defineComponent()

This test file is a great example of how to design compatibility: pick the exact subset you want, and encode it as executable assertions.

3.6 Whats not the same as React

Even though the syntax is familiar, semantics are Relaxs:

- Components are created via defineComponent().
- The event binding model binds handlers to the host component instance.
- Slots in Relax are a component feature (external/default content), not React children.

3.6.1 Practical limitations of the TSX subset

This repo intentionally does *not* try to be a full React clone. Two examples:

- The TSX runtime only rewrites "event props" when the value is a function. If you pass non-function values to onClick, you should expect it to behave as a normal prop.
- "React children" as a conceptual model is not the point. In Relax, children are just VNodes (or arrays of VNodes) passed to h().

The tests are the source of truth for what is supported. If you'd like to extend the supported subset, the safest workflow is:

1. Add a failing test first in src/$_{tests}$$_{/jsx-react-syntax.test.tsx}$.

1. Update relax-jsx/jsx-runtime.ts to normalize the new case.
2. Ensure mount/patch/destroy tests still pass.

3.7 Mini-exercise

Read src/$_{tests}$$_{/jsx-react-syntax.test.tsx}$$and answer$:

1. How are fragments compiled?
2. How are events passed?
3. How do components appear in JSX and how do props flow?

Then open relax-jsx/jsx-runtime.ts and locate the exact code that implements each answer.

3.8 Online resources (optional, citation-ready)

- TypeScript handbook: JSX (explains react-jsx and jsxImportSource): TypeScript: JSX.
- React JSX runtime reference (useful for understanding why jsx/jsxs/jsxDEV exist): React docs: jsx-runtime.
- Vite + Vitest module resolution (useful when JSX runtime imports fail): Vite: JSX, Vitest Guide.
- DOM events (what an onClick becomes): MDN: click event.

4. The VNode model (src/h.ts)

This chapter is the "data model" chapter of Relaxs renderer: before we can mount virtual UI into the DOM, we need a shape that can represent "text", "an element with attributes", "a component instance", and a few special cases (fragments, slots, and HRBR blocks).

Relax keeps this model intentionally small, and then makes the rest of the runtime depend on a single invariant:

> extbfInvariant: after creation, every child in VNode.children is a VNode (not a primitive) and every node has a stable type drawn from DOM_TYPES.

You can think of src/h.ts as the factory layer that enforces this invariant.

4.1 Why VNodes exist

A virtual DOM implementation needs a stable, serializable representation of UI. In Relax, that representation is the **VNode**. The renderer can then be split into three big phases:

1. **Create**: build a VNode tree (usually by calling h() directly, or indirectly via TSX as described in Chapter 3).
2. **Mount**: allocate real DOM nodes for that tree (src/mount-dom.ts).
3. **Patch**: reconcile an "old tree" with a "new tree" (src/patch-dom.ts), reusing as much DOM as possible.

The "create" step is deceptively important: the more regular the VNode tree is, the less branching the mount/patch code needs.

4.2 The VNode types: DOM_TYPES

Relax defines node kinds as string constants in DOM_TYPES (src/h.ts):

- TEXT: a text node (TextVNode).
- ELEMENT: an HTML/SVG element (ElementVNode).
- FRAGMENT: a list of siblings without an extra wrapper element (FragmentVNode).
- COMPONENT: a component constructor + props/children (ComponentVNode).
- SLOT: a placeholder for "external children" in component trees (SlotVNode).
- HRBR: a special vnode wrapping an HRBR "mount factory" (HrbrVNode).

Two details are worth calling out because they influence the rest of the runtime:

1. **Not every VNode has a DOM element.** TEXT uses a real Text node (el?: Text). ELEMENT uses a real HTMLElement (el?: HTMLElement). But FRAGMENT is special: it represents multiple siblings, so it cant point at a single "root node".
2. **Keys are part of identity.** Youll see props.key used later by the reconciliation algorithm to decide whether a node is "the same logical thing".

4.3 Contract: h(tag, props, children)

Relaxs primary constructor is h() (src/h.ts). Its intentionally close to "hyperscript" functions found in other VDOM systems.

Inputs

- tag: either a string (intrinsic element like div) or a component constructor/function.
- props: *must be a plain object*. Relax checks this using assert(...) and fails fast with a helpful message.
- children: *must be an array*. This is also asserted.

Output

- If typeof tag === 'string', h() produces an ELEMENT vnode.
- Otherwise it produces a COMPONENT vnode.

Even though Relax supports TSX, TSX still ends up calling this layer (via the relax-jsx runtime), so understanding h() is foundational.

4.4 Child normalization: withoutNulls() + mapTextNodes()

Relax does two child normalization passes inside h() and hFragment():

1. **Drop nullish children**: withoutNulls(children) removes null and undefined (see src/utils/arrays.ts). This means "conditional children" can be written as show ? node : null.
2. **Map primitives to TEXT nodes**: mapTextNodes() converts string | number | boolean | bigint | symbol into TextVNode via hString().

This is one of those design choices that pays rent everywhere:

- Mount code can check vNode.type and never needs to special-case "child is a string".
- Patch code can treat text nodes uniformly; it doesnt have to worry about coercion rules.

4.5 Fragments: hFragment(children) and extractChildren()

Fragments represent "multiple siblings" and are heavily used by TSX fragment syntax (<>...</>).

Constructing fragments

extthFragment(vNodes) asserts the input is an array, filters nullish children, and maps primitives to text nodes (exactly like h()).

Flattening fragments

Later, other parts of the runtime sometimes need to treat "a tree that may contain fragments" as a flat list (for example, when calculating insertion positions). extttextractChi provides that flattening:

- If it finds a fragment child, it recursively collects that fragments children.
- Otherwise it includes the child as-is.

The unit test extract children from a tree with fragments in src/$_{tests}$/$_{h.test.ts}$ $isyourexecutablespecforth$

4.6 Slots: hSlot(children) as a component-level placeholder

Slots are how Relax expresses "children passed into a component" without forcing the component to take opaque children and do manual array plumbing.

At the VNode level, a slot is just:

- type: DOM_TYPES.SLOT
- optional children: default content to use when no external children are provided.

Then, to connect slot VNodes to component semantics, src/h.ts keeps a small piece of module-scoped state: hSlotCalled.

Why the module flag exists

When rendering a component, Relax needs to know whether the component output contained at least one slot. Rather than scanning the render output tree after the fact, Relax uses this fast signal:

1. hSlot() sets hSlotCalled = true.
2. Component rendering checks didCreateSlot().
3. If a slot was created, the runtime calls the slot-filling algorithm (src/slots.ts) and then resetDidCreateSlot() to avoid leaking the flag into unrelated renders.

This is one of Relaxs "tiny global state" trade-offs: it keeps render fast and simple, but it also means slot creation is effectively a side-effect.

What the tests prove

The best way to internalize slot semantics is to read src/$_{tests}$/$_{component-slots.test.ts}$.$Afewkeycases$:

- **External content wins:** A component that renders hSlot() will receive the parents children in that position.
- **Default content:** hSlot([...default]) is used when the parent provides no children.
- **Empty slot:** if theres no external content and no default content, the slot becomes empty (the test asserts).
- **Conditional slots:** slot nodes can appear/disappear across re-renders, and patching handles it.

4.7 HRBR interop: hBlock(mount) and VDOM identity

HRBR (the compiled "block" runtime) needs to integrate into the same mounting/patching pipeline as regular VNodes. Relax does that by wrapping an HRBR mount factory in an HRBR vnode:

- hBlock(mount) returns { type: DOM_TYPES.HRBR, mount }.
- Mounting creates a host element region and calls mount(host).
- The returned instance exposes at least destroy(), and may expose update() / dispose().

Theres an important identity rule here, and the repo encodes it explicitly in areNodesEqual (src/nodes-equal.ts):

> extbfHRBR vnode identity is defined by mount factory identity: old.mount === new.mount.

That single line is the difference between:

- **Patch preserves instance:** stable mount function keep the same HRBR instance.
- **Patch remounts:** new mount function destroy and mount a fresh instance.

What the HRBR tests prove

Read src/$_{tests}$/$_{hrbr-integration.test.ts}$$asthecontract$:

- **Mount and destroy:** an HRBR block returned from a VDOM component mounts into the DOM and gets destroyed on destroyDOM.
- **Event semantics:** HRBR event slots bind handler this to the host Relax component instance (the test asserts calls[0].thisArg === vdom.component).
- **Patch behavior:** stable vs. changing mount factory identity changes whether the HRBR instance is preserved.

This is a great example of the books theme: treat tests as executable documentation.

4.8 VNode identity and diffing: a preview

This chapter is about creation, but VNode shape is designed with patching in mind. The function areNodesEqual() (src/nodes-equal.ts) shows you the identity rules Relax uses later:

- text nodes: always equal (Relax patches text by updating value rather than replacing node type)
- fragments: always equal
- elements: equal if tag matches and props.key matches
- components: equal if constructor matches and props.key matches
- HRBR: equal if mount factory function is the same

The tests in src/$_{tests}$$_{/node-equals.test.ts}$$enumeratethesecasesinsmall,sharpassertions.$

4.9 Mini-exercises

1. **Read like a maintainer:** In src/$_{tests}$$_{/h.test.ts}$$,locatethetestnamed$"h() maps 'number', 'boolean', 'bigi
1. **Write a new invariant test:** Add a small test that asserts h('div', , [null, undefined, 'x']).children.length === 1. (This should already be true, but the test helps lock the behavior down.)

4.10 Online resources

This chapter describes patterns that show up across many UI runtimes. These references help if you want to compare Relaxs choices to other ecosystems:

- Hyperscript and virtual node factories (conceptual background): https://github.com/hyperh
- DOM node types (Text vs Element) on MDN: https://developer.mozilla.org/en-US/docs/Web/API/
- Reacts reconciliation keys (why key exists): https://react.dev/learn/rendering-lists
- Web Components slots (a parallel mental model for "default vs external" content): https://developer.mozilla.org/en-US/docs/Web/API/Web$_{c}omponents/Using_{t}emplates_{a}nd_{s}lots$

5. Mounting DOM (src/mount-dom.ts)

This chapter is about the first half of the renderer: taking the data structures from src/h.ts and producing real DOM nodes.

In Relax theres a clean separation:

- **Creation** builds VNode trees (Chapter 4).
- **Mounting** turns VNodes into DOM and stores back-references (this chapter).
- **Patching** reconciles trees (Chapter 8).
- **Destroying** cleans up listeners / component instances / HRBR blocks (Chapter 9).

The entry point is mountDOM(vdom, parentEl, index?, hostComponent?) in src/mount-dom.ts.

5.1 Contract: mountDOM

Inputs

- vdom: a VNode (one of the DOM_TYPES).
- parentEl: the DOM element where the node should be inserted. This must be non-null.
- index (optional): insert position inside parentEl.childNodes.
- hostComponent (optional): the component instance that should become this for event handlers.

Outputs / side-effects

- Inserts DOM into parentEl.
- Mutates the vnode: for most node types, vdom.el becomes a back-reference to the created DOM.
- For elements, saves listener cleanup hooks on vdom.listeners.
- For components, creates and stores vdom.component.

Failure modes

- If parentEl == null, it throws [mountDOM] Parent element is null.
- If vdom.type doesnt match any known tag in DOM_TYPES, it throws Can't mount DOM of type: ....

The tests in src/$_{tests}$$_{/mount-dom.test.ts}$$includeacoupleof“guardrail”casesthatassertheseerrorsexist.$

5.2 The central pattern: dispatch by vdom.type

extttmountDOM() is a type-dispatcher. In code its a switch (vdom.type) that delegates to one helper per vnode kind:

- createTextNode for DOM_TYPES.TEXT
- createElementNode for DOM_TYPES.ELEMENT
- createFragmentNodes for DOM_TYPES.FRAGMENT
- createComponentNode for DOM_TYPES.COMPONENT
- createHrbrNode for DOM_TYPES.HRBR

This is not just an organizational trick: it is what makes patching symmetric later. Chapter 8 will mirror this shape with a patch dispatcher.

5.3 Text nodes: createTextNode

For TextVNode, mounting is:

1. Create a real text node: document.createTextNode(vdom.value).
2. Store a stable back-pointer: vdom.el = textNode.
3. Insert it into the parent using insert(textNode, parentEl, index).

What the tests prove

extttsrc/$_{tests}$$_{/mount-dom.test.ts}$$containstwosmallteststhatactlikeacontract$:

- mount a text element in a host element: asserts the DOM contains hello.
- save the created text element in the vdom: asserts vdom.el is a Text instance and points at the right content.

Read those tests as: "If we ever refactor mounting, these properties must remain true or patching/destroying will become unreliable."

5.4 Element nodes: createElementNode and the props/events split

For ElementVNode, Relax mounts in a deliberate order:

1. Create const element = document.createElement(tag).
2. Apply props and events via addProps(element, vdom, hostComponent).
3. Store vdom.el = element.
4. Mount children into element (recursively via mountDOM(child, element, null, hostComponent)
5. Insert the final element into the parent.

The most important concept here is that Relax treats events specially.

5.4.1 addProps: extracting attrs vs. events

extttaddProps() calls extractPropsAndEvents(vdom) (from src/utils/props.ts). The return value is split into:

- **attrs**: everything that becomes DOM attributes/properties (id, class, style, data-*, etc).
- **events**: the event map stored under props.on.

Then mounting wires two subsystems:

- addEventListeners(events, el, hostComponent) returns a map of cleanup functions; this is saved as vdom.listeners.
- setAttributes(el, attrs) applies the non-event attributes.

That vdom.listeners back-reference is critical later in Chapter 9: destroying uses it to remove listeners even if the vnode is no longer in the DOM.

What the tests prove

The tests in src/$_{tests}$/$_{mount-dom.test.ts}coverasmallbutmeaningfulmatrix$:

- **attributes**: id, class, list-of-classes
- **style**: a style object becomes an inline style string
- **events**: clicking the element calls the handler and vdom.listeners is populated

These are not "just tests"they are a spec for how Relax converts the VNode props object into real DOM behavior.

5.5 Event handler binding: the meaning of hostComponent

extttmountDOM() accepts an optional hostComponent argument, which gets threaded into createElementNode(...) and ultimately into addEventListeners(...).

The intent is: when a component mounts elements, those element event handlers should run with this === componentInstance so handler methods can mutate component state.

What the tests prove

The test where there is a host component, the event handlers are bound to it builds a fragment containing two buttons and passes a plain object comp = { count: 5 } as the host. Both click handlers increment this.count. The assertions show two things:

- the same host is used for nested mounts (a button nested inside a div still binds this correctly)
- binding is not limited to direct children

This is a key piece of "component semantics" even though its implemented entirely inside the mounting layer.

5.6 Fragments: `createFragmentNodes` and insertion indices

Fragments are the trickiest type to mount because a fragment is "many nodes".

Relaxs move is simple:

> A fragment vnode stores vdom.el = parentEl.

That is, a fragment doesnt own a DOM node. Instead, it keeps a reference to the container it was mounted in.

5.6.1 The index problem

When mounting into an explicit index, you want fragment children to appear starting at that position. But as soon as you insert the first child, the parents childNodes list changes, and the next childs target index must shift.

Relax solves this by maintaining a local counter idx:

- If idx == null, it just appends each child and doesnt care.
- Otherwise, after mounting each child, it increments idx by the number of DOM nodes that child contributed.

The "how many DOM nodes did this child contribute?" logic is where fragments and components matter:

- child fragment: contributes child.children.length
- child component: contributes child.component.elements.length
- other nodes: contribute 1

This design implies a subtle contract with the component implementation: component instances must maintain an elements array listing the DOM nodes they own. Youll revisit that in the component lifecycle chapters.

What the tests prove

extttsrc/$_{tests}{}_{/mount-dom.test.ts}containsseveralimportantfragmenttests$:

- mount a fragment in a host element and mount a fragment inside a fragment...: show that fragments flatten naturally at mount time.
- all nested fragments el references point to the parent element: asserts the design choice fragment.el === parentEl even for nested fragments.
- A parametrized test.each suite (later in the file) checks insertion-index behavior for complex nested fragment/component scenarios.

5.7 Components: `createComponentNode` and asynchronous `onMounted`

Component mounting bridges VDOM to component instances. The algorithm in createComponentNode() is:

1. Split vnode props into props and events via extractPropsAndEvents(...).
2. Allocate the instance: new Component(props, events, hostComponent).
3. Store external children: component.setExternalContent(vdom.children).
4. Delegate to the component to mount itself: component.mount(parentEl, index).
5. Store vdom.component = component.
6. Store vdom.el = component.firstElement.

The last step is important: a component vnode cannot point to "the component" as a DOM node, so Relax points it at the first DOM element the component produced. Patching uses this to locate and replace component output.

5.7.1 Lifecycle scheduling and the job queue

Immediately after creating a component node, mountDOM() enqueues a lifecycle callback:

```
extttenqueueJob(() => vdom.component.onMounted())
```

So onMounted() is intentionally asynchronous relative to "the DOM nodes now exist". This approach avoids re-entrancy hazards (mount causing mount causing mount) and aligns with how many UI systems schedule lifecycle effects.

What the tests prove

extttsrc/$_{tests}$/$_{mount-dom.test.ts}exercisescomponentcreationandparent-childlinks$:

- mount a component with props: shows that props are passed in and the instance is stored.
- mount a component with children: shows that children can be used to build deeper trees.
- mount a component with a parent component and child components keep a reference to their parent component: show that hostComponent threads the parent/child relationship.
- when a component with multiple elements is mounted, the vdom keeps a reference to the first element: locks down the vdom.el = firstElement rule.

5.8 HRBR interop: `createHrbrNode` and why `el` points to the host

HRBR nodes are mounted by allocating a dedicated host container element:

- Create a host.
- Mark it with a dataset flag: host.dataset.hrbr = '1'.
- Insert the host into the parent.
- Call the mount factory: instance = vdom.mount(host).
- Store vdom.instance = instance.
- Store vdom.el = host (and also vdom.host = host).

The comment in src/mount-dom.ts is worth repeating: **VDOM owns the host container, not the hosts first child.** Patch logic later locates nodes by oldVdom.el; if el pointed inside HRBR output, replacement/removal would get the wrong DOM index.

What the tests prove

While the mounting tests live in mount-dom.test.ts, HRBR mounting is exercised via integration tests in src/$_{tests}$$_{/hrbr-integration.test.ts}$.$ThosetesteststreatHRBRblocksas"justanothervnodetype"andassert$:

- the DOM output appears after mounting
- the block is destroyed when the VDOM is destroyed
- patching preserves vs remounts based on mount factory identity

5.9 Insertion mechanics: `insert(el, parentEl, index)`

Everything ultimately goes through insert(), which implements four rules:

- **append:** if index == null, append.
- **reject negative indices:** if index < 0, throw.
- **append out-of-range:** if index >= parent.childNodes.length, append.
- **insertBefore otherwise:** insert before childNodes[index].

The tests include a negative-index case (throws) and several index-sensitive fragment suites.

5.10 Mini-exercises

1. Add a focused test to mount-dom.test.ts proving that mounting with an out-of-range index appends for element vnodes. (Youll be asserting the same rule insert() implements.)
2. In mount-dom.test.ts, find the parameterized index tests and explain why nested components contribute component.elements.length to the index offset.

5.11 Online resources

- DOM basics: document.createElement (MDN): https://developer.mozilla.org/en-US/docs/Web/AP
- DOM basics: Node.insertBefore (MDN): https://developer.mozilla.org/en-US/docs/Web/API/Noc
- DOM basics: Node.appendChild / Element.append (MDN): https://developer.mozilla.org/en-US/
- Event targets and listener binding (MDN): https://developer.mozilla.org/en-US/docs/Web/API
- "Why schedule effects" background (Reacts mental model is a good reference even if implementation differs): https://react.dev/learn/synchronizing-with-effects

6. Attributes, properties, classes, and styles (src/attributes.ts)

This chapter is the "DOM edge" chapter. Mounting (Chapter 5) can create elements in a few lines, but *correctly configuring those elements* is where UI libraries accumulate complexity.

Relax keeps its attribute system intentionally small: a handful of rules, continuously validated by broad jsdom test coverage.

6.1 Why this chapter matters

Browsers expose multiple mechanisms that look similar but behave differently:

1. **DOM attributes**: strings stored in the elements attribute map (example: el.getAttribute(
2. **DOM properties**: live fields on the element object (example: (input as any).value).
3. **Special cased fields**: className and style have their own APIs and behavior.

If you set the wrong one, you can get subtle bugs. For instance, setting value as an attribute on <input> wont behave like setting the property in many scenarios.

Relaxs pragmatic rule is:

> extbfRule: if the name starts with data-, set an *attribute*; otherwise set a *property*.

You can see this rule directly in setAttribute() in src/attributes.ts.

6.2 Contract: setAttributes(el, attrs)

The top-level API is:

> extttsetAttributes(el: HTMLElement, attrs: ElementVNodeProps)

This function is used during mount (Chapter 5) and also during patches when props change (Chapter 8).

Inputs

- el: a real DOM element.
- attrs: Relaxs element props object (roughly: id, class, style, and anything else).

Behavior

extttsetAttributes() applies attributes in three phases:

1. split out class and style
2. delete key (VDOM identity only)
3. for each remaining entry: setAttribute(el, name, value)

6.2.1 Important invariant: key is not a DOM attribute

Relax uses key for diffing lists and stabilizing identity. It should never appear on the real DOM.

To enforce this, key removal happens in two places:

- src/attributes.ts: deletes otherAttrs.key
- src/utils/props.ts: deletes props.key when extracting props/events

The duplication is intentional: it keeps each layer correct even if used in isolation.

6.3 Class handling: replacement semantics

Relax exposes class as either:

- a string (example: "foo bar")
- an array of class tokens (example: ["foo", "bar"])

Implementation detail (setClass() in src/attributes.ts):

- it clears el.className = "
- if className is a string, it assigns el.className = className
- if its an array, it does el.classList.add(...className)

This makes class setting **replacement based**, not incremental diff. That's a conscious simplicity trade-off.

What the tests prove

The first three tests in src/$_{tests}$/$_{attributes.test.ts}$$providethecontract$:

- setting a class from a string
- setting a class from an array
- updating the class (proves replacement, not accumulation)

6.4 Style handling: direct assignment

Relaxs style prop is a plain object:

```
exttt{ style: { color: 'red', backgroundColor: 'blue' } }
```

For each [name, value] pair, setAttributes() calls setStyle(el, name, value), which does a direct assignment:

```
exttt(el.style as any)[name] = value
```

Removal goes through removeStyle() which sets the field to null.

One subtle consequence: setAttributes() does not diff old vs new style objects. If you call it twice, the second call only assigns the fields present in the second objectit does *not* remove fields that were present previously unless you explicitly remove them.

What the tests prove

extttsetting styles and updating styles assert that style fields are applied by assignment and can coexist across multiple calls.

6.5 The core rule: `setAttribute(el, name, value)`

The function setAttribute() is the heart of the module:

1. If value == null → call removeAttribute(el, name).
2. Else if name.startsWith('data-') → call el.setAttribute(name, String(value)).
3. Else → write as a property: (el as any)[name] = value.

This is where Relax chooses "property first" semantics. The benefit is correctness for form controls and boolean-ish fields.

What the tests prove: the matrix

extttsrc/$_{tests}$/$_{attributes.test.ts}$$isadeliberatelywidematrixofrealDOMbehavior.$

Highlights:

- **Generic element properties:** extttthidden, tabIndex (including tabIndex removal resulting in -1).
- **Input type=text properties:** extttvalue, placeholder, disabled, minLength, size, and more.
- **Checkbox / radio:** extttchecked is a property and removal resets to false.
- **Select:** setting value to a missing option results in ".
- **Anchors/buttons/spans:** properties like href, disabled, and textContent behave like live fields.

This test suite is doing an important job: it prevents "helpful refactors" from changing semantics that look equivalent but arent.

6.6 Removal: removeAttribute(el, name) and jsdom quirks

Removing an attribute sounds simple but isnt.

Relax tries to clear both levels:

1. it attempts to set the property to null inside a try/catch
2. it always calls el.removeAttribute(name)

The try/catch exists because some properties throw when assigned null (depending on element type and environment). When that happens, Relax logs a warning:

```
extttFailed to set "<name>" to null on <TAG>
```

That warning is not a test failure; its a visibility aid for cases where property clearing is not permitted.

6.7 Props vs events: extractPropsAndEvents (src/utils/props.ts)

Relax stores event handlers under props.on. Before mounting or constructing a component instance, it needs to split the two.

extttextractPropsAndEvents(vdom) does exactly that:

- events: vdom.props.on (defaulting to {})
- props: everything else (and it deletes key)

This split is used in two places youve already seen:

- element mounting: addEventListeners(events, el, hostComponent) and setAttributes(el, props)
- component mounting: new Component(props, events, hostComponent)

What the tests prove

extttsrc/$_{tests}/_{props.test.ts}isintentionallysmallandcrisp$:

- empty input yields empty outputs
- key is ignored
- non-event fields remain in props
- on: {...} becomes events

Treat these as invariants: if you ever change the VNode prop shape, these tests are the first alarm bell.

6.8 Design notes (trade-offs)

Relaxs attribute model is intentionally simple:

- it mostly treats prop names as property names
- it special-cases only data-*, class, and style
- it relies on a test matrix to ensure that simplicity doesnt break real DOM behavior

If you were extending Relax, likely next steps would be:

- richer SVG support (namespaces and attribute/property differences)
- style diff/removal semantics (tracking old style object)
- explicit boolean attribute reflection rules (attribute presence vs property value)

6.9 Mini-exercises

1. Add a new case to the matrix in src/$_{tests}$/$_{attributes.test.ts}$$for$<textarea>$_{valuebehavior.}$
1. Add a test that asserts setAttributes(el, { key: 'x', id: 'y' }) does not create a key attribute in the DOM.

6.10 Online resources

- Attributes vs. properties (MDN, conceptual background): https://developer.mozilla.org/en
- HTMLElement.dataset and data-* attributes (MDN): https://developer.mozilla.org/en-US/docs/
- Form control properties like value and checked (MDN): https://developer.mozilla.org/en-US/
- CSSOM style API (MDN): https://developer.mozilla.org/en-US/docs/Web/API/CSSStyleDeclaratio

7. Events and the Command Dispatcher

Relax has two different mechanisms that people often describe with the same word: "events".

- **DOM events** (click, input, change, ...): created by the browser (or by jsdom in tests) and delivered to listeners attached to real elements.
- **Commands** (named messages with payloads): dispatched programmatically through a tiny in-memory bus.

In the repo they live in two small modules:

- src/events.ts: turns Relaxs event props into real DOM listeners, with consistent this-binding.
- src/dispatcher.ts: a deterministic pub/sub for named commands.

And both are covered by focused tests:

- src/$_{tests}$/$_{events.test.ts}$
- src/$_{tests}$/$_{dispatcher.test.ts}$

7.1 The practical problem: handler identity and `this`-binding

If a UI library wants to support code like this:

```
<button onClick={this.onClick}>Hit</button>
```

it has to solve two problems:

1. **Binding:** when the event fires, should this inside onClick be the component instance?
2. **Removal:** if the tree updates or unmounts, the library must remove listeners using the *exact same function object* that was added.

Relaxs answer is to always register a wrapper listener and return it to the caller.

7.2 DOM event listener wiring (`src/events.ts`)

The central helper is:

```
extttaddEventListener(eventName, handler, el, hostComponent?)
```

7.2.1 Contract

- **Inputs:**
  - eventName: a DOM event name like "click".
  - handler: a function that should run when the event fires.
  - el: the real DOM element.
  - hostComponent: optional object used as the this binding.
- **Output:**
  - returns the concrete listener function object that was actually registered with el.addEventListener.

Returning the registered listener is not a convenienceits required for correctness. The DOM only removes listeners by identity.

7.2.2 Implementation: the wrapper listener

extttsrc/events.ts always creates a wrapper:

```
const listener = (event: Event) => {
  if (hostComponent) {
    handler.call(hostComponent, event)
  } else {
    handler(event)
  }
}
```

Then it registers listener and returns it.
This immediately gives Relax three desirable properties:

- **Stable binding semantics:** the wrapper decides whether to call with handler.call(hostCompo event).
- **No mutation of user handlers:** the original function object is untouched.
- **Reliable cleanup:** the wrapper function object can be stored and later removed.

7.2.3 Multiple listeners: addEventListeners(events, el, hostComponent?)

Mounting and patching usually deal with an event map (a record of event name to handler). Relax provides a convenience:

```
extttaddEventListeners(events, el, hostComponent)
```

Implementation is straightforward:

- iterate Object.entries(events)
- call addEventListener for each
- store returned wrappers in listeners: Record<string, EventListener>

The important artifact is the returned listeners map. In mount-dom.ts, Relax stores it on the vnode (vdom.listeners) so that unmounting can remove exactly what was added.

7.2.4 Cleanup symmetry: `removeEventListeners(listeners, el)`

Removal is the mirror image:

```
Object.entries(listeners).forEach(([eventName, listener]) => {
  el.removeEventListener(eventName, listener)
})
```

Note whats *not* present: the original user handlers. Only wrappers can be removed.

7.3 Tests for DOM event semantics (src/tests/events.test.ts)

The tests are small but they lock down two critical invariants.

7.3.1 Invariant 1: wrappers dont break sync or async handlers

extttevents.test.ts contains:

- synchronous event without host component, called with argument
- asynchronous event without host component

The async test uses flushPromises() (src/utils/promises.ts) after btn.click() to prove that:

- the wrapper does not swallow async behavior
- the handler is actually invoked and can await microtasks

This matters because event wrappers are a common source of subtle bugs where promises arent awaited or exceptions get lost.

7.3.2 Invariant 2: with a host component, `this` is correct

Two tests cover binding (sync and async):

- synchronous event with host component, the handler is bound to component
- asynchronous event with host component, the handler is bound to component

Both assert:

- actualBinding === hostComponent
- the handler argument is an Event

If you remember only one idea from this part of the codebase, make it this: **binding and cleanup depend on the existence of a wrapper listener.**

7.4 The command dispatcher (`src/dispatcher.ts`)

Where events.ts is about DOM integration, Dispatcher is a tiny in-process pub/sub.

7.4.1 A minimal contract

extttDispatcher has three public methods:

- subscribe(commandName, handler) → returns unsubscribe()
- afterEveryCommand(handler) → returns unsubscribe()
- dispatch(commandName, payload)

Internally it stores:

- #subs: Map<string, Array<(payload: unknown) => void»
- #afterHandlers: Array<() => void>

So you can subscribe to a particular command name (payload-aware) and/or attach a hook that runs after every dispatch (payload-agnostic).

7.4.2 Subscription semantics: dedupe by function identity

An important policy choice appears in subscribe():

> registering the same handler function twice is a no-op.

In code, thats handlers.includes(handler).
This makes the dispatcher deterministic and prevents accidental handler explosions if a component subscribes in a loop.

7.4.3 Dispatch semantics: warn on unhandled commands, always run after-hooks

extttdispatch(commandName, payload) does:

1. if there are handlers for commandName, call them with payload
2. otherwise, console.warn (if available): No handlers for command: <name>
3. always run #afterHandlers

That last rule is important: after-hooks run even if there were no command handlers. It allows patterns like global logging, tracing, or devtools hooks.

7.5 Tests for dispatcher semantics (src/tests/dispatcher.test.ts)

The dispatcher tests are a clean, readable spec for the API:

- **subscribe/unsubscribe works:** the handler is called once, then not called after unsubscribe.
- **dedupe works:** subscribing the same handler multiple times still results in one call per dispatch.

- **multiple handlers work:** two different handlers both fire, and both can unsubscribe.
- **afterEveryCommand works:** runs once per dispatch and unsubscribes cleanly.

Notice how these tests avoid implementation details (maps, arrays, indices) and instead assert the observable contract.

7.6 Common pitfalls and edges

- **Listener removal requires wrapper identity.** Never attempt to remove a handler by passing the original user function if you registered a wrapper.
- **Dedupe is per-function identity.** If you pass a new () => ... lambda each time, the dispatcher will treat it as a different handler.
- **Dispatcher is synchronous.** If a handler throws, that throw will propagate from dispatch().

7.7 Where this plugs into the renderer

Youve already seen the integration point in Chapter 5:

- element vnodes store props.on (after TSX mapping)
- mounting calls addEventListeners(...) and saves wrapper functions on vdom.listeners
- destroying later calls removeEventListeners(vdom.listeners, el)

The remaining questionshow listeners are updated during patch without leaking, and how host binding is preserved across updateslive in src/patch-dom.ts and src/destroy-dom.ts.

7.8 Online resources

- DOM events and bubbling (MDN): https://developer.mozilla.org/en-US/docs/Learn$_{w}eb_{d}evelopmen$
- EventTarget.addEventListener / removeEventListener (MDN): https://developer.mozilla.org/en
- Function call and this binding (MDN): https://developer.mozilla.org/en-US/docs/Web/JavaScr
- Message bus / pub-sub as a pattern (Wikipedia overview):

8. Patching and Diffing: Updating the DOM Efficiently

Mounting is the "first render" story. But real apps are dominated by *updates*: state changes, input events, list reorders, and component re-renders.

Relaxs patcher has one job: given an *old* VNode tree that is already mounted and a *new* VNode tree produced by the latest render, mutate the real DOM so it matches the new tree while reusing as much existing DOM as is safe.

In Relax the patcher is both:

- **small enough to read end-to-end** (in src/patch-dom.ts), and
- **heavily specified by tests** (especially src/$_{tests}$$_{/patch-dom.test.ts}$).

This chapter is long because patching is where most correctness and performance edge cases live.

8.1 patchDOM: the top-level operation

The public entry point is:

```
extttpatchDOM(oldVdom, newVdom, parentEl, hostComponent?)
```

It lives in src/patch-dom.ts.

8.1.1 Contract

- **Inputs**
  - oldVdom: the currently mounted VNode tree.
  - newVdom: the VNode tree produced by the latest render.
  - parentEl: the DOM parent that contains oldVdom.el.
  - hostComponent: optional component instance used to bind events and to account for fragment offsets.
- **Output**
  - Returns the patched newVdom (with newVdom.el pointing at the reused DOM).
- **Key invariant**
  - After patching, the returned tree (the *new* VDOM) must contain correct .el references.

8.1.2 Replace vs patch: areNodesEqual

The first decision is whether two nodes are "the same" (so we can patch in place) or different (so we must replace).

Relax defines "same" in src/nodes-equal.ts via areNodesEqual(oldNode, newNode).

This is not a philosophical definition of equality; it is *exactly* the predicate the patcher uses to decide between:

- patching in place (reuse), or
- destroying and remounting (replace).

The current identity rules are:

- If type differs: not equal.
- **TEXT**: always equal (text nodes are patched by content).
- **ELEMENT**: equal when tag matches and key matches.
- **COMPONENT**: equal when component function identity (tag) matches and key matches.
- **HRBR**: equal when mount factory identity matches.

If nodes are not equal, patchDOM does a replacement:

1. Find the current DOM index of oldVdom.el inside parentEl (findIndexInParent).
2. Destroy the old subtree (destroyDOM(oldVdom)).
3. Mount the new subtree at the same index (mountDOM(newVdom, parentEl, index, hostComponent)).

This behavior is covered early in src/$_{tests}{}_{/patch-dom.test.ts}{}^{by:}$

- change the root node
- change the root node, mount in same index

8.1.3 When nodes are equal: preserve el

If nodes are equal, patching starts by reusing the DOM reference:

```
newVdom.el = oldVdom.el
```

This is asserted by tests like no change and sets the el in the new vdom.

8.2 Node-type specific patching

After equality is established, patching dispatches on newVdom.type (see DOM_TYPES in src/h.ts).

8.2.1 Text nodes: patchText

patchText(oldVdom, newVdom) compares oldVdom.value and newVdom.value. If different, it updates Text.nodeValue.

Test: patch text.

8.2.2 Element nodes: patchElement

patchElement(oldVdom, newVdom, hostComponent) updates four categories of props:

- Attributes/properties (everything except class, style, on)
- class
- style
- on (DOM event handlers)

Implementation detail: it destructures props like:

```
const { class: oldClass, style: oldStyle, on: oldEvents, ...oldAttrs } = oldVdom.props
const { class: newClass, style: newStyle, on: newEvents, ...newAttrs } = newVdom.props
```

and then applies targeted patchers:

- patchAttrs(el, oldAttrs, newAttrs)
- patchClasses(el, oldClass, newClass)
- patchStyles(el, oldStyle, newStyle)
- patchEvents(el, oldListeners, oldEvents, newEvents, hostComponent)

Micro-optimization: object identity fast paths Notice the checks in patchElement:

```
if (oldAttrs !== newAttrs) { ... }
if (oldClass !== newClass) { ... }
if (oldStyle !== newStyle) { ... }
```

In large lists, applications often reuse the same props objects (or stable sub-objects) for unchanged rows. These identity checks avoid unnecessary diff work.

8.2.3 Attributes: patchAttrs uses objectsDiff

patchAttrs calls objectsDiff(oldAttrs, newAttrs) (from src/utils/objects.ts) to compute added, removed, updated.

It then:

- removes: removeAttribute(el, name)
- adds/updates: setAttribute(el, name, value)

Important: setAttribute and removeAttribute are the same helpers discussed in Chapter 6 (src/attributes.ts), including the property-vs-attribute rules.

8.2.4 Classes: patchClasses uses arraysDiff

patchClasses normalizes both old and new values into an array of class tokens using toClassList.

Then it runs arraysDiff(oldClasses, newClasses) to compute added and removed, and performs:

- el.classList.remove(...removed)
- el.classList.add(...added)

The behavior is tested extensively in patch class cases in src/$_{tests/patch-dom.test.ts}$.

8.2.5 Styles: patchStyles uses objectsDiff

patchStyles again uses objectsDiff and applies:

- remove: removeStyle(el, key)
- add/update: setStyle(el, key, value)

Test group: patch style.

8.2.6 Events: patchEvents

patchEvents compares old and new on objects using objectsDiff and performs two phases:

1. **Detach** removed and updated handlers by removing their old wrapper listeners.
2. **Attach** added and updated handlers using addEventListener from src/events.ts.

Why detach+attach for *updated*? Because the wrapper listener function identity changes when the handler changes.

Tests to read:

- patch event handlers in src/$_{tests/patch-dom.test.ts}$
- The binding rules in src/$_{tests/events.test.ts}$

8.2.7 Component nodes: patchComponent

patchComponent(oldVdom, newVdom) preserves the component instance and pushes changes into it.

Key steps:

- Extract external children and props (see extractPropsAndEvents in src/utils/props.ts)
- Call component.setExternalContent(children).

- Call component.updateProps(props).
- Copy instance fields:
  * newVdom.component = component
  * newVdom.el = component.firstElement

This is how Relax achieves the "component instance is preserved" semantics tested in patch component with props.

8.2.8 HRBR nodes: patchHrbr

8.2.9 A tiny but important helper: patchDOMSameElement

You will notice a second patch entry point inside src/patch-dom.ts:

```
extttpatchDOMSameElement(oldVdom, newVdom, hostComponent?)
```

It is identical to patchDOM except it *skips the replacement path* and therefore does not need parentEl.

This helper is used in child diffing paths where:

- we already know the node should be kept (for example, keyed nodes that exist in both old and new), and
- we want to avoid scanning parentEl.childNodes to find the current index for every child.

When youre reading patching code, seeing patchDOMSameElement is a clue: "were in an optimization path, and weve already committed to reusing the node identity."

Relax contains an experimental HRBR interop mode. In patch-dom.ts, patchHrbr(old, next):

- Reuses the existing mounted instance when old.mount === next.mount.
- Otherwise, disposes/destroys the old instance and remounts into the same host element.

This behavior matches the spec tests in src/$_{tests/hrbr-integration.test.ts}$.

8.3 Patching children: from fast paths to general diff

After patching the node itself, Relax patches children with:

```
patchChildren(oldVdom, newVdom, hostComponent)
```

This is where most performance work lives, and src/patch-dom.ts contains several layered strategies.

8.3.1 Step 0: child extraction

Children are normalized by extractChildren(oldVdom) / extractChildren(newVdom) from src/h.ts. This lets fragments and components present children uniformly.

8.3.2 Fast path A (opt-in): delegate keyed list reconciliation to HRBR reconciler

There is an optional integration with the runtime reconciler:

```
if (newVdom.props._reconcile === 'hrbr') { ... reconcileChildren(...) }
```

This path only applies when *all direct children are keyed* (keyed list). It builds ReconcileNode specs and lets reconcileChildren (from runtime/reconciler) patch/move nodes with a single keyed pass.

You can think of it as: "VDOM delegates list diffing to a specialized keyed reconciler".

8.3.3 Fast path B (opt-in): $_textOnlyrows$

For large lists of rows shaped like <li>label</li>, Relax supports an opt-in hint:

- If oldVdom.props.$_textOnly === trueandnewVdom.props._textOnly === true, and$
- both old and new children are exactly one TEXT node,

then patchChildren patches only the text node via patchDOMSameElement.

Test name: patchChildren row fast path (_textOnly).

8.3.4 Fast path C: keyed reorder via LIS

The LIS (Longest Increasing Subsequence) trick is a classic VDOM strategy:

- If you can identify the subsequence of children that are already in correct relative order,
- then you only need to move the remaining nodes.

Relaxs implementation is in patchKeyedChildrenReorder(...) and uses an internal helper lisIndices(arr) which computes LIS indices over a sequence of old positions (ignoring -1 markers for new nodes).

One very practical detail from the code: elementsForChild(child) returns *multiple* DOM elements for components (using child.component.elements) but only one for normal elements. Thats how the reorder path correctly moves a component that renders multiple siblings.

patchKeyedChildrenReorder(oldChildren, newChildren, parentEl, hostComponent) implements keyed reordering with minimal DOM moves:

1. Ensure both sides are fully keyed.
2. Remove old nodes not present in the new key set.
3. Patch existing nodes and mark new nodes with -1 in a sequence.
4. Compute the Longest Increasing Subsequence (LIS) over old indices.
5. Walk backwards, mounting new nodes and moving only nodes not in LIS.

This is a classic strategy: nodes in the LIS are already in relative order, so moving the rest yields the fewest inserts/reorders.

Test names to read:

- patchChildren keyed reorder fast path (LIS)
- The many keyed reorder cases in src/$_{tests}$/$_{patch-dom.test.ts}$

8.3.5 Fast path D: stable keyed same-order

If keys are present and the order is identical, Relax avoids diff work entirely:

```
if (isStableKeyedSameOrder(oldChildren, newChildren)) {
  for (i) patchDOMSameElement(oldChildren[i], newChildren[i])
  return
}
```

Test name: patchChildren fast path (stable keyed order).

8.3.6 General path: arraysDiffSequence

If none of the fast paths apply, Relax computes an edit script:

```
const diffSeq = arraysDiffSequence(oldChildren, newChildren, areNodesEqual)
```

extttarraysDiffSequence is a general-purpose utility in src/utils/arrays.ts. Its worth reading because it is one of the algorithmic cores of Relax.

It has extensive unit tests in:

- src/$_{tests}$/$_{diff-sequence.test.ts}$
- src/$_{tests}$/$_{diff-sequence.fuzzy.test.ts}$

It produces operations of types ADD, REMOVE, MOVE, NOOP (see ARRAY_DIFF_OP).

Then patchChildren executes them:

- **ADD**: mountDOM(item, parentEl, index + offset, hostComponent)
- **REMOVE**: destroyDOM(item)
- **MOVE**: move DOM nodes (component = multiple elements) and patch the moved subtree
- **NOOP**: patch the subtree in place

Component moves A subtle point: moving a component means moving *all* DOM elements owned by that component. That's why the code does:

```
const elementsToMove = isComponent(oldChild)
  ? oldChild.component.elements
  : [oldChild.el]
```

This matches the "component as a fragment" mental model: the component is a logical node that may span multiple DOM siblings.

8.3.7 What the diff-sequence tests prove

The patcher relies on arraysDiffSequence to produce a valid edit script. The contract is captured in two test files:

- src/$_{tests}$/$_{diff-sequence.test.ts}$:$ahuman-readablesetofexactexpectedoperationsequencesformanysmallcases$
- src/$_{tests}$/$_{diff-sequence.fuzzy.test.ts}$:$aproperty-styletestthatgeneratesrandomarraysandassertsthatapplyingtheoperationsrepro$

The workhorse assertion is essentially:

```
extttapplyArraysDiffSequence(oldArray, arraysDiffSequence(oldArray, newArray))
=== newArray
```

This is a great example of how Relax uses tests: not just to assert a single result, but to lock down the correctness of an algorithm under many variations.

8.4 Tests to lean on

If you want to understand Relax patch semantics quickly, scan these groups in src/$_{tests}$/$_{patch-dom.test.ts}$:

- Root replacement and .el preservation.
- Fragment patching (nested add/remove/move).
- Attribute/class/style patching.
- Event handler updates and removals.
- Children patching: keyed/unkeyed, moves, fast paths.
- Component patching and component list semantics.

8.5 Takeaways

- Relax's patcher is layered: correct general diff + several pragmatic fast paths for real-world performance.
- Keys are not just an optimization; they unlock better move semantics and fewer DOM operations.
- Correctness depends on two invariants:
  * areNodesEqual must be consistent across mount/patch.
  * newVdom.el (and component element arrays) must be updated to reflect reality.

8.6 Online resources

- React keys and list rendering (conceptual background; implementation differs): https://react.dev/learn/rendering-lists
- Longest Increasing Subsequence (LIS) algorithm overview: https://en.wikipedia.org/wi
- DOM operations cost model (MDN on Node.insertBefore / Node.removeChild): https://deve
- Virtual DOM diffing background reading (generic): https://increment.com/frontend/a-de

9. Destroying DOM and Unmounting Components

So far we've treated the renderer as a one-way pipeline:

extttVNode →mountDOM() →DOM →patchDOM()

But a real UI runtime lives and dies by teardown. If teardown is incomplete or inconsistent, it manifests as:

- **leaked event listeners** (handlers still firing after nodes are removed)
- **leaked component instances** (dispatch subscriptions never removed)
- **zombie DOM references** (stale vdom.el pointers used accidentally)
- **interop leaks** (third-party instances never disposed)

Relax centralizes teardown in destroyDOM(vdom) in src/destroy-dom.ts. The executable specs are primarily in:

- src/$_{tests}$$_{/destroy-dom.test.ts}$
- src/$_{tests}$$_{/component-lifecycle.test.ts}$
- src/$_{tests}$$_{/hrbr-integration.test.ts}$$(interop)$

9.1 A tiny contract for teardown

We'll use a tight contract so you can re-derive the implementation from the tests:

- **Input**: a *mounted* VNode subtree root.
- **Effects**: remove DOM nodes, dispatch cleanup, interop cleanup, and clear runtime references.
- **Idempotence**: not guaranteed. Calling destroy twice is a programmer error.
- **Success criteria**: after teardown, the DOM no longer contains the rendered subtree *and* the VNode tree no longer retains .el pointers.

That last bullet is important: removing DOM nodes is easy--cleaning the runtime state is the hard part.

9.2 destroyDOM(): the teardown dispatcher

The public entry point is: destroyDOM(vdom).

The implementation is built as a type-dispatcher over vdom.type (from DOM_TYPES in src/h.ts).

9.2.1 The shared post-step: deleting `vdom.el`

At the end of destroyDOM, Relax does:

```
delete vdom.el
```

This isn't cosmetic.

- It enforces a strong invariant in tests ("every destroyed node lost its .el").
- It makes stale VNodes fail fast if reused accidentally.
- It reduces accidental retention of DOM nodes by application code holding onto VNodes.

9.3 Type-based teardown

destroyDOM dispatches on vdom.type (DOM_TYPES is defined in src/h.ts).

9.3.1 Text nodes

For DOM_TYPES.TEXT, removeTextNode asserts el instanceof Text and calls el.remove().

Test: destroy a text element.

9.3.2 Element nodes: remove element + recurse + listener cleanup

For DOM_TYPES.ELEMENT, removeElementNode performs three distinct actions:

1. Remove the element from the DOM: el.remove().
2. Recursively destroy children: children.forEach(destroyDOM).
3. If there are DOM event listeners stored on the VNode, detach them:

   ```
   removeEventListeners(listeners, el)
   delete vdom.listeners
   ```

The listener removal uses removeEventListeners from src/events.ts, which removes by the wrapper listener identity that was created during mount/patch.

Tests:

- destroy an html element and its children
- destroy an html element and its children recursively
- remove an html element event listeners

Ordering note: remove first, then cleanup extttremoveElementNode() removes the element from the DOM *before* walking children. This ordering is deliberate and safe:

- In the DOM, removing a parent implicitly detaches descendants from the document.
- But Relax still needs to recursively teardown the VNode subtree (delete .el, detach listeners, unmount nested components).
- Running the recursion after removal avoids intermediate states where logic could observe partially-destroyed nodes in the live document.

9.3.3 Fragment nodes

Fragments are a logical grouping; they don't own a single DOM node. For DOM_TYPES.FRAGMEN removeFragmentNodes just destroys each child.

Tests:

- destroy a fragment
- destroy a fragment recursively

9.3.4 Component nodes: unmount + schedule `onUnmounted`

For DOM_TYPES.COMPONENT, destroyDOM delegates to the component instance:

```
vdom.component.unmount()
enqueueJob(() => vdom.component.onUnmounted())
```

Two important details are embedded here, and both are tested:

- **Unmount is synchronous structural cleanup.** It destroys the component's rendered subtree and unsubscribes internal dispatcher subscriptions.
- **onUnmounted is scheduled.** It runs later via enqueueJob (src/scheduler.ts), matching the mount side where onMounted is also scheduled.

This scheduling is why the component lifecycle tests await nextTick() after calling destroyDOM():

- a component reacts to being unmounted
- a component reacts to being unmounted asynchronously

9.4 Inside Component.unmount() (`src/component.ts`)

Components in Relax are created with defineComponent(...). The generated class maintains several private fields that matter for unmount:

- #vdom: the last rendered subtree returned by render()

- #dispatcher: a per-component Dispatcher used to implement emit() and event handler wiring
- #subscriptions: unsubscribe closures returned from dispatcher.subscribe()
- #isMounted and #hostEl: mount state and mount container

9.4.1 Unmount sequence

Look at unmount() in src/component.ts:

1. It throws if the component isn't mounted.
2. It destroys the rendered subtree: destroyDOM(this.#vdom).
3. It unsubscribes from internal dispatcher subscriptions: this.#subscriptions.forEach(u
4. It clears state: #vdom = null, #isMounted = false, #hostEl = null, #subscriptions = [].

This means after unmount, both DOM and book-keeping references are gone.

Why unsubscribe in unmount() and not in destroyDOM()? Because subscriptions are a property of the component instance, not a property of the rendered VDOM nodes. extttdestroyDOM() is intentionally "VNode-centric"; unmount() is "component-centri

9.5 HRBR nodes: dispose, destroy, and remove host

Relax also supports an experimental HRBR interop node type (DOM_TYPES.HRBR). These nodes represent "something else" mounting into a host container (a block, micro-framework component, etc.).

In removeHrbrNode(vdom) (src/destroy-dom.ts):

- If the instance provides dispose() and/or destroy(), Relax calls them (in that order).
- Then it removes the HRBR host element from the DOM.
- Finally, it deletes vdom.instance and vdom.host.

This mirrors the mounting/patching behavior from earlier chapters: HRBR nodes are treated like a mounted instance with an owned host container.

Interop rule of thumb Relax doesn't assume what "destroy" means. It only promises it will call well-known lifecycle hooks (dispose/destroy) if present, and remove the host container it created.

9.6 A practical mental model

It helps to treat teardown as the inverse of mounting:

- mounting creates DOM nodes and stores refs in vdom.el
- patching preserves DOM nodes and updates refs on the new VDOM
- destroying removes DOM nodes and deletes refs from the old VDOM

The explicit delete vdom.el makes it hard to accidentally keep using stale VNodes after tear-down.

9.7 Takeaways

- Relax teardown is explicit and recursive.
- Element teardown removes event listeners using the stored wrapper identities.
- Component teardown is two-phase: synchronous unmount + scheduled onUnmounted hook.
- Deleting .el and clearing instance fields is a core anti-leak strategy.

9.8 Walking the tests: what they prove

This section treats the tests as a teardown requirements document.

9.8.1 destroy a text element

From src/$_{tests}$/$_{destroy-dom.test.ts}$:

- Mount a text VNode via mountDOM and assert vdom.el is a Text node.
- Call destroyDOM(vdom).
- Assert the document is empty *and* allElementsHaveBeenDestroyed(vdom) returns true.

The helper allElementsHaveBeenDestroyed() recursively checks for the absence of .el. This is the strongest teardown invariant in this test suite.

9.8.2 destroy an html element and its children

This test ensures an ELEMENT node:

- removes the parent element from DOM
- still destroys (cleans up) its children VNodes

It's the "DOM removed" + "runtime graph cleaned" pairing again.

9.8.3 remove an html element event listeners

This is the leak-prevention test.

1. Mount a button with on: { click: handler }.
2. Trigger a click and assert the handler is called.
3. Destroy the VNode.
4. Trigger another click on the previously captured buttonEl and assert the handler isn't called again.

This works because during mount, Relax installs a wrapper listener and stores it in vdom.listeners. During destroy, removeEventListeners(vdom.listeners, el) uses those wrapper identities to detach correctly.

9.8.4 Recursive destruction and fragments

extttdestroy an html element and its children recursively, destroy a fragment, and destroy a fragment recursively collectively prove that:

- recursion always visits nested VNodes
- fragment nodes are teardown-neutral: they simply forward destruction to children

9.8.5 destroy a component with subcomponents

This test ensures that the recursion crosses component boundaries. The starting VNode is an element div containing a component. Destroying the root must unmount the component and destroy its rendered subtree, including nested child components.

9.8.6 Lifecycle scheduling tests (component-lifecycle.test.ts)

The component lifecycle tests focus on enqueueJob/nextTick() behavior:

- onMounted is not expected to run synchronously with mountDOM; the test awaits nextTick().
- Symmetrically, onUnmounted is not expected to run synchronously with destroyDOM; the test awaits nextTick().

This enforces a stable mental model: lifecycle hooks run in the scheduler, not "inline" within the DOM mutation operations.

9.8.7 HRBR teardown behavior (hrbr-integration.test.ts)

The HRBR integration suite demonstrates that destroyDOM() participates in interop cleanup.

- In the "can mount an HRBR block" test, the block's destroy() clears the host; afterwards, Relax removes the host itself.

- In "patch preserves HRBR instance", destroying triggers the instance's destroy() and the test asserts it was called.

9.9 Online resources

- MDN: Element.remove()
- MDN: removeEventListener() and listener identity
- React docs (conceptual parallel): Effects and cleanup
- Vue docs (conceptual parallel): Lifecycle hooks

10. Scheduling, nextTick, and Deterministic Work

Relax has *two* schedulers:

- A small microtask-based job queue used by the classic VDOM layer (src/scheduler.ts).
- A more capable lane-based scheduler used by the runtime/reactive layer (runtime/schedu

They serve different purposes, but they share one theme: **make UI work predictable**.

Scheduling discussions tend to get vague fast ("it's async, it'll happen later"). In Relax, scheduling is treated as part of the renderer's public contract: tests depend on it, and lifecycle semantics are defined by it.

This chapter is anchored in:

- src/scheduler.ts and src/$_{tests/scheduler.test.ts}$
- runtime/scheduler.ts and runtime/$_{tests/scheduler.test.ts}$

10.1 The VDOM microtask job queue (`src/scheduler.ts`)

The VDOM layer uses a minimal scheduling primitive: enqueueJob(job).

10.1.1 Why it exists

Several operations in Relax are intentionally deferred:

- component onMounted() is scheduled after mount completes
- component onUnmounted() is scheduled after unmount begins

Deferring hook execution avoids re-entrancy surprises and gives the renderer a clean boundary: "render/mount now, run hooks shortly after".

The hidden constraint: DOM consistency and re-entrancy If onMounted() ran inside mountDOM(), a component could synchronously call updateState(), which triggers patching while the initial mount is still underway. By scheduling hooks, Relax ensures the mount/patch/destroy primitives finish their structural work before user hook code runs.

10.1.2 Data model

src/scheduler.ts maintains:

- jobs: an array of queued callbacks
- isScheduled: a flag to ensure we schedule processing only once per tick

10.1.3 enqueueJob and queueMicrotask

enqueueJob(job) pushes the job and calls scheduleUpdate().

scheduleUpdate() is the key:

```
if (isScheduled) return
isScheduled = true
queueMicrotask(processJobs)
```

So no matter how many jobs you enqueue, you'll process them in a single microtask flush.

Batching rule The isScheduled latch is the batching mechanism:

- enqueue one job →schedule one microtask
- enqueue many more jobs before the microtask runs →still one microtask

This makes lifecycle hooks cheap even when many components mount in the same synchronous turn.

10.1.4 processJobs: ordering and async behavior

processJobs() drains jobs in FIFO order. For each job:

- it calls job()
- it wraps the result with Promise.resolve(result).then(...) to log async errors

Important nuance (documented in the code): **async jobs do not block the queue.** The queue keeps running other jobs; the promise continuation runs later.

Why the queue doesn't await The VDOM scheduler is designed for hook-style work ("start it soon, don't stall unrelated hooks, but surface errors"). If it awaited each job, one slow async hook could hold up every other component's hook.

This is tested in src/tests/scheduler.test.ts:

- Asynchronous jobs enqueue more jobs expects order [1, 3, 2].

That test shows:

1. async job pushes 1
2. it awaits a microtask, so it pauses
3. the next (sync) job runs and pushes 3
4. then the async job resumes and pushes 2

10.1.5 nextTick(): what it guarantees (and what it doesn't)

nextTick() is the testing and integration helper:

```
scheduleUpdate()
return flushPromises()
```

flushPromises() is implemented as setTimeout(resolve). That means **nextTick resolves after the microtask flush and one macrotask turn**.

The comment in src/scheduler.ts is explicit:

> If the jobs are asynchronous, the promise will resolve before all the jobs have completed.

So the contract is:

- It ensures the queued jobs have been *started* and synchronous effects have run.
- It does not await completion of async work inside a job.

Practical testing guidance extttnextTick() is a boundary for "scheduler-driven work started", not "all async completed". If you write a hook that does work after an await, you may need an additional wait (or design your test so the awaited promise resolves immediately).

Tests in src/$_{tests}$/$_{scheduler.test.ts}$cover:

- jobs run after nextTick
- jobs run in order
- jobs run after synchronous code

10.2 The runtime lane scheduler (runtime/scheduler.ts)

The runtime scheduler is built for reactive workloads and larger apps. Instead of a single FIFO queue, it manages multiple priority lanes:

- sync
- input
- default
- transition
- idle

This scheduler is deterministic by design. It uses stable tie-breakers, explicit budgets, and a promotion system (aging + deadlines) so low-priority work eventually runs.

This section is grounded directly in runtime/scheduler.ts and runtime/$_{tests}$/$_{scheduler.test.ts}$.

10.2.1 Creating a scheduler

The entry points:

- `createScheduler(options)`
- `createRequestFlush(strategy)` for how flush is triggered
- `createBrowserScheduler(...)` convenience wrapper

10.2.2 Budgets and time slicing

A scheduler flush is run with a budget (in milliseconds). The flush loop:

- picks the next task
- runs it
- checks elapsed time
- if budget is exceeded and work remains, schedules another flush

This behavior is tested in runtime/$_{tests/scheduler.test.ts}$:

- budget enforcement: flush splits work across multiple flush calls when budget is tight
- setFrameBudget caps the budget used by scheduled flushes

10.2.3 Lane priority and deterministic ordering

Lane priority is declared in LANE_PRIORITY and selection order is:

```
sync > input > default > transition > idle
```

Within a lane, tasks are sorted with deterministic tie-breakers:

1. earlier deadline
2. earlier timestamp
3. lower task id

Test: orders tasks within a lane by explicit deadline (then timestamp/id).

In code, the deterministic ordering is implemented by sorting each lane queue using:

```
(a.deadline - b.deadline) || (a.timestamp - b.timestamp) || (a.id - b.id)
```

The final tie-breaker (id) is the determinism anchor: it makes equal-time scheduling stable.

10.2.4 Aging and promotion (starvation prevention)

To prevent starvation, the scheduler promotes work that is overdue or old.

The implementation:

- defines AGING_MS = 250
- in pickNextTask(), checks queue heads in weaker lanes
- if a head task is overdue (deadline reached) or aged, it is promoted one lane upward

Tests:

- ages/pushes starving tasks upwards (idle -> transition)
- promotes a task when its deadline is reached (even if it is not old enough)
- starvation prevention: repeated promotions eventually let idle work run even under continuous input load

This is the scheduler equivalent of "don't let background work get stuck forever".

10.2.5 withBudget(): integrating user work

The scheduler also offers withBudget(budgetMs, fn). It executes user code and, if that work consumes the budget while tasks remain, schedules continuation.

Test: withBudget schedules continuation when user code exhausts budget and tasks remain.

10.3 How these schedulers relate to the book

In Part II and Part IV we saw that the VDOM layer uses enqueueJob to schedule lifecycle hooks.

In later parts of the book (signals, blocks, hydration), we'll lean more on the runtime scheduler. The runtime lane model is a foundation for:

- fair scheduling of effects
- batching updates
- making performance tunable (frame budgets, flush strategy)

10.4 Takeaways

- src/scheduler.ts is tiny but intentional: microtask batching + safe error logging.
- nextTick() is a boundary for tests and integration, but does not await async job completion.
- runtime/scheduler.ts adds lanes, deadlines, aging, and budgets for scalable reactive workloads.

10.5 Walking the tests: what they prove

You can treat these as the scheduler's executable specification.

10.5.1 VDOM scheduler tests (**src/**$_{tests/scheduler.test.ts}$)

Enqueued jobs run after nextTick Proves enqueueJob() is deferred and does not run synchronously.

Enqueued jobs run in order Proves FIFO execution order in processJobs().

Jobs run after synchronous code Proves microtasks run after the current synchronous call stack.

Asynchronous jobs enqueue more jobs Proves the "async doesn't block" rule: a job that yields allows later jobs to run, producing [1, 3, 2].

10.5.2 Runtime scheduler tests (**runtime/**$_{tests/scheduler.test.ts}$)

The runtime scheduler suite uses a manual clock (now()) and a manual flush trigger (requestFlush(cb) pushes cb into an array). That technique eliminates timing flakiness and makes time-based scheduling deterministic.

Key tests and the guarantees they enforce:

runs tasks by lane priority Proves lane selection order and that sync runs immediately.

budget enforcement / setFrameBudget Proves time slicing across multiple flush calls when the budget is tight.

ages/pushes starving tasks upwards Proves aging-based promotion (AGING_MS = 250).

promotes a task when its deadline is reached Proves deadline-based promotion even when a task isn't old.

withBudget schedules continuation Proves user code that consumes budget can trigger a continuation flush if tasks remain.

createRequestFlush(messageChannel/raf) Proves flush strategy selection when platform APIs exist.

10.6 Online resources

- MDN: queueMicrotask()
- MDN: JavaScript event loop (microtasks vs. macrotasks)
- WHATWG HTML: Microtask queue (spec)
- React (conceptual parallel): Queued updates and batching
- MDN (conceptual parallel): Cooperative background work

11. Slots and Content Projection

Components are most reusable when they can accept external structure. In Relax, *slots* are the mechanism for content projection.

Relax's slot model is intentionally minimal:

- there is exactly one slot "kind" (no named slots)
- there is no runtime slot registry
- slots are compiled away into normal VDOM shape before mount/patch does serious work

That constraint keeps the renderer small and makes the behavior easy to specify with tests.

Slots in the VDOM layer are intentionally small and test-driven:

- VNode marker: DOM_TYPES.SLOT built by hSlot() in src/h.ts
- Slot filling algorithm: fillSlots(vdom, children) in src/slots.ts
- Specs: src/$_{tests}$/$_{slots.test.ts}$and src/$_{tests}$/$_{component-slots.test.ts}$

11.1 Slot VNodes: `hSlot` and the SLOT marker

A slot is represented as a VNode whose type is DOM_TYPES.SLOT.

In src/h.ts:

- DOM_TYPES includes SLOT: 'slot'.
- hSlot(children = []) returns { type: SLOT, children }.
- hSlot also sets a module-level flag hSlotCalled = true.

That flag is read via didCreateSlot() and reset via resetDidCreateSlot().

11.1.1 Why track `didCreateSlot`?

Relax uses this flag as a cheap "did the render tree contain any slots?" signal. The component renderer (src/component.ts) uses it:

- if a slot was created during render(), it calls fillSlots(vdom, externalChildren)
- then it resets the flag

This avoids doing a full traversal to *search* for slots when there are none.

How the optimization is wired (`src/component.ts`) The component render pipeline is:

1. call user render() to produce a VNode tree
2. if any hSlot() call happened while rendering (module flag), run fillSlots(vdom, externalChildren)
3. reset the module flag so the next render starts clean

This is literally the code:

```
if (didCreateSlot()) {
  fillSlots(vdom, this.#children)
  resetDidCreateSlot()
}
```

So slot filling is a *conditional rewrite pass* executed only when a slot marker was created.

The tests explicitly check this optimization:

- when the component has slots, the "fillSlot()" function is called
- when the component has no slots, the "fillSlot()" function is not called

(See $\texttt{src/}_{tests/component-slots.test.ts}$).

11.2 fillSlots(): projecting external children into a component

The slot filling algorithm lives in src/slots.ts:

```
fillSlots(vdom, children)
```

At a high level:

- It traverses a VDOM tree.
- It finds SLOT nodes.
- It replaces a SLOT node with a FRAGMENT node containing either:
  - external content (children passed into the component), or
  - default slot content (children attached to the SLOT marker).

11.2.1 Contract

- Input: a rendered VDOM subtree vdom, and an array of children coming from the component call-site.
- Output: it mutates the tree in-place.
- Post-condition: every SLOT marker encountered is rewritten into a FRAGMENT node.

Why FRAGMENT? The renderer needs to splice *zero, one, or many* children into a single position in the parent. In Relax's VDOM model, fragments are the standard encoding for "a list of children with no wrapping element". By rewriting slots to fragments, everything downstream (mounting, patching, diffing) can treat the result as normal VDOM.

11.2.2 Choosing between default and external content

When fillSlots sees vdom.type === SLOT, it computes:

- defaultContent from vdom.children
- externalContent from the passed children

Then it uses a simple rule:

> If external content is provided, it wins. Otherwise the default content is used.

In code, this is:

```
const content = externalContent.length > 0 ? externalContent : defaultContent
```

This is exactly what src/$_{tests}$/$_{slots.test.ts}$$_{asserts}$:

- set default content if no external content
- set the external content if provided

11.2.3 Slots with no content are removed

If a slot has neither default nor external content, Relax removes it.

Implementation detail:

- A SLOT node is converted into a FRAGMENT.
- If the chosen content is empty, vdom.children = [].
- During the parent traversal pass, empty fragments that were slots are dropped.

Test: remove the slot if no content or default content.

A subtle implementation detail extttfillSlots() itself doesn't have access to a parent pointer. So it can't "remove the slot node" directly. Instead it uses a two-step approach:

1. rewrite SLOT →empty FRAGMENT (children = [])
2. during the parent's child-walk pass, drop empty fragments that were produced by empty slots

That design keeps the algorithm simple and makes behavior local to the traversal.

11.2.4 Shape detail: why `hSlot([defaultContent])` is special

There's a slightly odd but intentional shape rule documented in src/slots.ts:

> hSlot([defaultContent]) passes an array whose first element is itself an array.

So default slot content is often authored like:

```
hSlot([ h('span', {}, ['Hello']) ])
```

In this case the algorithm sees: "content is an array with a single element, and that element is also an array". It preserves that container shape so that: extbfa slot expands to exactly one fragment child in the parent, matching how the tests assert the resulting tree:

```
expect(vdom.children).toEqual([hFragment(defaultContent)])
```

11.3 Why fillSlots skips component children during traversal

The traversal logic in fillSlots explicitly ignores children whose type is COMPONENT.

This is important. Consider:

- You are filling a slot for component A.
- Inside A's subtree there might be a nested component B that also uses slots.

Relax wants **each component to fill its own slots** using its own external content. If the parent traversal reached into component children, it could fill the wrong slots with the wrong content.

This is a major boundary rule:

> Slot rewriting is done per-component and must not cross component vnode boundaries.

It also aligns with how the renderer treats components elsewhere: components are patched/mounted as black boxes whose internal VDOM is managed by the component instance.

This behavior is tested in src/$_{tests/slots.test.ts}$:

- ignores slots whose parent is a component (they are handled by the component)

11.4 Slots in real components (`component-slots.test.ts`)

The integration tests show what authoring looks like.

11.4.1 Basic projection

can have a slot where external content can be inserted:

- Component renders {slot} .
- Call site provides Hello.
- The mounted DOM contains the span.

11.4.2 Default content

when no external content is provided, the default content is used.

This is the standard "slot fallback" semantics.

11.4.3 Conditional slots

conditionally rendered slots verifies that slots can appear/disappear depending on component state, and the patcher will reconcile accordingly.

This is where slots meet the patching story from Chapter 8: slot output is just fragments and normal children after filling.

11.4.4 Updating slot content between renders

slot content updated between renders has a container component that alternates between:

- a single Hello
- or two children World !

The important point: slot external content is treated like regular children arrays and diffed normally after filling.

Practically:

- the container component re-renders and passes a different external children array
- the slotted component re-renders, sees a slot marker, and re-runs fillSlots
- after that rewrite, the patcher sees ordinary FRAGMENT/ELEMENT/TEXT children and reconciles them

11.4.5 Nested slots inside lists

slot in a nested component uses a table row component with a slot in a <td>. The parent maps titles ['One','Two'] into multiple TableRow instances.

This shows that slot content can be:

- a raw string child (which becomes a TEXT VNode via h() normalization)
- repeated many times across a list of nested components

11.5 Takeaways

- Slots are recorded as a VNode marker produced by hSlot().
- Slot filling is a tree rewrite: SLOT → FRAGMENT with chosen content.
- External content wins over default content.
- Empty slots are removed by dropping empty fragments.
- Each component fills its own slots; parent traversal does not cross component boundaries.

11.6 Walking the tests: what they prove

Slots are specified in two layers: a pure tree-rewrite unit suite and a DOM-level integration suite.

11.6.1 Unit semantics (src/tests/slots.test.ts)

remove the slot if no content or default content Proves the empty-slot removal behavior: a slot with no default and no external content is eliminated (via empty-fragment dropping).

set default content if no external content Proves fallback behavior and the default-content shape rule. The assertion expects [hFragment(defaultContent)].

set the external content if provided Proves the precedence rule: external content overrides default content.

ignores slots whose parent is a component Proves the traversal boundary rule: fillSlots must not rewrite nested component internals.

11.6.2 Integration semantics (src/tests/component-slots.test.ts)

can have a slot where external content can be inserted Proves end-to-end projection into mounted DOM.

default content is used Proves fallback behavior at DOM level.

slot is empty Proves empty slots become empty output (no wrapper nodes appear).

conditionally rendered slots Proves slots can appear/disappear and patch correctly when state changes.

slot content updated between renders Proves that changing the external children array results in normal DOM reconciliation after slot rewrite.

slot in a nested component Proves nested component usage with lists and primitive children (strings become text nodes via h() normalization).

fillSlots spy tests Proves the didCreateSlot() optimization works: slot rewrite only runs when hSlot() was actually used.

11.7 Online resources

- Web Components: MDN <slot> (conceptual comparison)
- Vue: Slots and content projection
- React: Passing JSX as children
- SolidJS (conceptual): Children utilities

12. HRBR Signals: a Small Deterministic Reactive Graph

Part III begins the HRBR runtime. Unlike the VDOM runtime (src/*), HRBR is a "compiled fast path" whose job is to:

- track fine-grained dependencies (signals)
- schedule updates deterministically (scheduler lanes)
- update DOM directly in blocks (later chapters)

This chapter focuses on the reactive graph implementation in runtime/signals.ts.

12.1 Public API: the contract we have to satisfy

The file runtime/index.ts exports these symbols from runtime/signals.ts:

- `createSignal(initial) -> [read, write]`
- `createEffect(fn, options?) -> { dispose() }`
- `createMemo(fn, opts?) -> readMemo`
- `batch(fn) -> result`
- `untrack(fn) -> result`

The executable specification lives in:

- runtime/$_{tests}$/$_{signals.test.ts}$
- runtime/$_{tests}$/$_{signals.determinism.test.ts}$
- runtime/$_{tests}$/$_{signals.fuzz.test.ts}$

12.2 Mental model

You can read runtime/signals.ts as a deliberately small version of a Solid-like graph:

- A **signal** is a node with a current value and a list of observers.
- An **effect** is a computation that re-runs when its dependencies (signals/memos) change.
- A **memo** is a computation that behaves like a derived signal.

Two design constraints keep showing up in the implementation and tests:

1. **Determinism**: observer notification runs in stable creation order.
2. **Leak-safety**: disposing a computation must detach it from all sources.

There's a third, more pragmatic constraint you'll see in the code:

- **Low ceremony**: everything is implemented with plain objects and a few intrusive linked lists.

This is not a "fully-featured" reactive system; it's a compact kernel meant to be embedded into compiled blocks.

12.3 Core data structures: intrusive lists for stable order

In runtime/signals.ts you will see a set of internal types:

- SignalNode<T> holds value and an intrusive doubly-linked list of observers: observersHead/observersTail.
- Computation represents an effect or memo computation.
- ObserverLink is the node stored in the signal's observer list.
- SourceLink is a reverse edge stored on the computation so it can detach from sources on re-run/dispose.

The key technique: when a computation reads a signal, addObserver(node, currentComputatio
inserts the computation into the signal's observer list *sorted by computation id*.

Computation ids come from a global counter:

```
let nextId = 0

comp._id = nextId++
```

Because ids are monotonically increasing, the observer list is stable and deterministic by construction.

12.3.1 Why two linked lists (observer list + source list)?

Each dependency edge is represented twice:

- a forward link in the signal's observersHead/observersTail list (who should be scheduled when the signal changes)
- a reverse link in the computation's sourcesHead/sourcesTail list (what should be detached before re-run/dispose)

This dual representation is what makes "conditional dependencies" safe: each run starts by detaching previous sources and then re-registering based on the reads that actually occur.

12.4 Dependency capture: `currentComputation`

Reactive systems need a way to know "who is currently running".

This file uses a single global pointer:

```
let currentComputation: Computation | null
```

12.4.1 Signal reads

In `createSignal()`, the reader is:

- `return node.value`
- `if currentComputation != null, also call addObserver(node, currentComputation)`

That means dependencies are collected by running the effect/memo function and letting reads "register".

No dedupe (by design) You may notice `addObserver()` explicitly does not deduplicate observers. This is safe because every run begins with `unlinkAllSources(comp)`, which removes all prior edges. So duplicates can only occur within a single run, and a well-behaved effect/memo typically reads each signal at most once.

12.4.2 `untrack(fn)`

`untrack(fn)` temporarily sets `currentComputation = null` while executing `fn`.

Spec (runtime/$_{tests/signals.test.ts}$):

- `untrack` prevents dependency registration

In other words: you can read a signal without subscribing.

12.5 Writes: schedule observers, preserve order

In `createSignal()`, the writer:

1. computes `nextValue` (supports "functional set" style)
2. bails out if `Object.is(nextValue, node.value)`
3. updates `node.value`
4. walks the observer list and schedules each observer

There's a subtle but important line in the loop:

> Capture next first, because running an observer can unsubscribe/resubscribe and mutate the list.

In code (comments preserved):

```
const next = cur.next
scheduleComputation(cur.comp)
cur = next
```

This ensures we traverse the original linked list in deterministic order.

What gets scheduled? When a signal is written, Relax schedules the dependent computations. Scheduling does not necessarily mean "run immediately" (see batching and lanes), but it always preserves the *creation order* contract.

12.6 Effects: cleanup, re-run, disposal

12.6.1 Cleanup semantics

Effects may return an optional cleanup function:

```
createEffect(() => {
  ...
  return () => cleanup()
})
```

The runtime guarantees cleanup runs:

- before an effect re-runs
- when the effect is disposed

Spec: effect cleanup runs before re-run and on dispose.

In code, the cleanup is stored on the computation object as comp.cleanup and executed in a try/finally block both in run() and dispose().

12.6.2 Detaching and resubscribing: `unlinkAllSources(comp)`

Before each run, the runtime calls unlinkAllSources(comp). That function walks the computation's sourcesHead list and unlinks each corresponding ObserverLink from the signal's observer list.

This gives the common reactive invariant:

> An effect's dependencies are exactly the signals it read during its last run.

It also enables conditional dependencies (read signal A in one run, read signal B in another) without leaking observers.

Leak-safety note extttunlinkAllSources() also deliberately nulls out fields on the link objects (with @ts-expect-error comments). That's not required for correctness, but it makes accidental retention less likely in this prototype implementation.

12.6.3 Dispose

Disposing an effect sets $_{d}isposedanddetachesfromallsources.$

Specs:

- disposed effects fully detach: further writes do not schedule work
- leak-ish checks in runtime/$_{t}ests_{/signals.determinism.test.ts}$

12.7 Re-entrancy: scheduling while running

In scheduleComputation(computation) there is an explicit re-entrancy rule:

- if the computation is $_{r}unning, donotrecursivelycall$run()
- instead, add it to a pending Set

Then flushPendingIfNeeded() runs pending computations after the current run unwinds.

12.7.1 Two reasons computations are delayed

The same pending set is used for two distinct delay cases:

- **re-entrancy**: a computation schedules itself while it's already running
- **batching**: writes should not trigger intermediate runs until the batch boundary

Both cases end up in the same deterministic flush code path.

Spec: re-entrancy: effect that schedules itself via writes does not recurse infinitely.

The test constructs an effect that reads a signal and writes it again until a fixed point is reached. The expected (deterministic) log is:

```
[0, 1, 2, 3]
```

12.8 Batching: a flush boundary

batch(fn) increments isBatching and makes writes collect work into pending. When the outer-most batch finishes, it flushes the pending set sorted by computation id.

Specs (runtime/$_{t}ests_{/signals.test.ts}$):

- batch groups multiple writes into a single effect re-run
- batch() is a flush boundary: nested batches still only produce one re-run after outer finishes
- batch() returns the callback result and preserves deterministic effect order after flush

This is the first place you see HRBR's theme: *deterministic scheduling is part of the user-facing contract.*

Implementation detail: deterministic flush At batch end, pending computations are flushed by sorting on $_id$:

```
Array.from(pending).sort((a,b) => a._id - b._id)
```

This is why the tests can assert stable order like ['a:2', 'b:2'].

12.9 Memos: derived signals with equality

createMemo(fn, opts?) creates a computation whose output is stored in a SignalNode<T>.

Important details:

- the memo lazily computes on first read (if (!hasValue) comp.run())
- it supports an equality function opts.equals (defaults to Object.is)
- it notifies observers only when the memo value actually changes

Spec: createMemo computes from signals and updates dependents.

Memo invalidation path When the memo value changes, it walks node.observersHead and schedules each observer computation. So effects that read a memo are notified exactly like effects that read a base signal.

12.10 Lanes: effects can be scheduled via the runtime scheduler

createEffect accepts options:
{ lane?: Lane, budgetMs?: number }.

If lane === 'sync' (the default), the initial run executes immediately.

If lane !== 'sync', createEffect creates a scheduler via createScheduler() (from runtime/scheduler.ts) and schedules runs into that lane.

12.10.1 How non-sync lanes are tested

In runtime/$_{tests/signals.test.ts, the tests override}$globalThis.setTimeout$_{to capture scheduled flush callbacks into an array. This}$

- createEffect(lane: 'default') does *not* run immediately
- the test manually drains the flush queue to trigger the run

Specs:

- createEffect(lane) schedules initial run (non-sync)
- createEffect forwards budgetMs to scheduler.schedule

The tests intercept setTimeout to make scheduler flushing deterministic in unit tests.

12.11 Determinism tests and fuzz tests

Determinism is explicitly tested at three levels:

- Unit test: effects run deterministically (creation order)
- Determinism suite: build a random-but-seeded graph and compare logs across fresh instances (signals.determinism.test.ts)
- Property/fuzz: randomly generate operations (set/batch/dispose) and ensure identical logs for the same seed (signals.fuzz.test.ts)

These tests are the reason the implementation invests in sorted observer lists and stable flush ordering.

12.12 Walking the tests: what they prove

This chapter is easiest to internalize by reading the tests as executable requirements.

12.12.1 Core semantics (**runtime/**$_{tests/signals.test.ts}$)

tracks dependencies and re-runs Proves that reads register dependencies and writes schedule effects. Also proves the Object.is bail-out on repeated values.

untrack prevents dependency registration Proves that untrack temporarily disables dependency capture.

batch groups writes Proves that multiple writes inside a batch yield a single re-run.

nested batch is a flush boundary Proves isBatching is a counter and only the outermost batch flushes.

deterministic effect order Proves flush order is creation order (based on $_id$).

cleanup runs before re-run and on dispose Proves cleanup semantics for both re-run and disposal.

disposed effects fully detach Proves unlinkAllSources works: disposed computations aren't scheduled by later writes.

re-entrancy convergence Proves the pending-set mechanism prevents infinite recursion and produces a deterministic sequence.

lane scheduling + budgetMs forwarding Proves non-sync effects are scheduled through the runtime scheduler and that budgetMs is passed through.

memos Proves memo recomputation + dependent effect notification.

12.12.2 Determinism suite (signals.determinism.test.ts)

This suite tests the strongest property: *same seed + same operation sequence yields the same log across fresh runtime instances.*

It also includes a practical leak-ish check: after disposing effect 0, it mutates signals repeatedly and asserts there are no new log entries starting with e0:.

12.12.3 Fuzz/property tests (signals.fuzz.test.ts)

The fuzz suite generates nested batches, writes, and disposals. The primary assertion is determinism (two independent runs produce identical logs).

It also has a smoke property about disposal effectiveness: if effects were disposed, the final mass-write should not trigger an absurd number of logs (catching obvious detach bugs).

12.13 Online resources

These systems are common in UI runtimes; the references below help connect the dots without assuming the exact same API surface.

- SolidJS (conceptual baseline): Fine-grained reactivity
- Preact Signals (conceptual): Signals and effects
- React (contrast): Render/commit model vs. fine-grained graphs
- A good "reactivity under the hood" talk: SolidJS reactivity deep dive (video)
- Linked list background (for intrusive lists): Linked list

12.14 Takeaways

- Signals store intrusive observer lists, sorted by computation creation id.
- Effects detach and re-subscribe each run, enabling conditional dependencies safely.
- Batching and re-entrancy are implemented with a pending set plus deterministic flush ordering.
- Lanes integrate with the scheduler so work can be delayed and budgeted.

13. HRBR Deterministic Scheduler: Lanes, Budgets, and Aging

Signals answer *what* should re-run. A scheduler answers *when* and in what order. In HRBR, scheduling is part of the public contract. We want:

- deterministic ordering (same inputs, same execution order)
- configurable latency via lanes (input vs transition vs idle)
- budget enforcement (avoid long flushes that block the main thread)
- starvation prevention (idle work should still eventually run)

This chapter is grounded in:

- Implementation: runtime/scheduler.ts
- Specs: runtime/$_{tests/scheduler.test.ts}$
- Devtools counters/events: runtime/devtools.ts (used by scheduler and signals)

13.1 The contract: schedules tasks, flushes deterministically

The scheduler exports three main concepts (see runtime/scheduler.ts):

- Lane: 'sync' | 'input' | 'default' | 'transition' | 'idle'
- ScheduledTask: { id, lane, priority, timestamp, deadline, budgetMs, run }
- a factory: createScheduler(options?) -> scheduler

The returned scheduler exposes:

- schedule(lane, run, budgetOrOptions?)
- flush({ budgetMs })
- hasPending()
- withBudget(budgetMs, fn)
- setFrameBudget(ms) and setDefaultBudget(ms)

The tests treat these as the source of truth.

A tiny scheduler contract We can summarize what the runtime scheduler promises in four bullets:

- **Pick order is deterministic**: same task creation order + same time evolution →same execution order.
- **Latency is explicit**: lane names are the only priority vocabulary.
- **Work is time-sliced**: every flush loop has a budget, and work may continue later.
- **Starvation is mitigated**: low-priority tasks can be promoted based on age/deadline.

13.2 Lanes: latency categories and fixed priority order

Lane priority is encoded as a small integer map:

```
const LANE_PRIORITY = {
  sync: 0,
  input: 1,
  default: 2,
  transition: 3,
  idle: 4,
}
```

But the scheduler's actual selection order is hard-coded as an array:

```
['sync', 'input', 'default', 'transition', 'idle']
```

Spec: runs tasks by lane priority (sync > input > default > transition > idle).

13.2.1 What is a lane in practice?

Lanes are not preemption. They're an ordering and continuation policy:

- tasks run to completion one-by-one
- lane selection chooses which queue head runs next
- aging/deadlines can temporarily override lane order via promotion

13.2.2 The sync lane

A sync task is special: schedule('sync', ...) triggers an immediate flush.
In schedule():

- the task is still enqueued
- then the scheduler calls flush({ budgetMs: Infinity })

This gives a "run now" lane while still reusing the same task machinery.

Why still enqueue sync tasks? Because determinism still matters. Enqueueing before flushing means sync work still participates in the same stable ordering rules if multiple sync tasks are scheduled.

13.3 Flush strategies: how work gets a time slice

The scheduler accepts requestFlush(cb) in its options. If you don't provide one, it uses createRequestFlush('timeout').

13.3.1 createRequestFlush(strategy)

`createRequestFlush()` supports:

- `'timeout' (universal fallback): setTimeout(cb, 0)`
- `'messageChannel': uses MessageChannel if available`
- `'raf': uses requestAnimationFrame if available`

Specs in runtime/$_{tests}$/$_{scheduler.test.ts}$$_{cover}$:

- `createRequestFlush(messageChannel) uses MessageChannel when available`
- `createRequestFlush(raf) uses requestAnimationFrame when available`

13.3.2 createBrowserScheduler() convenience

`createBrowserScheduler( strategy )` is a wrapper that wires the chosen strategy into `requestFlush`.

Spec: `createBrowserScheduler wires a strategy into requestFlush (smoke).`

Why these strategies exist The scheduler is designed to be embeddable:

- `in Node-like environments (setTimeout)`
- `in browsers where MessageChannel is often lower-latency than timeouts`
- `in UIs where requestAnimationFrame aligns work with the next frame`

13.4 Deterministic ordering inside a lane

Within a lane, tasks are sorted with explicit tie-breakers:

1. earlier deadline
2. earlier timestamp
3. lower id

This ordering is implemented by `sortQueueInLane()`:

```
q.sort((a, b) => (a.deadline - b.deadline)
  || (a.timestamp - b.timestamp)
  || (a.id - b.id))
```

Spec: `orders tasks within a lane by explicit deadline (then timestamp/id).`

13.4.1 Where deadlines come from

If you don't provide an explicit deadline, schedule() computes:

```
deadline = timestamp + budgetMs
```

So tasks with smaller budgets naturally get earlier deadlines and bubble forward within their lane. This is a useful "do small things first" heuristic that is still deterministic.

13.5 Budgets: splitting work over multiple flushes

There are two different "budget" knobs:

- defaultBudgetMs: the per-task budget used to compute default deadline and default budgetMs
- frameBudgetMs: a global cap for *scheduled flush loops*

13.5.1 What flush(budgetMs) does

flush({ budgetMs }):

- emits devtools events flushStart and flushEnd
- runs tasks in priority order until:
  - queues are empty, or
  - elapsed time exceeds budgetMs
- if budget is exceeded *and* work remains, it schedules a continuation flush

Specs:

- budget enforcement: flush splits work across multiple flush calls when budget is tight
- setFrameBudget caps the budget used by scheduled flushes

13.5.2 setFrameBudget(ms)

Even if tasks provide their own budgets, scheduled flushes are capped by frameBudgetMs.

The spec demonstrates this by setting frameBudgetMs = 1 and scheduling tasks that each consume 2ms; only one task runs per flush, and the second flush runs the remainder.

13.5.3 Frame budget versus task budget (important distinction)

In runtime/scheduler.ts, budgetMs is stored on each task, but it isn't directly used by flush(). Instead:

- **task budget** influences default deadlines (and therefore ordering)
- **flush budget** (flush({budgetMs}) or frameBudgetMs) limits how long the flush loop runs before continuation is scheduled

The test suite focuses on flush budgets because those directly affect responsiveness.

13.6 Deadlines and aging: starvation prevention via promotion

HRBR's scheduler includes an aging/promotion mechanism.

In the implementation:

- AGING_MS = 250
- lanes eligible for promotion checks: idle, transition, default
- promotion moves one step toward higher priority: idle -> transition -> default -> input

Promotion runs *before* picking the next task, and it only considers the **queue head** in each lane.

A task is promoted if:

- it is overdue: now() >= head.deadline, or
- it is old: now() - head.timestamp > AGING_MS

Specs:

- ages/pushes starving tasks upwards (idle -> transition)
- promotes a task when its deadline is reached
- starvation prevention: repeated promotions eventually let idle work run even under continuous input load

One subtlety the tests call out:

> Promotion happens before lane selection, so an aged/overdue idle task can execute before later-scheduled default work.

This is a deliberate fairness trade-off.

13.6.1 Fairness without nondeterminism

The promotion rule checks only **queue heads** and promotes at most one task per candidate lane per pick. That has two benefits:

- it's cheap (no scanning the entire queue)
- it stays deterministic (the head is a deterministic choice)

13.7 withBudget(): bridging user work and scheduler work

Real apps do work outside of the scheduler too (rendering, event handlers, parsing). withBudget(budgetMs, fn) is a helper that:

- runs fn
- if fn exceeds the budget and tasks remain, it schedules a continuation flush

Spec: withBudget schedules continuation when user code exhausts budget and tasks remain.

This is the "be nice to the main thread" glue between imperative work and queued reactive work.

13.7.1 Global helpers in `runtime/index.ts`

The runtime also exports shared budgeting helpers that use a module-level global scheduler:

- withBudget(budgetMs, fn)
- setFrameBudget(ms)

This is covered by the smoke test runtime withBudget() helper returns the callback result and is callable.

13.8 Devtools and instrumentation hooks

The scheduler calls emitDevtoolsEvent() (runtime/devtools.ts) when flushing:

- flushStart
- flushEnd{ durationMs }

These feed counters like flushes and flushTimeMs.

Signals also emit scheduleComputation events when a computation is scheduled.

(We'll use these counters later when measuring block update costs.)

13.9 Walking the tests: what they prove

The runtime scheduler suite is unusually readable because it uses a manual clock and a manual flush queue. That makes time and continuation behavior deterministic.

13.9.1 Core semantics (runtime/tests/scheduler.test.ts)

runs tasks by lane priority Proves the lane order and that sync executes immediately.

orders tasks within a lane by explicit deadline Proves the sort tie-breakers inside a lane.

respects budget and continues in a later flush Proves a scheduled flush loop may stop early and reschedule continuation.

budget enforcement: flush splits work Proves deterministic time slicing under a small budget.

setFrameBudget caps scheduled flush budget Proves continuation flushes use frameBudgetMs as a cap.

ages/pushes starving tasks upwards Proves age-based promotion and the fairness trade-off (promoted work can run before later default work).

promotes a task when its deadline is reached Proves deadline-based promotion independent of age.

starvation prevention under input load Proves repeated promotions eventually allow idle work to run.

createRequestFlush(messageChannel/raf) Proves strategy selection uses platform APIs when present.

createBrowserScheduler smoke Proves wiring strategy into requestFlush.

13.10 Online resources

- MDN: MessageChannel
- MDN: requestAnimationFrame
- MDN: Event loop (tasks, microtasks)
- React (conceptual parallel): Render/commit (scheduling intuition)
- Cooperative background work (conceptual): requestIdleCallback

13.11 Takeaways

- Lanes provide a small, explicit latency model with a deterministic priority order.
- Each lane is internally sorted by deadline, then timestamp, then id.
- Budgeting is enforced at flush time to time-slice work.
- Aging/promotion makes starvation unlikely without making scheduling non-deterministic.
- withBudget lets user code cooperate with the scheduler.

14. HRBR Blocks: Template + Slots (the Compiled Fast Path)

Chapters 12 and 13 gave us two halves of a reactive UI runtime:

- **Signals** decide *what* values changed.
- **Scheduler lanes** decide *when* to apply work.

Blocks are the third half: they decide *how* to touch the DOM.

A **block** is the "fast DOM" path: it describes a stable DOM template and a set of precise write locations (slots). Updates become *O(k)* slot writes instead of building a VDOM tree and diffing children.

This chapter is grounded in:

- Implementation: runtime/block.ts
- Specs: runtime/$_{tests/block.test.ts}$
- Slot matrix: runtime/$_{tests/block-slot-matrix.test.ts}$
- Input controls: runtime/$_{tests/block-input-controls.test.ts}$
- Disposal correctness: runtime/$_{tests/block-disposal.test.ts}$

14.1 A tiny contract (what a block guarantees)

Treat this as the "promise" the runtime makes to compiled code:

1. **Single stable root:** templateHTML must produce exactly one root element, which is cloned and appended to the host.
2. **Exact addressing:** each slot points to a node using a path: number[] walk over childNodes indices.
3. **Specialized patching:** slot kinds are patched by small, explicit routines (text/attr/prop/class/style/event).
4. **No needless DOM writes:** redundant values are skipped via Object.is.
5. **Safe lifecycle:** destroy() is idempotent, removes listeners, and after destroy further update() calls are ignored.

14.2 Block vocabulary

14.2.1 TemplateHTML

A block definition includes a templateHTML: string. This must produce exactly one root element.

Implementation detail: parseTemplate(templateHTML) uses a <template> element and asserts:

```
templateHTML must have a single root element
```

The implementation is intentionally strict:

- it does templateHTML.trim() before parsing
- it reads tpl.content.firstElementChild
- it asserts non-null and returns the root element

This makes "multiple root nodes" a loud error, which is what you want for compiled templates.

14.2.2 Slot

A **slot** is a metadata record describing *where* and *how* to write a dynamic value into a template instance.

Slots are keyed by string (slot key) and come in these kinds:

- text: write to a Text node
- attr: set/remove attributes
- prop: set DOM properties
- class: write class attribute
- style: write inline styles
- event: bind events once and swap handler references

Each slot includes a path: number[] which is a **childNodes index walk** from the template root.

Two practical consequences:

- childNodes counts text nodes and comments, so whitespace in templates matters.
- Slot paths are best generated by a compiler that controls the template serialization.

14.3 Why paths (and not querySelector)?

Paths are the "compiled fast path" address:

- they're *exact* (no ambiguities)
- they're cheap (depth walk, no subtree scan)
- they enable prefix caching across multiple slots

14.4 The public API (surface area)

The core exports in runtime/block.ts are:

- defineBlock(def): identity helper for typing and readability
- mountBlock(def, host, initialValues?, options?) -> MountedBlock
- mountCompiledBlock(def, host, compiledSlots, options?) -> MountedBlock & { dispose() }
- mountNestedBlock(parent, slotKey, def, initialValues?)
- mountNestedFallback(parent, slotKey, mount)
- resolvePath(root, path) and resolvePathCached(root, path, slotKey, dev?)

Reading guide:

- mountBlock is the non-reactive primitive (mount + update).
- mountCompiledBlock wires signals into update with effects and the HRBR scheduler.
- mountNestedBlock/mountNestedFallback are the structural escape hatches.

14.5 defineBlock(): the type boundary

extttdefineBlock(def) doesn't transform input. It exists so "compiled" code can be explicit and type-safe.

In runtime/block.ts, it is literally an identity function:

```
extttexport function defineBlock(def: BlockDef): BlockDef { return def
}
```

That sounds trivial, but it creates a clean "compiler boundary" and a stable type surface.

Tests use it in every case:

```
const block = defineBlock({ templateHTML: ..., slots: ... })
```

14.6 mountBlock(): instantiate template + resolve slot nodes

mountBlock() does the non-reactive mount:

1. parse template HTML into a root element
2. clone it to create an instance
3. append instance root to the host
4. resolve each slot path into a concrete DOM Node
5. apply initialValues via block.update(initialValues)

The key optimization is "resolve once, write many": paths are resolved at mount time and stored in slotNodes. Updates never traverse the DOM tree; they dereference and write.

The returned MountedBlock includes:

- root: the mounted root element
- slotNodes: mapping from slot key to resolved node reference
- update(values): apply slot writes
- destroy() / dispose() lifecycle

14.6.1 Path resolution and caching

resolvePath(root, path) is the simple implementation. But mountBlock uses resolvePathCach path, slotKey, dev).

The cache is a Map<string, Node> stored on the instance root as $_{h}rbrPathCache.Prefixesarestore$ $joinedstringslike$"1.0".

The caching algorithm is incremental: as it walks the path, it caches each prefix result ("1", then "1.0", etc.). Multiple slots that share a prefix benefit immediately.

Spec: mountBlock caches repeated path prefixes across slots.

14.6.2 Dev diagnostics

When options.dev === true, the assertions include the slot key in the error.

Spec: dev diagnostics: mountBlock includes slot key in invalid path errors (dev mode).

14.7 update(values): slot patching rules

The update loop iterates over provided keys only. If a key doesn't exist in def.slots, it is ignored.

Two important performance rules:

- The block stores prevValues[key] and skips redundant writes using Object.is.
- It emits a devtools event slotWrite for each actual write.

The use of Object.is is deliberate: it treats NaN as equal to itself and provides a cheap "same reference" check for objects (like style objects or class arrays).

Spec: skips redundant writes for unchanged values (text + attr).

14.7.1 text slots

Text slots must resolve to a real Text node. The runtime asserts:

```
slot 'x' expected a Text node
```

Writes use Text.nodeValue and normalize nullish values to the empty string.

Spec: mounts a template and updates text slots by path.

14.7.2 attr slots (including boolean attributes and SVG xlink)

Attribute writes route through setAttr(el, name, next):

- null | false removes the attribute
- boolean-ish attributes are present/absent (set to empty string)
- SVG xlink:* attributes use setAttributeNS

Implementation notes:

- SVG detection uses namespaceURI (not instanceof SVGElement) for broader runtime compatibility.
- Boolean attribute names are normalized (readOnly maps to readonly).

Specs:

- updates attribute slots by path
- treats boolean attributes as present/absent (attr slots)
- sets xlink:* attributes with the xlink namespace on SVG elements

14.7.3 prop slots (input correctness)

Property slots route through setProp(el, name, next).

Special-case correctness for inputs:

- value: written as a string property, nullish clears to ""
- checked: written as boolean

This matches standard controlled-input behavior: writing to properties updates the live control state, while attributes can drift.

Specs:

- updates property slots by path
- updates input value/checked via property semantics for prop slots
- input control suite in block-input-controls.test.ts

14.7.4 class slots

Class slots write the class attribute:

- null | false removes the attribute
- arrays are joined with spaces after filtering falsy values
- otherwise coerced to string

14.7.5 style slots

Style slots accept:

- null | false: remove whole style attribute
- string: set style attribute directly
- object: per-key style.setProperty / removeProperty

This small matrix is validated by block-slot-matrix.test.ts.

14.7.6 event slots: bind once, swap handler

Event slots are designed to avoid listener leaks and avoid churn:

- the first time a slot is set, the runtime binds a stable listener function
- the listener reads rec.current to find the latest handler
- updates replace rec.current without re-binding
- destroying the block removes the listener

Specs:

- supports event slots: binds once and updates handler without leaking listeners
- destroy removes event listeners (no further handler calls)
- skips redundant event listener churn when handler identity is unchanged

Matching Relax VDOM semantics: hostComponent binding

mountBlock(..., { hostComponent }) optionally calls handlers with this === hostComponent, matching Chapter 7's event binding behavior in the VDOM runtime.

14.8 Lifecycle: destroy vs dispose, and idempotency

mountBlock is not reactive by itself, so dispose() is just an alias for destroy().

destroy() is idempotent:

- it guards with a destroyed flag

- it destroys composed children
- it removes event listeners
- it removes the block root from the DOM

After destroy(), update() becomes a no-op (guarded by the same destroyed flag). That prevents stale scheduled work from mutating detached DOM.

Specs:

- destroy is idempotent even with event slots and children
- lifecycle: destroy() is idempotent and prevents further updates

14.9 mountCompiledBlock(): wiring signals to slot writes

mountCompiledBlock is the bridge between the reactive graph and the block DOM writer.

Input: CompiledSlot[] where each entry is:

```
{ key: string, read: () => unknown }
```

Important: read() can access signals; that's how dependency tracking happens.

Mechanism:

- compute initial values by running each read()
- mount a non-reactive block with those initial values
- create one createEffect() per compiled slot
- coalesce multiple invalidations into a single scheduled flushPending
- execute block.update(values) inside batch() to keep effects consistent

The coalescing logic (pending + scheduled flag) is the performance heart: many effects can invalidate in one turn, but only one scheduled task drains pending and commits to the DOM.

Scheduling goes through the deterministic lane scheduler from Chapter 13:

```
extttscheduler.schedule(lane, flushPending)
```

Spec: changing a signal updates only the intended DOM node (mountCompiledBlock).

That test also checks identity stability:

- the static DOM elements are not recreated
- text node identity remains stable; only nodeValue changes

Spec: mounting does not recreate static DOM during updates.

14.9.1 Disposal stops reactive updates

mountCompiledBlock returns a dispose() method that:

- disposes all effects
- destroys the block DOM
- is idempotent

Specs:

- mountCompiledBlock.dispose stops reactive updates and removes DOM
- lifecycle: mountCompiledBlock.dispose() stops reactive updates and is idempotent

14.10 Composition: nested blocks and fallback regions

Blocks are stable and fast, but real UIs still need dynamic structure. runtime/block.ts provides two small composition helpers:

- mountNestedBlock(parent, slotKey, def): mount a child block into an Element slot
- mountNestedFallback(parent, slotKey, mount): mount a structural fallback region into an Element slot

Both helpers:

- clear the host element's existing children
- mount into it
- register a child destructor so destroying the parent destroys the child

Specs in runtime/$_{tests}$$_{/block.test.ts}$:

- block composition: can mount a nested block ... and destroys it with parent
- block composition: can mount a nested fallback region ... and disposes it with parent

This is HRBR's interop story:

> Compiled stable subtrees use blocks; dynamic subtrees route to fallback reconciliation.

Implementation detail: composition is tracked with a small internal child registry. extttmountBlock exposes an internal $_{h}rbrTrackChild function, andthecompositionhelpersuseittoregiste$

14.11 Walking the tests (the executable spec)

14.11.1 runtime/$_{tests/block.test.ts}$

This test file locks down the baseline behavior of each slot type and of the compiled/reac
mount.

- **text slots:** the slot must resolve to a real Text node; updates rewrite nodeValue; destroy removes DOM.
- **attr slots:** null removes the attribute.
- **prop slots:** input value is updated via property semantics.
- **class/style:** null removes attrs, arrays join for class, style objects set properties.
- **event slots:** click once, swap handler, click again; after destroy() clicks do nothing.
- **boolean attributes:** present/absent behavior is asserted with disabled.
- **SVG xlink:** uses getAttributeNS to validate namespace behavior.
- **path caching:** asserts $_{hrbrPathCache}containsprefixkeys(e.g.$"0", "0.0").
- **dev diagnostics:** invalid path errors include the slot key in dev mode.
- **mountCompiledBlock:** changing one signal updates only its text node, and node identity is stable across updates.

14.11.2 runtime/$_{tests/block-slot-matrix.test.ts}$

This is a matrix-style smoke test that mounts one template with **all slot kinds** and asserts both initial values and updated values, including event swapping.

14.11.3 runtime/$_{tests/block-input-controls.test.ts}$

This suite documents the intended controlled-input semantics for value and checked when patched as properties. It also includes a note about radio group behavior in jsdom.

14.11.4 runtime/$_{tests/block-disposal.test.ts}$

This suite locks down lifecycle safety:

- destroy() removes event listeners (clicking a detached element must not call handlers).
- destroy() is idempotent.
- mountCompiledBlock.dispose() stops reactive updates and removes DOM permanently.

14.12 Online resources

- MDN: <template> (template parsing and cloning)
- MDN: Node.childNodes (why paths include text/comment nodes)
- MDN: addEventListener (listener identity + removal)
- MDN: HTMLInputElement (property semantics for value and checked)
- MDN: xlink:href (SVG namespaced attribute background)
- Svelte: "Virtual DOM is pure overhead" (compiler-driven DOM updates intuition)

14.13 Takeaways

- Blocks are "template + slots": a stable DOM instance plus precise write locations.
- Slot patching is specialized per kind, with correctness for input controls and SVG namespaces.
- Updates skip redundant writes via Object.is.
- Event slots bind once and swap handler references; destroy removes listeners.
- mountCompiledBlock wires signals to slots and coalesces updates into scheduled flushes.
- Composition helpers let blocks nest blocks and fallback regions safely.

15. Hydration: Attaching HRBR to Server HTML

A block runtime has a natural relationship with server rendering:

- The compiler produces a stable template shape (templateHTML).
- The server can render that template to HTML.
- The client can *attach* to the existing DOM instead of rebuilding it.

In Relax HRBR v1, hydration is deliberately conservative. If anything looks wrong, it bails out and remounts client-side.

This chapter is grounded in:

- runtime/hydration.ts (hydrate implementation)
- runtime/ssr.ts (string rendering helper used in tests)
- runtime/$_{tests}$/$_{hydration.test.ts}$
- runtime/$_{tests}$/$_{ssr.test.ts}$

15.1 A tiny contract (what hydrateBlock promises)

Hydration is the moment where two worlds meet:

- The **server** produced HTML as a string.
- The **client** wants a live block instance with slot node references.

In v1, Relax uses a strict approach that is easy to reason about and hard to get subtly wrong. You can treat hydrateBlock(def, host, initialValues) as enforcing this contract:

1. If there is no DOM to attach to (host.firstElementChild == null), hydration simply mounts a fresh block.
2. If the root element tag does not match the template's root tag, hydration clears the host and mounts fresh.
3. If slot paths resolve to the wrong node types or to the wrong element tag names (compared to the parsed template), hydration clears the host and mounts fresh.
4. If everything matches, hydration returns a block-like object whose root and slotNodes refer to the server DOM nodes.

The headline design choice is: **when in doubt, remount**. That makes correctness obvious at the cost of being less clever than incremental/partial hydration systems.

15.2 Hydration contract (v1)

hydrateBlock(def, host, initialValues?) returns a HydratedBlock (type alias of MountedBlock).

Its docstring states three key assumptions:

1. host contains exactly one root element matching the template.
2. Hydration does not mutate DOM; it only resolves slot node references.
3. On mismatch, hydration bails out and remounts client-side.

This is a strict, "safety first" strategy: it prefers correctness over partial hydration tricks.

It's also worth noticing what the contract does *not* promise:

- It does not try to recover from mismatches by patching individual nodes.
- It does not attempt to reconcile text/comment whitespace differences.
- It does not do event delegation or partial listener attachment.

15.3 The fast path: attach without rewriting

Given server markup like:

```
<div><span>A</span></div>
```

and a matching block definition:

```
templateHTML: `<div><span>A</span></div>`
slots: { a: { kind: 'text', path: [0, 0] } }
```

The hydration algorithm is:

1. Read serverRoot = host.firstElementChild.
2. Validate root tagName against the template (cheap check).
3. Resolve every slot path against serverRoot.
4. Validate slot node types, and element tagNames against the template DOM.
5. Return a HydratedBlock whose update() writes into those resolved nodes.

Two details in runtime/hydration.ts make this efficient:

- Root tag validation is a fast regex over templateHTML (getRootTagName). This allows an early bail-out without parsing a template DOM.
- Slot resolution is just a childNodes index walk (resolvePath). It's similar to runtime/block.ts, but hydration keeps its own copy with different error strings.

Spec: hydrates without rewriting DOM when markup matches.

That test captures node identities before hydration and asserts identity stability after calling hydrateBlock:

- same root element
- same inner
- same text node

Then it updates slot values and checks the content changed without replacing nodes.

15.4 Mismatch detection: when to bail out

Hydration needs a mismatch policy. In v1 it is intentionally conservative.

15.4.1 Root tag mismatch

The first check is easy: compare the root tagName.

Implementation: getRootTagName(templateHTML) uses a small regex for the first tag. If root tags don't match, hydration does:

```
host.innerHTML = ''
return mountBlock(def, host, initialValues)
```

This is the first key safety property: mismatched DOM never becomes a "half-hydrated" zombie tree. We either attach cleanly or we tear down and replace.

Spec: mismatch triggers remount for subtree.

15.4.2 Slot path mismatch: element tagName mismatch

Root tag matching is not enough. You can have the same root element but different internal structure.

The conservative check in runtime/hydration.ts is:

- Parse the expected template root into a detached DOM node.
- For each non-text slot:
  - resolve the same path against the expected template DOM
  - assert both are Elements
  - assert expected.tagName === actual.tagName

Why compare tag names at slot paths (instead of deep-walking the entire tree)?

- Slots are where we will write later, so validating those boundaries catches many mismatches that would otherwise crash later.
- It avoids paying for a full tree diff during hydration.

- It keeps the rule extremely easy to explain and test.

If any assertion fails, hydration catches and bails out with client remount.

Spec: slot path element mismatch triggers bailout remount (conservative).

The test intentionally renders:
X
but the template expects:

So hydration remounts and you end up with a .

15.5 Hydrated update semantics: same slot kinds as mountBlock

Hydration returns an object that looks like a mounted block:

- root is the server root element
- slotNodes maps slot keys to resolved DOM nodes
- update(values) writes into those nodes
- destroy() removes hydrated listeners and removes root

The update logic supports the full set of slot kinds:

- text: write Text.nodeValue
- attr: boolean attribute semantics and SVG xlink namespace
- prop: input correctness for value / checked
- class: string or array join
- style: string or object diff
- event: bind event handler(s)

Spec: hydrates with full slot semantics (event/attr/prop/SVG) without rewriting existing DOM.

Most of the slot patching logic is a copy of the mount-time block patching rules (Chapter 14), but hydration differs in one important place: event slots.

That test verifies:

- DOM identity is preserved for root/button/input/link
- clicking calls the hydrated handler
- boolean attribute hidden is present
- checked property toggles
- SVG xlink:href is set in the xlink namespace

15.5.1 Redundant write skipping

Hydration tracks prevValues and skips updates when Object.is(prev, next). This matches the block runtime's "skip redundant writes" optimization.

This is a subtle performance win because it also prevents unnecessary listener churn (given hydration's v1 event strategy below).

15.6 Event slots: v1 difference vs mountBlock

There's an intentional simplification in hydration:

- mountBlock binds once and swaps rec.current (no listener churn).
- hydrateBlock removes the previous listener and re-adds on update.

You can see this directly in runtime/hydration.ts:

- On event updates, it looks up listenersByKey[k].
- If present, it calls removeEventListener(prev.type, prev.handler).
- If the new value is a function, it creates a fresh wrapper handler and calls addEventListener(slot.name, handler).

This is not as optimal as the bind-once approach in mountBlock, but it is still correct and easy to reason about.

It still avoids leaks because destroy() removes any listeners it attached.

This choice keeps hydration simpler while still correct. If hydration becomes a performance hot spot, it can be refactored to reuse the bind-once strategy.

15.7 SSR helper: renderBlockToString()

runtime/ssr.ts provides a minimal string renderer used mainly for tests.

Contract:

- create a detached host element
- mountBlock(def, host, initialValues)
- read host.innerHTML
- destroy the mounted block

This is explicitly a v1 teaching tool:

- It uses the browser's DOM implementation to do serialization.
- It's not the kind of streaming SSR engine you'd use in production.

But it has an important property: it guarantees that the HTML string shape matches mountBlock semantics, because it's literally produced by the same code.

Spec: renders a block to HTML string, hydrates it, and updates via mountCompiledBlock.

This gives a full lifecycle demo:

1. render HTML string from a block
2. insert it into the document
3. hydrate it without rewriting
4. mount a compiled block on top of the hydrated DOM and update via signals

15.8 Takeaways

- Hydration v1 is conservative: it validates structure and bails out on mismatch.
- When markup matches, hydration preserves DOM identity (no rewrites).
- Hydrated updates support the full slot matrix (text/attr/prop/class/style/event + SVG xlink).
- SSR v1 uses the DOM to produce strings for tests; it's simple and correct, not optimized.

15.9 Walking the tests (the executable spec)

Hydration and SSR are notoriously easy to get "mostly right" and still ship broken edge cases. Relax locks down the v1 behavior with two test files.

15.9.1 runtime/tests/hydration.test.ts

This suite defines the hydration safety model.

- **hydrates without rewriting DOM when markup matches** captures root/span/text node identities before hydration, hydrates with an initial value, and asserts:
  - the same elements/text nodes are still present (identity preserved)
  - only the content changes
- **hydrates with full slot semantics (event/attr/prop/SVG) without rewriting existing DOM** verifies the "slot matrix" works even when DOM is pre-existing:
  - event handler is attached and called
  - boolean-ish attr (hidden) is present
  - checked is set as a property
  - SVG xlink:href is set in the xlink namespace
- **mismatch triggers remount for subtree** uses the wrong root tag (<section>) and asserts hydration replaces the entire subtree.
- **slot path element mismatch triggers bailout remount (conservative)** uses the same root tag but the wrong inner tag at a slot path (vs expected). The test asserts that hydration bails out and remounts so the template structure is restored.

15.9.2 runtime/tests/ssr.test.ts

This test is a full lifecycle demo "in miniature":

1. **render** a block to an HTML string via renderBlockToString
2. insert the HTML into a host
3. **hydrate** without changing root/text node identities
4. **mountCompiledBlock** to wire signals and update the hydrated DOM

It's intentionally pragmatic: rather than relying on timing, it calls mounted.update(value: 'B') to assert the DOM is updated. The key idea being demonstrated is: *hydration produces a block-shaped surface that the rest of HRBR can build on.*

15.10 Online resources

- MDN: <template> (hydration compares template structure by parsing)
- MDN: firstElementChild (the attach point)
- MDN: Node.childNodes (path-based DOM addressing)
- MDN: events + listener removal
- MDN: xlink:href (namespaced SVG attribute semantics)
- A broader hydration overview (framework-agnostic): web.dev: Rendering on the Web.

16. SSR Authoring (v1): Rendering Blocks to HTML Strings

Server-side rendering (SSR) is how you produce initial HTML on the server so the browser can show an app quickly.

In HRBR v1, SSR is intentionally simple and primarily exists to support tests and examples. The important property is not performance; it's **compatibility** with hydration.

This chapter is grounded in:

- runtime/ssr.ts
- runtime/$_{tests}$/$_{ssr.test.ts}$
- and the follow-on behavior in runtime/hydration.ts (Chapter 15).

16.1 A tiny contract (what SSR v1 guarantees)

SSR is not "render an app" in the abstract. In HRBR v1 it is narrowly scoped:

1. You provide a BlockDef (template + slots).
2. You optionally provide initialValues (a slot-key → value map).
3. The runtime returns an HTML string that matches the same template structure the client will mount/hydrate.

Put differently:

> SSR v1 is *mountBlock + serialize*. Hydration v1 is *validate + attach, else remount*.

This tight coupling is deliberate. It makes SSR and hydration "round-trip compatible" by construction.

16.2 The contract: `renderBlockToString(def, options?) -> string`

The only exported function is:

```
renderBlockToString(def: BlockDef, options?: { initialValues?: Record<string, unknown> }):
```

The options only support one thing today: initialValues.

This mirrors the core block update contract:

A block update is just a map { [slotKey]: value }.

SSR is simply the "apply values, then serialize" version of that idea.

One way to think about it:

- **Client mount:** template → DOM instance + slot node references.
- **Client update:** slot values → DOM writes.
- **Server render:** slot values → DOM writes → serialize.

16.3 Implementation: DOM-based SSR for correctness

runtime/ssr.ts implements SSR using the platform DOM:

1. create a detached host element: document.createElement('div')
2. mount the block into that host via mountBlock(def, host, initialValues)
3. read host.innerHTML
4. destroy the mounted block to avoid leaks
5. return the HTML string

This is documented directly in the function comment:

- it uses a detached DOM
- it serializes via innerHTML
- it's primarily intended for tests/examples
- a future version could emit HTML without a DOM

The crucial correctness choice is that SSR applies initialValues by calling mountBlock(def host, initialValues). That means:

- all the slot semantics from Chapter 14 apply (boolean attributes, SVG namespaces, input properties, etc.)
- the generated markup shape matches what the client expects to attach to

16.3.1 Why DOM-based SSR is acceptable in v1

For a production SSR renderer, involving a full DOM implementation server-side can be expensive.

But for v1 HRBR, this has two big advantages:

- it is hard to get HTML escaping and attribute semantics wrong
- it produces markup exactly consistent with the same block runtime used in the client

So correctness comes for free.

There's also a subtle lifecycle advantage: because SSR uses mountBlock, tearing down is just mounted.destroy(). That ensures listeners are removed and the temporary root is cleaned up, even in a test environment.

16.4 SSR must round-trip with hydration

SSR isn't useful unless the client can hydrate it.

Hydration v1 makes strong assumptions (Chapter 15):

- the root element tagName matches templateHTML
- slot paths resolve to the expected node types
- element slots match expected tagNames along the path

Therefore, SSR output must preserve the template's structure. Because SSR uses mountBlock(), it naturally meets the same invariant:

> SSR uses the same template parser and slot patching semantics as client mount.

This is the important "authoring rule" for SSR v1:

> If you change templateHTML shape, you must change both server render and client hydrate together.

You should treat templateHTML as a stable ABI between server and client.

16.5 The end-to-end spec: render → hydrate → update

The key test is runtime/$_{tests}$/$_{ssr.test.ts}$:

1. Define a block with templateHTML and a text slot.
2. Render HTML with renderBlockToString(block, { initialValues }).
3. Insert the returned string into the DOM.
4. Call hydrateBlock(block, host).
5. Verify DOM identity is preserved (same root, same text node).
6. Mount a compiled block via mountCompiledBlock(...) and update.

This accomplishes two things:

- it proves that SSR output is "hydrate compatible"
- it shows how hydration and compiled blocks can be composed (a realistic client story)

One implementation-level nuance in the test is worth calling out. After updating the signal with setCount('B'), it also calls:

```
extttmounted.update({ value: 'B' })
```

That makes the assertion deterministic and keeps the test independent of scheduler timing. It's modeling the fact that mountCompiledBlock can schedule work, and test environments often differ in how timers/microtasks flush.

16.6 Limitations and the intended direction

The HRBR whitepaper explicitly lists a non-goal for v1:

> A DOM-free string renderer (v1 SSR uses a DOM-based approach for tests/examples)

A future SSR implementation could:

- parse templateHTML once into a token stream
- apply slot writes into a string builder
- avoid constructing DOM nodes entirely

Even if SSR becomes DOM-free later, the invariants from hydration still apply:

- the root tag must match
- the structure along slot paths must match
- text nodes must exist where text slots expect them

But even then, the contract remains:

> SSR output must match the template structure so hydration can do a no-rewrite attach.

16.7 Takeaways

- SSR v1 is intentionally minimal: mount a block in a detached DOM and serialize innerHTML.
- initialValues uses the same slot update contract as client updates.
- The primary goal is hydration compatibility, not server performance.
- The end-to-end test shows SSR + hydration + compiled updates working together.

16.8 Walking the tests (the executable spec)

SSR v1 has one test file, but it covers the entire story.

16.8.1 runtime/tests/ssr.test.ts

The test renders a block to HTML string, hydrates it, and updates via mountCompiledBlock enforces these guarantees:

- **SSR applies initial values:** renderBlockToString(..., { initialValues }) must produce HTML containing 'A'.
- **Hydration preserves identity:** after hydrateBlock, the root element and the text node under t are the same objects as before hydration.

- **Compiled updates work on hydrated DOM:** wiring a signal through mountCompiledBlock and then updating should change text to 'B'.
- **Cleanup is explicit:** the test disposes the compiled mount and destroys the hydrated block.

Because this file touches SSR, hydration, signals, and compiled blocks, it acts as a integration-level "safety rail" for the whole pipeline.

16.9 Online resources

- HTML serialization via DOM: MDN: Element.innerHTML.
- Server rendering and hydration overview: web.dev: Rendering on the Web.
- Template element (used by mounting/hydration): MDN: <template>.
- A compiler-first argument for skipping VDOM in the steady state: Svelte: "Virtual DOM is pure overhead".

17. Fallback Reconciler: Dynamic Structure When Blocks Aren't Enough

Blocks are fast because structure is stable: the template doesn't change, only slot values do.

But JSX in the real world has loops, conditionals, fragments, and dynamic lists. HRBR needs a safe escape hatch for those regions.

That escape hatch is the **fallback reconciler**:

- a runtime mount: runtime/fallback.ts
- a child-list reconciler: runtime/reconciler.ts

This chapter is grounded in:

- runtime/fallback.ts
- runtime/reconciler.ts
- runtime/$_{tests}$/$_{fallback.test.ts}$
- runtime/$_{tests}$/$_{reconciler.test.ts}$
- runtime/$_{tests}$/$_{reconciler.fuzz.test.ts}$

17.1 A tiny contract (what fallback + reconciler promise)

The fallback reconciler is the HRBR answer to "dynamic structure":

- Blocks are optimal when structure is stable.
- Fallback reconciliation is the safe, general mechanism when structure changes.

The contract is intentionally narrow:

1. reconcileChildren(host, next) only patches **direct children** of host.
2. Each child spec is responsible for building/patching its own subtree (if any).
3. In keyed mode, keys must be unique among siblings.
4. In keyed mode, DOM identity is preserved by key when possible; the algorithm minimizes moves, but *correctness* is the first goal.

extttmountFallback(host, render) is the reactive wrapper:

- render() can read signals.
- When those signals change, fallback schedules a reconcile pass in a scheduler lane.
- Disposal stops further reactive updates.

17.2 Contracts

17.2.1 The reconciler contract: direct children only

reconcileChildren(host, next, options?) patches host.childNodes so they match next.

Important: it reconciles **direct children only**. Each child spec is responsible for patching its own subtree.

That design choice keeps the reconciler small and composable:

- A "child spec" can represent a component, a fragment, or even an embedded compiled block.
- Nested reconciliation happens inside that child spec's own patch() function.

Types (runtime/reconciler.ts):

- ReconcileNode: a logical node that maps to exactly one DOM node
- ReconcileRange: a logical item that maps to multiple DOM nodes
- ReconcileItem = ReconcileNode | ReconcileRange

Each item supports:

- create() to create DOM
- patch(existing) to patch in place
- optional destroy(existing) to cleanup when removed

17.2.2 Keyed vs unkeyed mode

reconcileChildren chooses mode:

- if options.keyed is provided, use it
- else infer keyed mode if any next item has a key

It also emits dev-only warnings for suspicious usage patterns when opts.dev === true:

- mixing keyed and unkeyed siblings (unkeyed nodes become "unstable")
- replacing an entire keyed list with a disjoint set of keys (often unstable keys like array index)

The two algorithms are:

- reconcileChildrenUnkeyed(host, next)
- reconcileChildrenKeyed(host, next, dev)

17.3 Why ranges exist (one logical item => multiple DOM nodes)

In UI rendering, one logical item can expand to multiple nodes:

- a fragment
- a component that returns multiple roots
- a compiled block or template that expands into siblings

HRBR models this with ReconcileRange:

```
{ kind: 'range', key?, create(): Node[], patch(nodes: Node[]) }
```

The reconciler stores the range length on the first DOM node:

- $_{h}rbrKey : keyofthelogicalitem(onthestartnode)$
- $_{h}rbrRangeLen : numberofDOMnodesintherange(onthestartnode)$

This metadata is the core trick that lets the reconciler treat DOM as a list of *logical items* rather than raw nodes. The helper buildCurrentRanges(host) reads childNodes and groups them into ranges by repeatedly:

- read len = $_{h}rbrRangeLen??1 fromthenextstartnode$
- slice len nodes and treat them as one logical item
- advance by len

This is also why correctness requires normalization when items change shape (node ↔ range).

Spec: keyed: supports range items (logical item => multiple DOM nodes).

17.4 Unkeyed reconciliation: simple and predictable

reconcileChildrenUnkeyed relies on positional patching:

1. patch common prefix (min(oldLen, newLen))
2. remove extra old nodes
3. append new nodes

Spec: unkeyed: patches in order and truncates/appends.

Unkeyed mode is fast and simple, but it can't preserve identity across reorders.

Unkeyed mode is best when:

- lists are small
- items rarely reorder
- you don't care about preserving DOM identity across reorder (e.g. no input state)

17.5 Keyed reconciliation: preserve identity and minimize moves

Keyed mode tries to:

- reuse DOM nodes by key
- patch in place instead of recreating
- move existing ranges/nodes to new positions
- minimize DOM moves using a Longest Increasing Subsequence (LIS)

Specs in runtime/$_{tests}{}_{/reconciler.test.ts}coverthebasicexpectations$:

- keyed: moves DOM nodes instead of recreating them
- keyed: removes nodes that are no longer present
- keyed: inserts new nodes at the right position

17.5.1 Two-ended scan + LIS

The reconciler uses a standard sequence:

1. Build a map from key to current range (start..end, nodes[]).
2. Fast two-ended scan for common prefix/suffix keys.
3. For the middle region:
   - build an "old index" sequence for new keys
   - compute LIS indices
   - move only the ranges not in the LIS

Why LIS?

- If a subsequence of items already appears in increasing old-index order, those items can stay where they are.
- Everything else must move (or be created), but LIS keeps that set minimal.
- The implementation uses lisIndices(seq) over old indices, ignoring -1 (new items).

Implementation cues:

- buildCurrentRanges(host) groups DOM nodes into logical ranges by $_{h}rbrRangeLen$.

- lisIndices(seq) computes LIS over old indices (ignoring -1).
- moveRangeBefore(host, range, anchor) moves nodes end-to-start to preserve order.

17.5.2 Dev diagnostics for suspicious usage

Keyed mode includes two dev-only warnings:

- Mixed keyed and unkeyed children in the same list.
- Next keys have no overlap with previous keys (often indicates unstable keys).

Specs:

- dev diagnostics: warns on mixed keyed/unkeyed children in keyed mode
- dev diagnostics: warns when next keys have no overlap with previous keys

It also asserts that keys are unique among siblings.

That uniqueness assertion is a hard error, not a warning:

```
extttassert(!seen.has(k), '[hrbr/reconciler] duplicate key ...')
```

This is intentional: duplicate keys make identity preservation ill-defined.

17.5.3 Shape-change safety: node ↔ range

A tricky edge case: a key might render as a singleton node in one render and as a range in another.

The reconciler handles this explicitly:

- if key matches but logical shape changes (node vs range), it removes the old nodes and recreates
- it aggressively removes any *unkeyed trailing nodes* that might remain after a range collapses into a singleton

This is where the metadata invariants matter. The code contains normalization passes that:

- remove unkeyed nodes owned by keyed mode
- delete stale $_{hrbrRangeLen}whenthelogicalitemisnowasingleton$

The fuzz test exists largely to guard these invariants.

One specific implementation detail to notice is cleanupTrailingRangeNode(start, key). When a keyed item collapses from a range into a singleton node, the DOM may still contain trailing unkeyed nodes that used to be part of that range. The reconciler aggressively removes any contiguous unkeyed siblings immediately after the start node.

This is later reinforced by a final normalization pass that removes any unkeyed nodes interleaved between keyed logical items.

17.6 Property/fuzz testing as an executable oracle

runtime/$_{tests}$$_{/reconciler.fuzz.test.ts}$$runs250seededsteps.$

It compares reconcileChildren(host, ...) to a reference renderer that rebuilds the list each step but reuses nodes by key.

The key correctness check is not "minimal moves". It is:

> For the same list of logical items, the resulting DOM represents the same logical structure and text content.

The fuzz harness reads the logical structure with:

- start key: $_{h}rbrKey$
- range length: $_{h}rbrRangeLen(default1)$

Then it compares the logical tuples (key, len, texts[]) against the reference.

The fuzz test is doing something important for maintenance:

- It doesn't claim the reconciler is optimal.
- It claims that for any seeded sequence of updates, the reconciler produces the *same logical output* as a simpler reference "rebuild" strategy.

That's exactly the right kind of oracle for reconciliation code: optimality is a performance concern, but correctness is a binary property.

17.7 Fallback mount: reconciling signal-driven dynamic children

runtime/fallback.ts provides a reactive mount function:

```
mountFallback(host, render, { lane?, scheduler?, keyed? })
```

Contract:

- render() may read signals.
- When signals change, schedule a reconcile pass in the requested lane.
- The render result is { children: ReconcileItem[], keyed? }.

Implementation uses two passes:

- An effect that runs render() just to track dependencies, then schedules.
- A runOnce() pass that actually calls reconcileChildren().

This split is subtle but intentional:

- The tracking effect should be very cheap; it exists to subscribe to signals.
- The DOM work should be scheduled and batched (via batch(() => runOnce())).
- A pending flag ensures multiple invalidations coalesce into one scheduled reconcile.

In other words, mountFallback mirrors the same "coalesce and commit" philosophy used by mountCompiledBlock (Chapter 14): collect changes reactively, then perform one DOM commit.

Spec: reconciles structural changes when signal-driven list changes (fallback.test.ts).

That integration test verifies that:

- reorders preserve node identity by key
- inserts create new nodes
- removals shrink the DOM
- disposing stops reactive updates

17.8 Takeaways

- The fallback reconciler gives HRBR a safe strategy for dynamic structure.
- Keyed reconciliation preserves identity and minimizes moves via two-ended scan + LIS.
- Ranges let one logical item expand to multiple DOM nodes, enabling fragments and multi-root outputs.
- Metadata invariants $({}_{h}rbrKey, {}_{h}rbrRangeLen)arecrucial; fuzztestsenforcethem.$
- mountFallback bridges signals/scheduler to reconcileChildren for real apps.

17.9 Walking the tests (the executable spec)

17.9.1 runtime/$_{tests/reconciler.test.ts}$

This suite locks down the core expectations:

- **Dev warnings are emitted:** by temporarily replacing console.warn, the tests prove warnings fire for mixed keyed/unkeyed children and for disjoint key sets.
- **Keyed reorder preserves identity:** the "moves DOM nodes" test captures original nodes a0/b0/c0 then reorders and asserts node identity is preserved.
- **Keyed removals:** nodes that disappear from next are removed, while survivors keep identity.
- **Keyed inserts:** missing keys are created and inserted at the correct position.
- **Unkeyed mode is positional:** patches first, then truncates/appends, without identity guarantees across reorder.
- **Ranges work:** a range key produces two DOM nodes and behaves as a single logical item during moves/inserts/removals.

17.9.2 runtime/$_{tests/reconciler.fuzz.test.ts}$

This is the long-term safety net. It runs 250 seeded steps and compares the reconciler output against a reference renderer.

Key idea: the test reads *logical structure* from the DOM via $_{hrbrKey} and_{hrbrRangeLen}, thencompa$

17.9.3 runtime/$_{tests/fallback.test.ts}$

This test shows the reconciler in its intended reactive setting:

- a signal produces a keyed list
- changing the signal reorders, inserts, and removes
- identity is preserved for reused keys
- dispose() stops further reactive updates

The test uses `await new Promise((r) => setTimeout(r, 0))` to wait for the scheduled reconcile to run.

17.10 Online resources

- A classic keyed diff discussion (and why identity matters): React docs: Rendering Lists (keys).
- Longest Increasing Subsequence (the core move-minimization idea): Wikipedia: Longest increasing subsequence.
- A VDOM implementation reference with LIS-based keyed diff (for historical comparison): Vue docs: Maintaining State with key.
- DOM child manipulation primitives used by the reconciler: MDN: Node.insertBefore and MDN: Node.removeChild.

18. VDOM ↔ HRBR interop (mixed trees)

This chapter covers the experimental interop layer that allows compiled HRBR output to live *inside* a Relax VDOM tree. The goal is practical: you can adopt HRBR incrementally (or only for hot paths) without rewriting your whole app.

18.1 The tiny contract: an HRBR vnode is just a mount factory

In the VDOM layer, HRBR is represented by a new vnode type, DOM_TYPES.HRBR, implemented in src/h.ts.

18.1.1 VNode shape

The HRBR vnode type is defined as HrbrVNode:

- type: DOM_TYPES.HRBR
- mount(host: Element) -> instance
- internal bookkeeping fields: host, instance, el

The important part is that mount is referentially meaningful. Two HRBR vnodes are treated as the "same" subtree iff:
extttold.mount === new.mount.

In other words, *mount factory identity is the key*. This identity flows through three separate places that must agree:

- src/nodes-equal.ts: HRBR equality is old.mount === new.mount.
- src/patch-dom.ts: HRBR patching reuses the mounted instance on that same condition.
- src/destroy-dom.ts: HRBR teardown calls the instance hooks and removes the host wrapper.

This isn't an accident: it lets the VDOM runtime treat HRBR the same way it treats components. "Same function identity" means "keep state/instance".

18.1.2 HRBR instances: a minimal lifecycle interface

The VDOM runtime does *not* understand HRBR internals. It only requires that mount(host) returns an object with the following hooks:

- destroy(): void -- required (DOM and internal cleanup).
- dispose?: () => void -- optional (reactivity teardown, unsubscribe).

- update?: (values: any) => void -- optional (to push new values).

This interface matches the objects returned from the HRBR runtimes (compiled blocks and fallback mounts), but it's also easy to fake in tests.

18.1.3 Creating HRBR vnodes: `hBlock()`

The helper hBlock() in src/h.ts wraps a mount factory into an HRBR vnode:

```
hBlock((host) => mountCompiledBlock(def, host, compiledSlots))
```

This is the shape the compiler produces (or that you can use manually for experiments).

18.2 Mounting: why HRBR gets its own host container

Mounting happens in src/mount-dom.ts. The main mountDOM() switch includes a new case:

```
case DOM_TYPES.HRBR: createHrbrNode(vdom, parentEl, index)
```

18.2.1 `createHrbrNode()`: mount into a dedicated `<span>`

createHrbrNode() creates a dedicated host element (currently a span) and inserts it into the parent:

- It sets host.dataset.hrbr = '1' to make the host easily recognizable in the DOM.
- It stores the host on the vnode: vdom.host = host.
- It calls vdom.mount(host) and stores the returned instance.
- It sets vdom.el = host.

The decision *to set vdom.el to the host container* is subtle but critical.

Why `vdom.el = host` matters

Relax's patcher has a generic "replace" path:

```
if nodes aren't equal ⇒ find index of oldVdom.el in parentEl.childNodes
⇒ destroy old ⇒ mount new at the same index.
```

That index scan only works if oldVdom.el is actually a direct child of parentEl. For HRBR, the direct child is the wrapper host, not the inner DOM nodes produced by the HRBR runtime.

So createHrbrNode() intentionally makes the wrapper the unit of ownership:

- moves are moves of the wrapper

- removals remove the wrapper
- inner churn stays encapsulated in HRBR

Even though the vnode type includes an el?: Element field described as "first element produced by the block" in src/h.ts, the mount code currently treats el as the wrapper host. This aligns with "VDOM owns the wrapper, HRBR owns the inside".

A small but useful marker: `data-hrbr="1"`

The wrapper uses host.dataset.hrbr = '1'. This isn't required for correctness, but it's very handy when inspecting mixed trees in DevTools: you can quickly spot the boundaries between VDOM-managed and HRBR-managed DOM.

18.3 Patching: stability by mount factory identity

Patching happens in src/patch-dom.ts. The HRBR case delegates to patchHrbr(old, next).

18.3.1 Fast path: stable mount function means reuse

patchHrbr() first checks referential identity:

```
if (old.mount === next.mount) { copy host/instance/el; return; }
```

This mirrors component identity rules: if the "type" of the subtree hasn't changed, we preserve the mounted instance.

In the actual implementation, the fast path copies internal bookkeeping from the old vnode to the new vnode:

- host
- instance
- el

That copy step is important because the patcher returns the *new* vnode object, and subsequent patches/destroys must be able to reach the original HRBR host and instance.

18.3.2 Slow path: mount changed means dispose+destroy+remount

When mount changes, patchHrbr() treats it as a subtree replacement *inside the same outer HRBR vnode type*:

- reuse the same host container
- if the instance supports dispose(), call it
- call destroy() (required)
- clear the host's children

- call next.mount(host) and store the new instance

Note the interplay with areNodesEqual():

- If areNodesEqual(old, next) returns *false*, patchDOM() goes through the full replace path (destroy old vnode + mount new vnode at the same index).
- If equality says they are the "same type" (both HRBR), then patchHrbr() is responsible for remounting when the mount factory changes.

This makes HRBR a first-class citizen in the patcher without teaching the patcher HRBR internals.

18.3.3 A note on where remounts happen

There are two remount scenarios and they get handled in different layers:

- **VNode type changes** (or HRBR mount identity changes per areNodesEqual()): handled by the generic replace-path in patchDOM() (destroy + mount at the same index).
- **Same vnode type** but HRBR mount changes: handled by patchHrbr() (dispose/destroy/clear into the same wrapper).

The integration tests (Section ??) mostly focus on the second path, because the first isn't HRBR-specific.

18.4 Destroying: one call site, two optional hooks

Destroying happens in src/destroy-dom.ts via removeHrbrNode(vdom).

The HRBR instance contract is designed to work for both compiled blocks and fallback runtime objects:

- destroy(): void is required.
- dispose(): void is optional (signals cleanup / reactive graph unsubscribe).

The destroy path:

1. call instance.dispose() if present
2. call instance.destroy() if present
3. remove the host element itself from the DOM

Because the host wrapper is removed, the HRBR subtree can't leak DOM nodes even if the inner runtime forgets to clean up.

18.4.1 Why both `dispose()` and `destroy()`?

This pattern matches the rest of the HRBR runtime in this book:

- dispose() is for disconnecting from reactive graphs (signals/effects) and releasing scheduled work.
- destroy() is for deterministic DOM cleanup, including nested mounts.

The VDOM runtime calls both (when present) so HRBR implementations can keep those concerns separate.

18.5 Tests-as-spec: the integration suite

The file src/__tests__/hrbr-integration.test.ts is the executable contract for everything above. It contains four key behaviors.

18.5.1 Test harness: what this suite is actually exercising

This isn't a unit test of the HRBR runtime itself; it's testing the integration points:

- hBlock() produces a vnode with type: DOM_TYPES.HRBR.
- mountDOM() correctly creates the wrapper host and calls the mount factory.
- patchDOM() decides reuse vs remount using mount identity.
- destroyDOM() tears down and removes the wrapper.

18.5.2 1) Mount an HRBR block returned from a VDOM component

The first test constructs a component whose render returns hBlock(mount). The mount function appends a child to the provided host and returns an object with destroy().

Assertions:

- mounting the VDOM mounts the HRBR region (DOM contains HRBR)
- the mount function is called exactly once
- destroying the VDOM destroys the HRBR region (DOM becomes empty)

18.5.3 2) Event semantics: bind to the host component

The second test verifies the *host component binding* convention. An HRBR block is mounted from inside a VDOM component, and the block defines an event slot: { kind: 'event', path: [], name: 'click' }.

The key detail is the mountBlock(..., { hostComponent: this }) option. It ensures the handler's this matches the VDOM component instance, mirroring Relax's VDOM event behavior.

Assertions:

- clicking the button calls the handler
- the handler sees this === vdom.component
- the argument is an Event

This test is a key "mixed tree" guarantee: HRBR doesn't invent a new event binding model when embedded inside Relax VDOM; it can reuse Relax's convention of binding handlers to the component instance.

18.5.4 3) Patch: stable mount identity preserves instance

The third test mounts a component that always returns hBlock(mount) where mount is stable. It then patches the VDOM with another render of the same component.

Assertions:

- mount was called only once
- the instance array length remains 1
- DOM remains hello

This is the "no remount" fast path.

Notice what is (and isn't) asserted:

- It doesn't require HRBR to implement any diffing--the integration layer simply promises not to tear it down.
- It asserts the VDOM runtime carries the old instance forward by copying host/instance into the new vnode.

18.5.5 4) Patch: changing mount identity remounts

The fourth test swaps currentMount from MountA to MountB and triggers a component re-render via updateState().

Assertions:

- MountA called once
- MountB called once
- DOM changes from A to B

The mechanism used here is also important: the test triggers a component re-render via updateState(), which causes the component's render() to run again and return an HRBR vnode with a different mount reference. This matches real application code where mount factories can accidentally change due to closures.

18.6 Practical guidance (and foot-guns)

- **Keep mount factories stable** if you want patch-time reuse. In practice, that means hoisting the mount function out of render (or memoizing it).
- **Always implement destroy()** in your HRBR instance. Without it, detaching the host wrapper still removes DOM, but reactive resources and event hooks can leak.
- **Pass hostComponent** when you mount blocks from components if you want event handlers to behave like VDOM events.
- **Don't assume DOM ownership beyond the host.** The VDOM layer treats the host wrapper as the unit of reconciliation.

18.6.1 Common failure modes in mixed trees

- **Accidental remounts:** if you create the mount function inline (a new closure every render), it will remount every patch. Prefer a stable function identity.
- **Forgetting to clear:** patchHrbr() explicitly clears host before remount. If your custom HRBR instance assumes an empty host but doesn't clean on destroy(), you'll get surprising leftover nodes in custom setups.
- **Destroying the wrong thing:** the integration treats the wrapper host as the owned node. Removing inner nodes manually from outside HRBR can break assumptions inside the HRBR runtime.

In Part IV's later chapters, we'll connect this interop layer to the compiler output: the compiler emits hBlock(() => mountCompiledBlock(...)) for static templates and hBlock(() => mountFallback(...)) when it detects dynamic structure.

18.7 Online resources

- DOM node insertion and wrapper ownership patterns:
  MDN: Node.appendChild,
  MDN: Node.insertBefore,
  MDN: ChildNode.remove.
- "Identity is state" in UI runtimes (keys and function identity are the rough analogs used here):
  React docs: list keys,
  Vue docs: maintaining state.
- Event handler binding and the meaning of this in JavaScript:
  MDN: this.
- Data attributes used as lightweight, inspectable markers:
  MDN: Using data attributes.

19. Compiler architecture overview (prototype + Babel transform)

This chapter explains how Relax's HRBR compiler takes TSX/JSX input and produces output that mounts *compiled blocks* (the fast path) or routes to the *fallback runtime* when the template's structure is dynamic.

The compiler currently exists in two forms:

- A TypeScript-AST prototype compiler in compiler/index.ts (used as an executable spec for template extraction).
- A Babel plugin scaffold in compiler/jsx/transform.ts (used by build tooling to transform real TSX/JSX files).

The purpose of having both is pedagogical and practical:

- The TS compiler is a small, deterministic "reference" that focuses on *template + slot-path extraction*.
- The Babel plugin is the real integration point for user codebases.

19.1 The output contract (what the runtime needs)

Before we talk about parsing and visitors, it's worth making the compiler/runtime boundary explicit. The compiler's job is to emit *data and closures* that satisfy a small runtime API. Everything in the implementation is organized around making that boundary stable.

19.1.1 Tiny contract (2 bullets)

Given a JSX expression, the compiler must be able to produce one of:

- **Block mode (fast path):** defineBlock(templateHTML, slots) plus a mount factory that calls mountCompiledBlock(def, host, compiledSlots).
- **Fallback mode (safe path):** a mount factory that calls mountFallback(...) and delegates structural reconciliation to the runtime.

The runtime intentionally does not know how compilation works; it only requires these shapes.

The HRBR runtime expects two main pieces of data.

19.1.2 1) A `BlockDef`: stable template + slot metadata

Defined in runtime/block.ts, a BlockDef contains:

- templateHTML: string the stable HTML string that will be parsed once.
- slots: Record<string, BlockSlot> the mapping from a slot key to its write kind and a DOM path.

A BlockSlot describes *what* to write (text/attr/class/style/prop/event, etc.) and *where* to write it (a childNodes index path).

19.1.3 2) A `CompiledSlot[]` list: how to read live values

At runtime, mountCompiledBlock(def, host, compiledSlots) needs an array of objects shaped like:

```
{ key: string, read: () => unknown }
```

The compiler can't always inline the read closures in a pure data file. In particular, the TypeScript prototype compiler deliberately returns no reads; it records expression source code so a caller with lexical scope can build closures.

In contrast, the Babel transform *can* emit read: () => expr arrow functions directly into the output program, precisely because it is already transforming a file with the correct lexical scope.

19.2 Routing decision: block vs fallback

The compiler's core architectural decision is: *can this TSX be represented as a stable HTML template?* If not, we must use the fallback reconciler.

19.2.1 Dynamic structure

The term **dynamic structure** means "the set/shape/order of DOM nodes can change as state changes." Examples:

- conditionals: cond ? : null
- loops/lists: items.map(...)
- fragments that change child count
- dynamic tag names or component tags
- spreading attributes (often implies unknown keys)
- event handlers (in the current compiler phase) if they can't be represented as static bindings

Compiled blocks are designed for *stable structure* with dynamic *values*. When structure is dynamic, we let the fallback runtime reconcile children.

19.2.2 How "block vs fallback" relates to Chapter ??

The compiler output doesn't mount directly into the VDOM. Instead, it produces a *mount factory* that matches the HRBR vnode contract described in Chapter ??. That is why the VDOM type DOM_TYPES.HRBR stores a mount(host) function: it is the common denominator between compiled blocks and fallback regions.

19.3 TypeScript prototype compiler (Phase 5 reference)

The TypeScript implementation lives in compiler/index.ts. It provides two public APIs:

- compileTSXToBlock(source, opts?) -> CompileResult
- compileTSXOrFallback(source, opts?) -> CompileOrFallbackResult

19.3.1 compileTSXToBlock(): parse extract render template

Inputs

- source: string: a single-root TSX expression (or a statement containing one).
- opts.fileName?: helps TypeScript provide useful source locations.
- opts.strictTemplate? (default true): reject patterns that can't be expressed as a static HTML template.

Main pipeline

1. Parse the input with TypeScript: ts.createSourceFile(..., ScriptKind.TSX).
2. Find the first expression via findFirstExpression().
3. Ensure the root is JsxElement or JsxSelfClosingElement.
4. Build a TemplateBuild using buildTemplate().
5. Return { block: {templateHTML, slots}, slots: [], meta }.

19.3.2 Determinism: why the TS compiler is a good executable spec

The implementation in compiler/index.ts is deliberately small and deterministic:

- It parses with TypeScript's parser (so syntax + TSX rules match what users write).
- It walks the syntax tree into a simple string renderer that produces templateHTML.
- It computes slot paths using an explicit childIndex counter.

This makes it ideal for teaching and for pinning down invariants (especially path semantics) before committing to a more complex Babel visitor implementation.

19.3.3 Paths: the compiler and runtime must agree

The most important shared invariant between compiler and runtime is how a "path" is interpreted.

In this prototype compiler, each element maintains a childIndex counter, and each child node increments the counter as it is rendered. When the compiler emits a slot, it records the current path (a list of child indices) that identifies a target node under the template root.

Why this works The block runtime later parses templateHTML into a DOM tree and walks childNodes[index] repeatedly to locate the slot target.

The critical detail is that the compiler counts *every child node* (elements *and* text nodes), because childNodes includes text nodes. In compiler/index.ts this is represented by PathState{ childIndex }.

This is also why the compiler uses explicit sentinels for some dynamic expressions: it needs to ensure a real, countable node exists at the path.

19.3.4 Text expression handling: sentinel + slot path

In renderJSX(), when a child is JsxExpression:

- the compiler marks meta.hasDynamicStructure = true
- it allocates a slot key sN
- it emits a unique sentinel string $_{h}rbr_{s}N_{intothetemplate}$
- it records a text slot whose path points to that text node
- it records the expression's source text as a "read" entry

This is a deliberately conservative design: any child expression could be a conditional or a list, so the prototype treats it as structure-changing. The Babel compiler (later in this book) uses heuristics to allow some expressions as "text-like" without falling back.

19.3.5 Void elements and template correctness

The TS compiler treats HTML void tags specially via the VOID_TAGS set. For these tags it emits <input ...> rather than <input ...></input>. This matters because the runtime parses templateHTML through <template> and relies on the browser's HTML parser to build a stable node tree. Producing non-HTML-canonical markup can shift the parsed tree and break slot paths.

19.3.6 Attributes: literals vs expression slots

The prototype supports:

- boolean attributes: <div hidden />

- string literal attributes: `<div id="x" />`
- expression attributes: `<div className={count()} />`

Expression attributes allocate slot keys and record kind and name:

- `className` normalizes to `class`
- `style` maps to a style slot
- anything else becomes an attr slot

The compiler also records lane hints (per slot) and "signal-like" hints:

- `meta.lanesByKey`: a compiler-to-runtime hint used by schedulers in later phases
- `meta.signalLikeByKey`: a heuristic that recognizes `foo()` style getters

Even when later phases choose to ignore (or reinterpret) these hints, keeping the meta separate from the structural template makes it possible to evolve scheduling policy without changing the fundamental block ABI.

19.3.7 Fallback decision wrapper: `compileTSXOrFallback()`

`compileTSXOrFallback()` runs `compileTSXToBlock()` and then returns:

- `{ kind: 'fallback', reason: 'dynamic-structure' }` if `meta.hasDynamicStructure` is true
- otherwise `{ kind: 'block', ...result }`

This is the high-level policy the real Babel transform is evolving toward.

Note the important nuance: `compileTSXToBlock()` still returns a BlockDef even when `meta.hasDynamicStructure` is true. That mirrors how the Babel plugin can still hoist block defs for other expressions in the same file even if one subtree needs fallback.

19.4 Babel transform scaffold (real-world integration)

The Babel plugin entrypoint is exported as `hrbrJsxTransform` from `compiler/jsx/index.ts` and re-exported from `compiler/index.ts`. The implementation lives in `compiler/jsx/transfor`

19.4.1 Roles of the Babel plugin

Compared to the TS prototype, the Babel plugin has additional responsibilities:

- Operate on full program files, not just an isolated expression string.
- Inject runtime imports: `defineBlock`, `mountCompiledBlock`, and optionally `mountFallbac`
- Generate wrapper helpers (e.g. $_{h}rbrBlock)thatconnectemitteddataliteralstoruntimecalls.$
- Derive stable slot keys in dev mode (optionally) so diffs are readable.

19.4.2 "Block mode" is best-effort: return `null` to fall back

The internal function `buildTemplateFromJsxElement()` tries to convert a JSX element into:

- `templateHTML`
- `slots map`
- `compiledSlots array literal (with read: () => expr)`

If it sees unsupported patterns (events, dynamic elements, structural expressions), it returns null and marks that fallback imports are needed.

19.4.3 Babel transform: what "best-effort block mode" means in code

In `compiler/jsx/transform.ts`, block mode is essentially:

- Try to turn a `JSXElement` into `{ templateHTML, slots, compiledSlots }` using `buildTemplateFromJsxElement()`.
- If any unsupported structure appears, return null.
- The caller then chooses a fallback mount factory instead of a compiled-block mount factory.

The current transform is strict on purpose (e.g. it rejects events and spreads in block mode), because the project is trying to keep the block subset crisp while the fallback runtime is still evolving.

19.4.4 Slot key strategies: sequential vs stable (dev mode)

The Babel compiler contains two different slot key allocation styles:

- **Default (prod-like):** sequential keys `s0, s1, ....`
- **Dev mode:** location-based keys like `s_text_12:18` derived from source positions.

Location-based keys are helpful because they reduce "slot churn" in diffs and make debugging invalid-path errors more readable.

19.4.5 Lane pragmas (compiler-to-scheduler hint)

The Babel plugin recognizes comment pragmas in expression containers:

```
className={/*@lane transition*/ cls()}
```

It extracts the lane and stores it on the slot metadata. Later chapters will connect this to the deterministic scheduler.

Concretely, `extractLanePragmaFromExpressionContainer()` scans leading comments for a pattern like:
`exttt@lane transition`. When found, the compiler attaches `lane: "transition"` to the slot metadata object.

19.5 Tests-as-spec

The compiler has three layers of tests:

- Unit compilation tests (shape of emitted template/slots): compiler/__tests__/compiler
- Snapshot tests for stability: compiler/__tests__/compiler.snapshot.test.ts
- End-to-end tests that mount the result in jsdom:
  - compiler/__tests__/compiler.e2e.test.ts (TS prototype -> mountCompiledBlock)
  - compiler/__tests__/babel-transform.e2e-jsdom.test.ts (Babel output -> evaluate -> mountCompiledBlock)

In the rest of this section, we'll treat the tests as a specification of the compiler/runt boundary. We won't run them here, but reading them tells you what changes are allowed without breaking the public contract.

19.5.1 Unit tests (TS): compile + mount sanity

extbfFile: compiler/__tests__/compiler.test.ts

This test anchors the smallest happy path:

- compileTSXToBlock() accepts a simple intrinsic element with a dynamic attribute expression.
- The caller supplies CompiledSlot{ key: 's0', read: () => name() }.
- mountCompiledBlock() produces DOM matching the template, and the dynamic slot lands in the right place.

One subtle detail: the test disposes the mounted block via mounted.dispose(). This keeps the contract clear: compiled mounts are still reactive, and disposal is part of the lifecycle.

19.5.2 TS end-to-end: compile -> mount -> update without recreating root

In compiler/__tests__/compiler.e2e.test.ts:

1. Create a signal: const [count, setCount] = createSignal(1).
2. Compile TSX: (Count).
3. Mount via mountCompiledBlock(block, host, { key: 's0', read: () => count() }).
4. Update the signal, wait for the scheduler (microtask + setTimeout(0)).
5. Assert the root element identity is preserved and only the class changed.

This is the key end-to-end contract for compiled blocks.

This test also documents a scheduling assumption: it comments that HRBR scheduling uses setTimeout(0) by default in tests, so updates require waiting at least one timer tick.

19.5.3 Babel transform unit tests: shape of emitted program

extbfFile: compiler/__tests__/babel-transform.test.ts

These tests assert that the plugin is producing the intended *program shape*:

- it injects an import from the configured runtime path (default relax/hrbr)
- it references defineBlock and mountCompiledBlock
- it hoists a const __hrbr_block_0 = defineBlock(...)
- it emits a wrapper mount function of the form host => __hrbrBlock(host, def, slots)

The snapshot test in this file is especially important: it pins down the exact emitted structure (templateHTML string, slots map, and compiledSlots array with read: () => expr).

The suite also asserts two "developer experience" guarantees:

- lane pragmas are propagated to slot metadata
- dev mode can emit location-based keys

19.5.4 Babel transform errors: the intentional block-mode subset

extbfFile: compiler/__tests__/babel-transform.errors.test.ts

This suite documents what is currently *rejected* in block mode:

- event handlers (e.g. onClick)
- spread attributes (<div {...props} />)
- fragments (<>...</>)
- non-intrinsic tags (components like <Foo />)

Even if future versions route these to fallback instead of throwing, the existence of these tests is useful: it makes the current compilation subset explicit and prevents accidental expansion that would silently generate incorrect templates.

19.5.5 Babel transform E2E (jsdom): evaluate emitted literals + mount

extbfFile: compiler/__tests__/babel-transform.e2e-jsdom.test.ts

This test checks something subtle but crucial: the compiler output is not just syntactically valid--its emitted *data* can actually be evaluated and mounted.

Notable aspects:

- It extracts the object literal passed to defineBlock(...) from the emitted text.
- It extracts the compiledSlots array literal from the wrapper call.

- It evaluates those two pieces with new Function(...) while providing cls() and name() from the test scope.
- It mounts using mountCompiledBlock and asserts identity preservation across updates.

This approximates how "real code" will behave: the compile-time output must work with runtime scope (signals) and produce stable DOM.

19.5.6 Babel fallback E2E (jsdom): structural expression routes to mountFallback

extbfFile: compiler/__tests__/babel-transform.fallback.e2e-jsdom.test.ts

This suite pins down the fallback routing policy:

- A conditional expression in children is treated as **dynamic structure** (unsafe for block mode).
- The transform output must include mountFallback.
- The emitted module is evaluated, and calling App() returns a mount function.
- Updates are reactive: signal changes trigger DOM updates without throwing.

The test includes a comment that the MVP fallback spec doesn't yet clear conditional text in all cases; the contract being asserted is deliberately smaller: the pipeline compiles, mounts, and updates safely.

19.6 Where we go next

Chapter ?? will zoom in on the Babel visitor model and show how the plugin emits the full final shape that user code will run. We will also connect the "fail block mode" paths to the fallback runtime and reconciler described earlier.

19.7 Online resources

- TypeScript compiler API: parsing source and working with AST nodes:
  TypeScript Wiki: Using the Compiler API.
- Babel plugin authoring and visitor model:
  Babel docs: plugins,
  Babel docs: plugin API.
- JSX specification details (useful when deciding what constitutes "dynamic structure"):
  JSX spec (historical reference).
- Why HTML parsing matters for template correctness (void elements, tree building, etc.):
  MDN: HTML elements reference.
- Source maps basics (relevant to dev mode and debugging transforms):
  Source Map Revision 3 Proposal.

20. Babel plugin in detail (JSX transform)

This chapter explains the current HRBR Babel transform in compiler/jsx/transform.ts. It is intentionally a *scaffold*: it covers a useful subset, has crisp failure modes, and is wired end-to-end in jsdom tests.

The transform's job is to recognize TSX/JSX expressions that can be compiled into a stable BlockDef + CompiledSlot[] and to rewrite them into a mount(host) => instance factory that the VDOM interop layer can consume (see Chapter ??).

When it can't safely compile a JSX expression into a stable template, it routes to mountFallback().

20.1 Public entrypoint and options

The plugin is exported as hrbrJsxTransform:

- compiler/jsx/index.ts: re-exports the default plugin
- compiler/index.ts: re-exports it as part of the public compiler API

20.1.1 Tiny contract: what this plugin must guarantee

The transform is successful if it guarantees two things:

- The emitted program is **self-contained**: all runtime references are imported (or intentionally omitted when not needed).
- Any transformed JSX expression becomes a **mount factory** of the form (host: Element) => MountedInstance, matching the HRBR vnode contract from Chapter ??.

Everything else (slot heuristics, fallback subset size, key readability) is an implementation choice that can evolve as long as the tests keep passing.

The options type HrbrJsxTransformOptions (in compiler/jsx/transform.ts) controls a few key behaviors:

- runtimeImport: module path for runtime helpers (default "relax/hrbr")
- mountWrapperName: helper function name used to call mountCompiledBlock (default $_hrbrBlock$)
- dev: enables more readable output and debug-oriented metadata
- stableSlotKeys: when true, derive slot keys from source locations (defaults to dev)

20.2 High-level rewrite shape

Given a JSX expression like:

```
const App = () => <div className={cls()}>Hi {name()}</div>
```

The plugin emits three conceptual things:

1. An import of runtime functions:
   import { defineBlock, mountCompiledBlock } from "relax/hrbr"
2. A module-scope const ${}_{h}rbr_{b}lock_{N}=defineBlock(...)containing$:
   3. templateHTML: "<div ...>..."
   4. slots: { key: { kind, path, name?, lane? }, ... }

- A replacement of the JSX expression with a mount factory:
  host => ${}_{h}rbrBlock(host,{}_{h}rbr_{b}lock_{N},[\{key,read\},...])$

This shape matches the interop contract in src/h.ts: an HRBR vnode is just a mount function that returns an instance.

20.3 Program-level state and synthesis (imports + wrappers)

The plugin keeps per-file state variables:

- needsRuntimeImport: whether to inject defineBlock/mountCompiledBlock
- needsFallbackImport: whether to also import mountFallback
- needsWrapper: whether to emit the wrapper function ${}_{h}rbrBlock$
- nextBlockId: a counter used to generate ${}_{h}rbr_{b}lock_{0}, {}_{h}rbr_{b}lock_{1}, \cdots$
- currentFileName/currentFileCode: used for context-rich error messages

These are reset in the Program.enter visitor and materialized in Program.exit:

- ensureRuntimeImport(programPath) uses unshiftContainer('body', ImportDeclaration).
- ensureWrapper(programPath) appends function ${}_{h}rbrBlock(host,def,slots)\{returnmountCompiledBlock(def,hos$

Implementation note: this scaffold avoids depending on Babel "types" helpers by emitting raw AST node objects.

20.3.1 Why inject at Program.exit?

The plugin defers emissions until Program.exit so it can answer questions like:

- Did we ever successfully compile a block? If so, we need defineBlock and mountCompiledBlock.

- Did any subtree route to fallback? If so, we need mountFallback too.
- Did we generate a wrapper call? If so, we need to emit the wrapper function body.

This design keeps the output minimal: files that contain no transformable JSX don't get imports they don't use.

20.4 The core visitor: JSXElement(path)

The plugin's heart is the JSXElement visitor. For each JSX element in an expression position, it calls:

```
const built = buildTemplateFromJsxElement(path)
```

20.4.1 Success path: compile into a block

When built is non-null:

1. Mark needsRuntimeImport = true and needsWrapper = true.
2. Insert a module-scope const ${}_{h}rbr_{b}lock_{N}=defineBlock(\{templateHTML,slots\})$.
2. Replace the JSX expression with:
   host => ${}_{h}rbrBlock(host,{}_{h}rbr_{b}lock_{N},built.compiledSlots)$

The slot objects are turned into AST literals through literalizeSlot(slot).

Hoisting: why defineBlock(...) lives at module scope

The plugin uses unshiftContainer('body', ...) on the program node to emit:

extttconst ${}_{h}rbr_{b}lock_{N}=defineBlock(templateHTML,slots)$

Hoisting matters for two reasons:

- **Performance:** we want template parsing and slot-path validation to happen once per module, not once per render.
- **Identity:** the mount factory will refer to a stable ${}_{h}rbr_{b}lock_{N}constant.$

20.4.2 Failure path: route to fallback

When built is null, the plugin attempts a *minimal fallback transform*.

Why fallback exists Block mode requires a stable template. Expressions that can produce different node shapes at runtime (conditionals, JSX inside expressions, etc.) must be executed by the fallback runtime, which uses the reconciler.

Current fallback subset The fallback routing in transform.ts is intentionally small:

- It only supports intrinsic lowercase root tags.
- It only supports children that are JSXText or JSXExpressionContainer.
- It allows only static string attributes (and ignores dynamic attrs).
- It rejects spread attributes and event handlers.

In this subset, the plugin synthesizes a render() function (a closure-free data + expression composition) and emits:

```
host => mountFallback(host, () => ({ children: [spec] }))
```

Each child spec contains:

- create(): Element creates the element and sets attributes/text
- patch(node) updates textContent based on the latest expressions
- optional key if a key= attribute is present

This is enough to prove the end-to-end pipeline and to connect the compiler to the fallback runtime from earlier chapters.

20.4.3 Fallback synthesis: what gets captured (and what doesn't)

The fallback compiler intentionally avoids capturing complex structures. It builds a renderFn with the shape:
exttt() => (children: [spec])

Where spec.create() and spec.patch(node) both close over the user expressions.

Important implications:

- **Reactivity comes from runtime:** mountFallback re-runs renderFn under tracking and schedules reconciliation.
- **This is not a general JSX renderer:** it is a deliberate minimal bridge used to validate the "block or fallback" architecture.

20.5 Template extraction: buildTemplateFromJsxElement()

The function buildTemplateFromJsxElement() is a best-effort compiler for a supported subset. It returns either a full block build or null.

20.5.1 Supported subset (today)

In block mode, the transformer supports:

- intrinsic lowercase tags only (e.g. div, section)
- nested elements
- static string attributes
- expression attributes -> slot kinds class/style/attr
- dynamic text children only when the expression is "text-like"

It explicitly rejects (in block mode):

- event handlers (onClick= etc.)
- fragments (<>...</>)
- non-intrinsic tags (components, capitalized)
- spread attributes ({...props})

20.5.2 How paths are computed in the Babel transform

Like the TS prototype compiler (Chapter ??), this transform must produce childNodes-based paths.

In renderElement(...) the transform:

- keeps a per-parent counter childState.i that increments for each child node
- uses pathToHere plus the current index to form paths like [3,1]
- emits a placeholder text node (a single space) when it needs to guarantee the existence of a text node for a dynamic text slot

This is why the snapshot test shows a template that contains *extra whitespace* in places where text slots are inserted. That whitespace isn't an aesthetic choice; it's a structural anchor.

20.5.3 Slot keys: sequential vs location-based

Slot keys are allocated in allocSlotKey().

- Default: sequential keys s0, s1, ...
- Dev mode (stableSlotKeys=true): keys include source location: s$_{<}kind>_{<}line>:<col>$

This is validated in compiler/__tests__/babel-transform.test.ts.

One nuance: location-based slot keys depend on Babel preserving loc information and comments; this is why dev mode is framed as "debug output" rather than a semantic requirement.

20.5.4 Lane pragmas: /*@lane transition*/

The helper extractLanePragmaFromExpressionContainer() scans leading comments and looks for:

```
@lane (sync|input|default|transition|idle)
```

If present, the emitted slot metadata includes lane: "...". The tests verify that the transformed output contains the lane strings.

20.5.5 Text-like safety

In isTextLikeExpr(), the transform tries to reject expressions that can yield JSX elements or otherwise change DOM structure.

Examples treated as *not text-like*:

- returning JSX elements/fragments
- conditionals/logicals that could yield elements (conservatively rejected)

When an expression is accepted as text-like, the transform:

- emits a sentinel space in the template (so a text node exists)
- records a text slot pointing to that position
- emits a compiled slot read: () => expr

The predicate isTextLikeExpr(expr) is the key safety gate. It explicitly rejects:

- JSXElement and JSXFragment (obviously structural)
- ConditionalExpression and LogicalExpression (conservatively structural for now)

This conservative choice is why the fallback e2e test uses show() ? name() : " to force fallback routing: it is not considered "text-like" yet.

20.6 Failure modes are part of the API

This compiler phase aims for clarity and determinism. Unsupported syntax produces explicit errors via fail(path, message) which attempts to build Babel code-frame errors.

The file compiler/__tests__/babel-transform.errors.test.ts locks in these failure modes:

- event handlers should throw
- spread attributes should throw
- fragments should throw

- non-intrinsic tags should throw

While these errors might eventually become "route to fallback" behaviors, they're still valuable today: they prevent the compiler from silently generating templates that would be incorrect (for example, by treating an event handler like an attribute).

20.7 End-to-end tests: jsdom execution proves output shape

Two e2e suites validate that the emitted output is not just syntactically correct but *runnable*.

20.7.1 Block mode e2e: transformed output mounts and updates

compiler/__tests__/babel-transform.e2e-jsdom.test.ts compiles a small component, extracts the generated defineBlock(...) literal and the compiledSlots array literal, evaluates them with new Function(...) and mounts the result with mountCompiledBlock.

The key invariant: updating a signal updates the DOM without recreating the root element.

This suite also proves a deeper property: the emitted output can be *partially evaluated*. The test extracts only the defineBlock(...) object literal and the compiled slots array literal, then evaluates them with new Function. So the plugin isn't just emitting runnable code; it's emitting code whose pieces behave like data.

20.7.2 Fallback routing e2e: structural text expression goes to `mountFallback`

compiler/__tests__/babel-transform.fallback.e2e-jsdom.test.ts compiles an expression with a structural conditional:

```
<div>{show() ? name() : "}</div>
```

This should include mountFallback in output and remain reactive under signal updates.

The spec here is intentionally small:

- it asserts that mountCompiledBlock should *not* be called
- it asserts that mounting succeeds and subsequent signal changes cause updates
- it does *not* assert perfect minimal DOM operations yet

That matches the current purpose of fallback in this compiler phase: correctness-first safety for dynamic structure.

20.8 What to improve next (guided by tests)

The current transform is intentionally conservative. The next incremental steps typically look like:

- add safe event slot support (mapping onClick to HRBR event slots)
- support fragments by lowering them to concatenated children in block mode
- expand fallback routing to support nested children specs (not just a single root spec)
- improve text-like analysis so common conditionals can stay in block mode when both arms are text

The rule is always: add a targeted test first (unit + e2e), then widen syntax support.

When you widen the subset, remember the three invariants all previous chapters rely on:

- slot paths must match childNodes
- stable mount function identity enables reuse in VDOM interop
- fallback must remain safe when structure can change

20.9 Online resources

- Babel plugin authoring and visitor lifecycle:
  Babel docs: plugins,
  Babel docs: plugin API.
- JSX AST node shapes (useful when reading JSXElement, JSXText, and JSXExpressionContai
  visitor logic):
  Babel parser AST spec.
- Source maps and why transforms try to preserve locations:
  Source Map Revision 3 Proposal.
- "Static template + dynamic slots" as a general pattern (useful for intuition):
  Svelte docs: compiler-first model.

21. Devtools and instrumentation (optional)

Relax's HRBR runtime is designed to be deterministic and debuggable. This chapter documents the small *optional* devtools surface in runtime/devtools.ts and how the rest of the runtime reports events into it.

The design goals are:

- **Low overhead by default**: counters for heavyweight instrumentation (DOM ops, allocations) are disabled unless explicitly enabled.
- **Always hookable**: events can be observed via a hook regardless of whether counting is enabled.
- **Stable semantics**: the runtime emits a small number of event types at well-defined points.

21.1 The API surface: five functions and two helpers

All of devtools lives in runtime/devtools.ts. The public API is:

- setDevtoolsHook(hook | null) register/unregister a global event sink
- getDevtoolsCounters() get a defensive copy of counters
- resetDevtoolsCounters() zero all counters
- setInstrumentationEnabled(boolean) enable counting for expensive instrumentation
- isInstrumentationEnabled() query the flag

And the helper emission functions used across the runtime:

- emitDevtoolsEvent(event) core increment + forward logic
- emitDomOp(op) and emitAlloc(kind, count?) convenience wrappers

The reactivity and scheduling layers re-export these through runtime/index.ts so "public HRBR" users can import them.

21.1.1 Tiny contract (what you can rely on)

The devtools module is deliberately small, but it still has a stable contract:

- If you install a hook with setDevtoolsHook(), it will receive *every* emitted event.
- Counters are best-effort metrics intended for debugging/perf investigation, not correctness.
- Expensive counters (DOM ops, allocations) are disabled by default and must be explicitly enabled.
- getDevtoolsCounters() returns a copy (so the internal counter object can't be mutated externally).

21.2 Event model

Devtools events are a tagged union DevtoolsEvent:

- slotWrite: emitted when a block slot write is attempted
- scheduleComputation: emitted when a reactive computation is scheduled
- flushStart / flushEnd(durationMs): emitted by the deterministic scheduler
- domOp(op): emitted by the block runtime around DOM mutations (counted only when instrumentation is enabled)
- alloc(kind, count?): emitted by allocation sites (counted only when instrumentation is enabled)

The idea is that you can build either:

- simple counters (the built-in ones), or
- richer tooling (log streams, flamegraphs, dashboards) via the hook.

21.3 Counters: what is tracked and when

The internal counters live in a module-local object $_counters$:

- slotWrites
- scheduledComputations
- flushes
- flushTimeMs
- domOps (instrumentation gated)
- allocs (instrumentation gated)

21.3.1 Defensive reads: getDevtoolsCounters()

getDevtoolsCounters() returns { $..._counters\}.Thisisimportant : testsanduserscan'tmutateinternals$

The corresponding reset function, resetDevtoolsCounters(), mutates the internal object in-place. That choice keeps object identity stable for any internal references while still providing deterministic test setup.

21.3.2 Why DOM ops and allocs are gated

Counting DOM operations and allocations often requires adding emission calls in hot loops. The devtools system separates *event forwarding* from *counter increments*:

- Events are always forwarded to the hook.
- Counter increments for domOp and alloc only happen when $_instrumentationEnabledistrue.$

This split is implemented directly in runtime/devtools.ts:

- emitDevtoolsEvent() always forwards to $_{h}ook?.(event)$.
- Only the domOp and alloc branches check $_{i}nstrumentationEnabledbeforetouchingcounters$.

21.4 Emission: increment + forward

emitDevtoolsEvent(event) does two things:

1. updates counters based on the event type
2. forwards the event to $_{h}ookifpresent$

In pseudo-code, the key policy is:

```
always forward to hook; only count domOp/alloc when enabled
```

The convenience wrappers emitDomOp() and emitAlloc() add one more optimization:

- if instrumentation is disabled *and no hook is installed*, they return immediately

This prevents needless event objects from being allocated on the hot path.

21.4.1 A subtle performance point: "no hook" short-circuit

Notice the early return:

```
extttif (!enabled && !hook) return
```

This means there are three distinct operating modes:

- **Default mode (recommended for production):** no hook; instrumentation disabled. Most hot-path emitters become no-ops.
- **Streaming mode:** hook installed; instrumentation disabled. Events still flow, but counters don't track domOps/allocs.
- **Instrumentation mode:** instrumentation enabled (hook optional). Counters track domOps/allocs.

21.5 Where events come from in the runtime

The core runtime emits events from a few key paths:

21.5.1 Blocks: slot writes and DOM operations

In runtime/block.ts:

- Every slot write emits { type: 'slotWrite', slotKind, slotKey }.
- Many DOM mutations are annotated with emitDomOp('setText'), emitDomOp('setAttribute'), etc.

This provides a cheap "what changed" signal even without full DOM diffing.

Slot writes: one event per attempted update

In runtime/block.ts, every slot application includes:

```
extttemitDevtoolsEvent({ type: 'slotWrite', slotKind: slot.kind, slotKey:
key })
```

This is intentionally emitted even when the runtime later decides to skip a DOM write (for example when values are Object.is equal). That makes slotWrites a measure of "reactive updates attempted" rather than "DOM updates performed".

DOM operations: a vocabulary of `domOp` strings

The block runtime calls emitDomOp(...) around operations like:

- setText
- setAttribute / removeAttribute
- setProperty
- styleSetProperty / styleRemoveProperty
- addEventListener / removeEventListener

The string payload keeps the event cheap to allocate and flexible to extend.

21.5.2 Scheduler: flush timing

In runtime/scheduler.ts, the flush loop emits:

- flushStart at the beginning
- flushEnd(durationMs) at the end

This is the basis for simple performance dashboards.

In runtime/scheduler.ts, these events are emitted inside flush(...):

- flushStart at the beginning
- flushEnd(durationMs) at the end (duration computed using the scheduler's now() source)

Because the scheduler is designed to be deterministic (queues sorted by lane/priority and stable promotion rules), these timing events are also useful when you want to prove "same inputs produce the same flushing pattern".

21.5.3 Signals: scheduled computations

In runtime/signals.ts, scheduling a non-sync computation emits scheduleComputation. The devtools counter is intentionally ‘how many computations were scheduled‘, not ‘how many ran‘.

The payload includes optional name and lane metadata taken from the computation object. That metadata isn’t required for semantics, but it makes event streams much more useful in practice (you can group by lane or by effect name when debugging).

21.6 Tests-as-spec

There are two test files that define the intended behavior.

21.6.1 runtime/__tests__/devtools.test.ts

This suite checks two high-level properties:

- **slotWrites are counted** when mountBlock() applies initial values and when update() applies new values.
- **scheduledComputations and flush metrics are counted** when the scheduler runs and when signals schedule work.

In the scheduler portion, the test uses createScheduler(requestFlush: (cb) => cb()) to make flushing deterministic.

Reading this test as a contract gives you the intended semantics:

- Slot writes must be observed during both mount and update.
- Scheduler flush events must be emitted even when a custom requestFlush runs synchronously.
- Signal scheduling should increment a counter even if the actual effect execution happens immediately (depending on lane/batching).

21.6.2 runtime/__tests__/instrumentation.test.ts

This suite locks in the *gating* behavior:

- without instrumentation enabled, domOps stays at 0 even though the runtime emits DOM op events
- after calling setInstrumentationEnabled(true), DOM-related operations increment domOps

This is a very important behavioral line for performance: many hot paths call emitDomOp() all the time, and the test documents that these calls must not inflate counters unless instrumentation is explicitly enabled.

21.7 How you might use this in practice

Two common patterns:

- **Counters dashboard**: poll getDevtoolsCounters() and display deltas per second.
- **Event stream**: set a hook via setDevtoolsHook((e) => { ... }) and record or filter events.

A practical rule of thumb:

- enable instrumentation only in debug builds or perf investigations
- leave it off in production unless you have a carefully designed sampling strategy

21.7.1 A simple pattern: hook + manual aggregation

If you want detailed breakdowns without touching the built-in counters, install a hook and aggregate in userland:

- Count per-domain domOps (e.g. setText vs setAttribute)
- Track flush durations into histograms
- Group schedule events by lane

This works even when instrumentation is disabled, because the hook still receives events.

21.8 Online resources

- Performance measurement pitfalls and best practices (timers, main-thread work, long tasks):
  web.dev: RAIL performance model,
  MDN: performance.now().
- Observability patterns for UI runtimes (profilers, tracing, counters):
  OpenTelemetry: overview.
- Source maps and error locations are why devtools often care about source positions:
  Source Map Revision 3 Proposal.

22. Benchmarks methodology (repeatability + fairness)

Benchmarks are where performance stories can go wrong fast. This chapter documents Relax's benchmark harness under benchmarks/ and the methodology guidelines captured in docs/benchmarks.md.

The core goals are:

- **Repeatability**: seeded randomness, stable profiles, and a clear warmup/measure separation.
- **Fairness**: compare like-for-like work (same DOM outcomes, similar update patterns), and control build modes (dev vs prod).
- **Interpretability**: report metrics that explain tail latency (p95) and not just averages.

22.1 Benchmarks run in a real browser

The harness is designed to run in a browser environment:

- Uses requestAnimationFrame to align work with frames.
- Uses performance.now() for timing.
- Bundles through Rollup configs that target browser: true.

See benchmarks/index.html and the entry script benchmarks/main.ts.

22.1.1 A small but important detail: Node/jsdom fallback exists

The harness is designed for browsers first, but benchmarks/harness.ts contains a compatibility fallback:

- now() prefers performance.now() but falls back to Date.now().
- nextFrame() prefers requestAnimationFrame but falls back to setTimeout(..., 16).

This is *not* intended to make Node-based benchmarking "accurate". It's there so that small smoke runs and tooling can still function in non-browser environments.

22.2 The harness contract

The core harness is benchmarks/harness.ts.

22.2.1 Benchmark case shape

A benchmark case is a BenchmarkCase:

- name: stable identifier for reporting
- setup(host): returns a controller with
  tick() perform one unit of work
  teardown() clean up resources and DOM
- optional commit(): a barrier that resolves only once the framework has committed for the tick

The commit() hook is how the harness stays fair across frameworks that batch work differently (e.g. React flush/paint boundaries).

22.2.2 Tiny contract for a fair BenchmarkCase

To keep comparisons honest, the benchmark cases should follow a simple contract:

- tick() should perform one repeatable unit of work and avoid random extra allocations.
- teardown() must remove event handlers / timers / subscriptions (avoid cross-case leaks).
- If a framework batches updates asynchronously, provide commit() that resolves after work is actually committed.

22.2.3 Warmup vs measure windows

runBenchmark() splits execution into two windows:

- **Warmup frames** (default 30): let JIT, memoization, layout caching, and internal queues settle.
- **Measure frames** (default 120): record per-tick durations.

Implementation details in runBenchmark():

- For each tick: await nextFrame() record t0 run tick() await commit() compute dt.
- A **frame drop** is counted when dt > frameBudgetMs (default 16.7ms).

22.2.4 Reported metrics: avg + p95 + drops

The harness reports:

- avgMs average time per tick
- p95Ms 95th percentile tick time (tail latency)
- frameDrops number of frames over budget

Percentiles are computed on sorted samples with a simple index selection. This is good enough for stable comparisons at the harness scale.

The harness computes p95 with:
extttidx = floor(p * (n - 1)) (clamped). This is a simple deterministic estimator, which is a feature here: the same sample list always produces the same percentile.

22.3 Profiles and seeded runs

Repeatability relies on standard profiles and seeded randomness.

22.3.1 Profiles via URL parameters

The profile parser lives in benchmarks/profile.ts.

It reads query params from benchmarks/index.html:

- profile: quick | default | stress
- seed: integer seed for randomized update patterns
- warmup, frames, budget: overrides for fine-grained control

Defaults are intentionally "reasonable and stable" for local runs.

Implementation note: the profile parser clamps overrides to avoid accidental runaway configurations:

- warmup is clamped to [0, 10000]
- frames is clamped to [1, 100000]

Seed values are forced into uint32 range using >> 0, which keeps the PRNG behavior consistent.

22.3.2 Seeded RNG

Randomized patterns (e.g. which list indices change each tick) should be deterministic. Benchmark code uses a seeded RNG in benchmarks/rng.ts.

A key methodology rule: the seed must be part of the report output so results can be reproduced.

The benchmark RNG implementation is Mulberry32 (benchmarks/rng.ts). It is simple, fast, and deterministic--perfect for selecting indices or toggling boolean flags in a way that remains repeatable across runs.

22.4 The benchmark runner UI

The main entrypoint benchmarks/main.ts wires a button click to sequential benchmark runs.

Key details:

- It resets host.innerHTML = " between runs so each case starts clean.
- It prints formatted results to a <pre>.
- It runs both HRBR and VDOM variants, and includes adapters for React/Solid.

Because different frameworks have different setup/teardown costs, the harness keeps the per-tick loop stable and encourages cases to isolate the steady-state behavior.

22.4.1 Why host.innerHTML = " between runs matters

Clearing between runs gives you two strong properties:

- **Isolation:** DOM nodes and event listeners from a previous framework/case do not bleed into the next.
- **Repeatability:** each case starts from the same initial DOM shape.

This is also why safe teardown() behavior is part of the case contract: the harness clears DOM, but frameworks may have timers/schedulers that need explicit disposal.

22.5 Cases: what they measure

Cases live under benchmarks/cases/. The intent is to cover:

- **Pure text throughput:** text-1m
- **Large lists with sparse updates:** list-10k-1pct
- **Many small components/widgets:** widgets-200
- Extra comparisons in cases/more.ts

Important fairness guideline: each case should produce the same DOM output across frameworks (or as close as is reasonable).

22.5.1 Case study: text-1m

In benchmarks/cases/text-1m.ts, the HRBR variant builds the smallest possible compiled block:

- templateHTML: ‘ ‘ includes an explicit space so there is a real text node.

- Slot path [0,0] means: root child 0 is , and its child 0 is the Text node.
- tick() updates a signal in chunks of 10,000 so work remains frame-visible.

This isn't a "real app" workload, but it is an excellent microbenchmark for measuring pure slot-write throughput.

22.5.2 Case study: `list-10k-1pct`

In benchmarks/cases/list-10k-1pct.ts, the cases delegate to example mount helpers:

- HRBR: mountList10k1pctHRBR(host, 10000)
- VDOM: mountList10k1pctVDOM(host, 10000)

The contract is the same for both: return a controller with update1pct() and dispose(). Methodologically, that matters: both implementations are being asked to do the same "one tick" of work.

22.6 Build modes matter (dev vs prod)

Some frameworks change scheduling and checks based on process.env.NODE$_E NV.Thebenchmark Ro$

- Rollup config: rollup.benchmarks.config.mjs
- It uses @rollup/plugin-replace to define process.env.NODE$_E NV.$

NPM scripts in package.json expose common modes:

- build:bench:dev
- build:bench:prod

To be fair, comparisons should run with comparable modes (usually prod vs prod).

22.6.1 What the Rollup configs actually do

The benchmark bundles are built through two rollup configs:

- rollup.benchmarks.config.mjs (full UI runner)
- rollup.benchmarks.smoke.config.mjs (perf-smoke gate)

Both use @rollup/plugin-replace to inject:
extttprocess.env.NODE$_E NV =' production' bydefault.$

This matters for fairness because some libraries (notably React) perform extra checks and different scheduling policies in development builds.

22.7 Perf smoke (CI-friendly)

For CI and "obvious regression" detection, the repo includes a reduced entry:

- benchmarks/perf-smoke.ts
- benchmarks/perf-smoke.html

It builds via rollup.benchmarks.smoke.config.mjs into benchmarks/dist-smoke/.

22.7.1 What perf-smoke is actually specifying

You can read benchmarks/perf-smoke.ts as an executable specification of the harness:

- it obtains a profile via getBenchmarkProfile()
- it forces short defaults unless overridden (default profile becomes quick)
- it runs a small fixed set of cases (currently text-1m and list-10k-1pct, HRBR variants)
- it emits a JSON report that includes profile, timestamp, and result rows
- it enforces very wide thresholds to catch catastrophic regressions

Even though this isn't a Vitest file, it plays the same role as tests in earlier chapters: it defines what "performance sanity" means for this repository.

22.7.2 Why the thresholds are wide

perf-smoke.ts prints a JSON report to console.log and then checks very wide thresholds:

- fail only when avgMs > 250 or p95Ms > 500

This is intentional: CI machines are noisy. The smoke test is designed to catch catastrophic regressions (infinite loops, accidental O(n^2)$behavior), notmicro-optimizations.$

This is also why the gate uses two different lenses:

- avgMs: for sustained runaway behavior
- p95Ms: for severe tail latency spikes

Tail latency matters because user experience is dominated by occasional slow frames.

22.8 Instrumentation and benchmarks: keep them separate

Optional runtime instrumentation exists in runtime/devtools.ts (Chapter ??).

Methodology rule: keep instrumentation disabled during benchmarks unless you are explicitly measuring it. Otherwise, the measurement includes the measurement.

In practice, if you want to correlate benchmark results with DOM op counts:

  - run once with instrumentation off to measure baseline
  - run again with instrumentation enabled (and accept the overhead) to measure the profile of operations

Comparing these two runs helps you understand whether a regression is dominated by DOM work, scheduling, or allocation overhead.

22.9 Checklist: how to run a fair comparison

A practical checklist grounded in this repo's structure:

1. Build benchmark bundle in a known mode (dev or prod) and record it.
2. Use a named profile (quick/default/stress) so warmup and frames are consistent.
3. Fix a seed and include it in the report.
4. Run multiple times and compare distributions (p95 and drops), not just averages.
5. If a result looks surprising, isolate the case and profile it (browser performance tools).

22.10 Online resources

- Measuring performance in the browser and why alternatives to Date.now() matter: MDN: performance.now().
- About frame budgets and why 16.7ms is the common baseline: web.dev: rendering performance.
- Why warmup matters (JIT, inline caches, tiered compilation): V8 blog (background on optimization and warmup effects).
- Percentiles and tail latency as a reporting best practice: Wikipedia: percentile, Google SRE: tail latency (The Tail at Scale).
- General benchmark hygiene and common pitfalls: benchmarking.sites (community notes).

23. Release-quality discipline: tests, typecheck, lint, builds

This chapter is about the habits that make a codebase shippable. Relax is a teaching framework, but it also treats its test suite as an executable spec and its build pipeline as part of correctness.

The goal of this chapter is not to list commands--it's to explain *why* each quality gate exists, what it catches, and how to debug failures without guessing.

If you only remember one thing: each gate is a different lens on correctness. Typecheck defends your *data model*, lint defends your *human factors* (consistency and obvious mistakes), tests defend your *observable behavior*, and builds defend your *published artifact*.

23.1 The quality-gates contract

A "release-quality" change in this repo is expected to be green on:

- **Typecheck**: TypeScript compilation with -noEmit
- **Lint**: ESLint rules for consistency and obvious mistakes
- **Unit/integration tests**: Vitest (jsdom)
- **Build**: Rollup bundles for ESM + CJS, plus type bundle generation on publish

These are not redundant; they fail on different classes of bugs.

23.1.1 A tiny contract (inputs, outputs, failure modes)

It helps to think about the pipeline as a function that takes a working tree and returns publishable artifacts:

- **Input**: TypeScript source under src/, plus experimental surfaces under runtime/, compiler/, and the TSX runtime under relax-jsx/.
- **Output**: compiled JS bundles in dist/esm/ and dist/cjs/ and a declaration bundle in dist-types/.
- **Success**: consumers can import from the package exports map and get the runtime semantics the tests describe.
- **Failure modes**: "works in tests" but fails to typecheck; "works locally" but fails to bundle; "bundles" but exports the wrong files.

23.2 Typecheck: make illegal states unrepresentable

The repo's primary TypeScript config is tsconfig.json. A few settings worth noticing:

- strict: true -- the baseline.
- noUncheckedIndexedAccess: true -- forces you to handle missing keys.
- exactOptionalPropertyTypes: true -- optional means "may be absent", not "may be undefined".
- jsx: react-jsx and jsxImportSource: relax-jsx -- enables the TSX layer.

Two other settings are quietly important for how the codebase is built and consumed:

- moduleResolution: Bundler: TypeScript resolves imports similarly to modern bundlers. In practice, this aligns the typechecker with how Rollup (and consumer bundlers) will interpret ESM.
- paths for relax-jsx: TS can typecheck TSX code that imports the JSX runtime via the virtual specifier relax-jsx. This is mirrored on the test side by the Vitest alias for relax-jsx.

The typecheck script (in package.json) runs:

```
tsc -p tsconfig.json -noEmit
```

23.2.1 Common typecheck failures (and how to debug)

- **Indexing errors**: caused by noUncheckedIndexedAccess. Fix by narrowing or checking bounds.
- **Optional property mismatches**: caused by exactOptionalPropertyTypes. Fix by removing explicit undefined assignments or changing types.
- **DOM typing issues in tests**: Vitest uses jsdom; make sure lib: ["DOM"] is present (it is).

23.2.2 Why there are multiple `tsconfig` files

This repo uses a small family of configs to keep responsibilities separate:

- tsconfig.json: the "truth" for developer typechecking. It enables declarations and maps, targets modern JavaScript (ES2022), and includes the DOM library.
- tsconfig.build-types.json: extends the base config but turns on *declaration-only* emit. This produces the publishable .d.ts surface under dist-types/.
- tsconfig.rollup.json and tsconfig.rollup.cjs.json: extend the base config but disable declaration emission. These exist to keep Rollup builds focused on JavaScript output (ESM to dist/esm/, CJS to dist/cjs/) without creating duplicate .d.ts trees.

This split is release discipline in practice: build tooling stays fast and predictable, while types remain a first-class published artifact.

23.3 Lint: keep the surface area consistent

Lint config lives in eslint.config.mjs. It uses the ESLint 9 flat config model and @eslint/js recommended rules.

The lint scripts are:

- npm run lint -- runs eslint on src
- npm run lint:fix -- applies safe auto-fixes

Note: linting only src is a choice. HRBR runtime and compiler code are primarily guarded by tests and typecheck, but could be added to lint scope later.

23.3.1 What lint catches that typecheck doesn't

TypeScript proves things about types, but lint is often better at catching:

- accidental unused variables or imports
- inconsistent patterns that make reviews harder
- "looks correct" code that has a known foot-gun shape

Even with a mostly-recommended ESLint config, getting a clean lint run is part of keeping diffs small and reviewable.

23.4 Tests: executable specs with jsdom

All tests use Vitest (see vitest.config.js). The environment is jsdom, which means:

- DOM APIs are available (document, elements, events)
- you can test DOM manipulation deterministically without a real browser

Test inclusion patterns:

- **/tests/**/*.test.ts
- **/tests/**/*.test.tsx

23.4.1 The test suite structure

This repo uses tests in three roles:

- **Unit tests**: small functions like diffs and helpers.
- **Integration tests**: end-to-end flows (mount → patch → destroy, SSR → hydrate, compiler → mount).
- **Property/fuzz tests**: seeded tests that exercise many update sequences (e.g. reconciler and signals).

If you change invariants (like reconciliation semantics), expect to update both unit tests and fuzz oracles.

23.4.2 Tests-as-spec example: TSX wiring

The TSX layer is a good example of how "typecheck + tests" combine into a stronger guarantee. The typechecker is configured to understand TSX (jsx: react-jsx and jsxImportSource: relax-jsx), but tests still need to verify the *runtime* behavior.

In src/$_{tests}$/$_{jsx-tsx.test.ts}$,thetestsetsupahostelementinjsdom,runsanexample,andassertsontheresultingHTML:

- it writes </div> into document.body
- it locates the host element and passes it into runHelloExample
- it asserts that the returned HTML matches Hello Relax</div>

The important release-quality point is the scope of the guarantee: this test doesn't just say "TSX compiles"--it says "the TSX runtime integrates with the DOM mounting layer and produces the expected structure".

23.4.3 Debugging failing tests (a workflow)

A practical workflow:

1. Re-run only the failing file (Vitest supports file targeting; use the repo scripts).
2. Read the failure output and find which invariant changed.
3. Locate the corresponding runtime function and its other tests.
4. Add a minimal regression test that isolates the bug.
5. Only then, fix the implementation and re-run the focused test.

When failures are timing-related (scheduler, signals), prefer deterministic schedulers and explicit ticking (the repo already uses this pattern).

23.4.4 What to do when a test failure seems "impossible"

The fastest way to debug tough failures is to make the implicit invariant explicit:

1. Find the smallest assertion that fails.
2. Name the invariant in one sentence (e.g. "patch preserves node identity when keys and tag match").
3. Search for other tests that mention the same concept.
4. Read the corresponding implementation with the invariant in mind.

This avoids the common trap of chasing symptoms across unrelated modules.

23.5 Builds: ESM, CJS, and types

Relax ships multiple entrypoints (see package.json exports):

- "." -- the VDOM library
- "./hrbr" -- experimental HRBR runtime
- "./compiler" -- experimental compiler

Rollup builds:

- ESM: rollup.config.mjs -> dist/esm/
- CJS: rollup.config.cjs.mjs -> dist/cjs/

Type declaration builds:

- npm run build:types uses tsconfig.build-types.json (emit declarations only)
- npm run prepack runs both JS builds and types before publishing

23.5.1 Why builds are a quality gate

A green test suite doesn't guarantee:

- exports are correct
- bundling doesn't pull in unexpected dependencies
- generated output is structurally valid for consumers

For example, the compiler bundle is large; build output sizes help you notice unexpected growth.

23.5.2 Build discipline: match exports to outputs

From package.json, Relax publishes multiple entry points (".", "./hrbr", "./compiler"). Release discipline means each export must be correct in three dimensions:

- **ESM path** points at something in dist/esm/
- **CJS path** points at something in dist/cjs/ (with .cjs entry file names)
- **Types path** points at something in dist-types/

It's easy to accidentally break this mapping when renaming files, changing entry points, or restructuring module boundaries. That is why prepack builds both JS bundles and types before publish.

23.6 Release checklist (practical)

A minimal release checklist aligned with this repo:

1. **Typecheck**: ensure TS still models your invariants.
2. **Lint**: ensure code style and basic safety checks pass.
3. **Tests**: run the suite; if you're changing HRBR internals, keep an eye on fuzz determinism.
4. **Build**: confirm ESM + CJS outputs compile.
5. **Bench smoke (optional but recommended)**: build:bench:smoke to catch catastrophic perf regressions.

23.7 Evidence from this repo

This chapter is grounded in real scripts and configs:

- package.json scripts: typecheck, lint, test, build, prepack
- vitest.config.js (jsdom environment)
- eslint.config.mjs (flat config)
- Rollup configs for bundling
- TypeScript configs for declaration and rollup builds

In the next section of the book (appendices), we'll turn these practices into a learner-friendly "build it from scratch" checklist and a test-reading guide.

23.8 Online resources

These references mirror the key moving parts in this chapter:

- TypeScript tsconfig reference (compiler options, strictness flags): https://www.types
- exactOptionalPropertyTypes (why "optional" is not the same as "possibly undefined"): https://www.typescriptlang.org/tsconfig#exactOptionalPropertyTypes
- noUncheckedIndexedAccess (making missing keys explicit): https://www.typescriptlang.c
- TypeScript JSX / jsxImportSource: https://www.typescriptlang.org/docs/handbook/jsx.ht
- ESLint flat config (ESLint 9+): https://eslint.org/docs/latest/use/configure/configur
- Vitest docs (jsdom environment, include patterns, reporters): https://vitest.dev/guic
- jsdom project (what it simulates and what it doesn't): https://github.com/jsdom/jsdor
- Rollup configuration and multi-entry builds: https://rollupjs.org/configuration-optic
- Node.js package.json exports field (how entry points are resolved): https://nodejs.or

Appendix: notation

This book uses file/path.ts to reference source files. Symbol names are written like h() or mountDOM().